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Stability study of a bipedal robot subject to strong streams

Bachelor's Degree in Mathematics

Bachelor's Degree in Aerospace Engineering Technologies

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*To Lola,
to the Deer,
to the Aeros-In-A-Row,
and to you.*

Abstract

This project addresses the challenge of stable robotic locomotion in environments with significant external perturbations, such as strong streams. The main objective is the development of an integral architecture able to detect and react to these forces, to keep balance.

First, a simplified hydrodynamic force model was proposed that avoids the complexity of CFD; second, a momentum observer was theoretically implemented and validated, able to estimate external wrenches acting on the center of mass of the robot; third, the *Whole Body Control* framework was improved to take into account external forces, ankle offsets and torque limits; and fourth, a strategy to react to the forces was implemented based on LIP and DCM theories.

Experimental results show the effectiveness of the observer, achieving a resistance increment to external forces of over a 100% with respect to the previous framework state. This work lays the foundation of future investigations in humanoid operation in environments with strong perturbations.

Keywords: Humanoids, Whole-Body Control, Momentum Observer, Hydrodynamics, LIPM, DCM

AMS codes [\[38\]](#): 34H05, 49M25, 65L20

Resumen

Este proyecto afronta el desafío de locomoción robótica estable en entornos con perturbaciones externas significativas, como riadas intensas. El objetivo principal es el desarrollo de una arquitectura integral capaz de detectar y reaccionar a estas fuerzas para mantener el balance.

Primero, un modelo hidrodinámica simplificado fue propuesto que evita la complejidad del CFD; segundo, un observador de momento fue teóricamente implementado y validado, capaz de estimar fuerzas y momentos externos sobre el centro de masas del robot; tercero, el marco de *Whole Body Controller* fue mejorado para incluir fuerzas externas, offsets en los tobillos y límites de torque; y cuarto, una estrategia para reaccionar a las fuerzas fue implementada basada en teorías de LIP y DCM.

Los resultados experimentales muestran la efectividad del observador, consiguiendo incrementos de resistencia de sobre un 100% con respecto al estado anterior del controlador. Este trabajo sienta las bases de futuras investigaciones en operación de humanoides en entornos con fuertes perturbaciones.

Palabras clave: Humanoides, Control de Cuerpo Completo, Observador de Momento, LIPM, DCM

Códigos AMS [\[38\]](#): 34H05, 49M25, 65L20

Resum

Aquest projecte afronta el desafiament de la locomoció robòtica estable en entorns amb pertorbacions externes significatives, com ara riuades intenses. L'objectiu principal és el desenvolupament d'una arquitectura integral capaç de detectar i reaccionar a aquestes forces per mantenir l'equilibri.

Primer, es va proposar un model hidrodinàmic simplificat que evita la complexitat del CFD; segon, es va implementar i validar teòricament un observador de moment, capaç d'estimar forces i moments externs sobre el centre de masses del robot; tercer, el marc del *Whole Body Controller* es va millorar per incloure forces externes, desplaçaments als turmells i límits de parell motor; i quart, es va implementar una estratègia per reaccionar a les forces basada en les teories de LIP i DCM.

Els resultats experimentals mostren l'efectivitat de l'observador, aconseguint increments de resistència de més d'un 100% respecte a l'estat anterior del controlador. Aquest treball estableix les bases per a futures investigacions en l'operació d'humanoides en entorns amb fortes pertorbacions.

Paraules clau: Humanoides, Control de Cos Sencer, Observador de Moment, LIPM, DCM

Codis AMS [\[38\]](#): 34H05, 49M25, 65L20

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Chapter 1

Introduction

1.1 Context and motivation

Bipedal robotics is one of the most promising and challenging branches of research within current research. Its ability to keep balance and move in complex environments poses an essential requirement for its application in many tasks, from trivial navigation jobs to search and rescue (SAR) missions.

Nonetheless, and unlike controlled lab or simulation environments, real scenarios are ruled by external, unpredictable perturbations that risk stability and the correct working of the robot. In this context, studying the response of the robot to external intense disturbances is a necessity from theoretical as well as practical points of view.

Motivated by recent storms and floods hitting Valencia (Spain) [13] and Austin (USA) [47], a new branch of research is proposed: robots walking in flooded environments. Indeed, one of the causes that are pointed out as an aggravating factor of the aftermath effects is the lack of quick response by authorities [23]. If a *army* of robots could be deployed in these kinds of demanding and challenging environments, many tasks, such as reconnaissance or high-priority missions that might be too dangerous for a human, could be performed. Although many types of platforms can be proposed to perform certain kinds of missions [16], the present work opts to propose a humanoid as the study type of robot. Precisely, Unitree's g1 humanoid. The fact that it is a humanoid and not another type of robot (such as an UAV, a snake-typed or a floating robot) gives it unique locomotion abilities to maneuver through complex and varied terrains, as well as more similarities to humans that can trigger better acceptance either among crew members and rescued victims.

1.2 State of the art

Stability study has been, from its origins, one of the main challenges in humanoid robotics, due to its delicacy. Unlike fixed robots, humanoids need to rely just on their feet (which are normally a small point of contact) to keep balance. External perturbations, such as unexpected contacts or wind, may destabilize the whole system and cause failures.

Throughout last decades, many improvements have been achieved. First, simple concepts and models such as the *Zero Moment Point* (ZMP) were introduced [51], and then more complex theories regarding capturability, trajectory planning and multicontacts were developed [21, 37, 46].

However, many of these models lacked unpredicted force estimation and were designed just for controlled environments. In this sense, many approaches were followed to tackle the problem. The first one is the use of momentum observers combined with *Model Predictive Control* (MPC) [26]. Another vanguard approach is the use of reinforcement or deep learning techniques [54, 55]. This last option has been the favorite one for the past years, given its great adaptability and generalization capabilities. The only problem that reinforcement learning and similar methods pose is that it is probabilistic, and even though the robot can behave well if it has been properly trained, there still exist a chance of catastrophic failure if it faces a complete unknown scenario. Model-based control ensures that the robot takes steps and moves exactly where the planner states, which is crucial in dangerous situations where a false step can cause the full loss of a hull.

Regarding fluid perturbations, the main object of study has been wind, and as expected, the main problem is the unpredictable nature of the fluid. Therefore, reinforcement learning techniques have been recently used to model the phenomenon [55]. Robots walking in water also have been studied, though only still waters were used and not rough waters [33]. What really has been a subject of study was human resistance against currents [6, 43, 53].

Therefore, a perfect gap arises between these two issues. Robot stability thresholds have never been deeply studied and leaves a perfect opportunity for research.

Finally, and as a remark, current state of CFD simulations show that a real-time run of fluid-solid interaction is not currently possible, though advanced GPU and ML techniques are getting it closer [40]. For the moment, one needs to stick to simpler and more traditional approaches to solve fluid tasks in real time.

1.3 Objectives and structure

The main objective of the project has been to study and improve the stability and locomotion of a robot subject to forces mimicking strong water streams, through the use of force observers and force rejection mechanisms.

From this general objective, the following are proposed:

1. To model external forces acting on the robot.
2. To implement a momentum observer able to detect precisely magnitude and direction of forces and moments acting on the robot's center of mass.
3. To integrate these estimations into an *Whole Body Controller* (WBC) framework.
4. To validate hypotheses and architectures via simulations and experiments, either in static and dynamic conditions.

Along the report, all of the necessary theory is presented for a clear understanding of what is being done.

There is a section corresponding to each of the objectives. In chapter 2, robot dynamics and external forces are modeled. In chapter 3 the observer is built. Due to the mathematical nature of the observer, its convergence and consistence are characterized for the simulation environments. In chapter 4, the WBC framework is presented, as well as the necessary changes that must be made to improve it. In chapter 5, robot locomotion is explained and disturbance rejection mechanisms are implemented. Finally, chapter 6 validates all the studied methods and architectures.

Chapter 2

Robot dynamics and external forces

The first task to address is to describe the dynamical model of the robot, as well as to estimate the forces acting on it. The main forces that are (externally) being applied are: its weight, contact, and fluid forces.

This chapter aims to describe and characterize all of the dynamics that are involved in the problem, from the robot's equations to these external forces.

2.1 Robot's dynamic model

First, the robot's dynamic model is introduced. This is the basis to later compute the theoretical momentum and compare it to the actual state of the robot.

Let $\{W\}$ denote the world frame, and

let $\{B\}$ denote the floating base of the robot. For g1, the latter is located at the hip.

Let x_b be the position of the origin of $\{B\}$ with respect to $\{W\}$. Accordingly, $\dot{x}_b$ and $\ddot{x}_b$ can be defined as the velocity and acceleration of such point.

Let ϕ_b represent the orientation of $\{B\}$ with respect to $\{W\}$. Again, ω_b and $\dot{\omega}_b$ can be defined as the angular velocity and acceleration of the floating base.

Lastly, let $\mathbf{q}$ be the joint positions of the robot. $\dot{\mathbf{q}}$ and $\ddot{\mathbf{q}}$ can be defined as well.

The g1 robot model has 27 joints, therefore $\mathbf{q} \in \mathbb{R}^{27}$ as well as $\dot{\mathbf{q}}$ and $\ddot{\mathbf{q}}$. It is obvious that $\mathbf{x}_b$, $\dot{\mathbf{x}}_b$, $\ddot{\mathbf{x}}_b$, $\boldsymbol{\omega}_b$ and $\dot{\boldsymbol{\omega}}_b$ are all elements of $\mathbb{R}^3$.

A vector $\mathbf{v}$ can be defined by concatenating the following:

$$\mathbf{v} := \begin{bmatrix} \dot{\mathbf{x}}_b \\ \boldsymbol{\omega}_b \\ \dot{\mathbf{q}} \end{bmatrix} \in \mathbb{R}^{33} \quad (2.1)$$

This is called the *generalized velocities* of the system.

With all of these elements, the dynamics equation for the robot can be expressed [26]:

$$\underline{\mathbf{M}}\dot{\mathbf{v}} + \underline{\mathbf{C}}\mathbf{v} + \mathbf{G} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \boldsymbol{\tau}_{joint} \end{bmatrix} + \boldsymbol{\tau}_{cf} + \boldsymbol{\tau}_{ext} \quad (2.2)$$

Where $\underline{\mathbf{M}}$, $\underline{\mathbf{C}}$ are the inertia and coriolis matrices; $\mathbf{G}$ is the gravity vector; $\boldsymbol{\tau}_{joint}$ is the joint torque vector; $\boldsymbol{\tau}_{cf}$ is the generalized external force vector due to contacts; and $\boldsymbol{\tau}_{ext}$ is the generalized external force due to any other reason.

2.2 Contact and external forces

The mapping between generalized external forces ($\boldsymbol{\tau}_{cf}$ and $\boldsymbol{\tau}_{ext}$) and the actual external wrench is given by the jacobian of the force-application point with respect to the floating base. For contact forces:

$$\boldsymbol{\tau}_{cf} = \sum_{i=1}^{n_c} \underline{\mathbf{J}}_{ci}^T \mathbf{w}_{ci} \quad (2.3)$$

n_c represents the number of points where contact forces are applied. For simplicity, only the centers of the two feet have been considered, so $n_c = 2$, although each foot can sense four forces in four different points. Let $\mathbf{f}_{ij} \in \mathbb{R}^3$ represent the force sensed in the j -th sensor of i -th foot, located in $\mathbf{x}_{ij} \in \mathbb{R}^3$ in world coordinates. Let $\mathbf{c}_i \in \mathbb{R}^3$ be the position of the reference point of the foot in world coordinates. Then, the total external wrench for foot i is computed as:

$$\mathbf{w}_{ci} = \sum_{j=1}^4 \left\{ \begin{bmatrix} \mathbf{f}_{ij} \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{f}_{ij} \times (\mathbf{c}_i - \mathbf{x}_{ij}) \end{bmatrix} \right\} \in \mathbb{R}^6$$

Finally, $\underline{\mathbf{J}}_{ci} \in \mathbb{R}^{6 \times 33}$ is the jacobian of the i -th foot with respect to the floating base.

For external forces:

$$\boldsymbol{\tau}_{ext} = \sum_{i=1}^{n_{ext}} \underline{\mathbf{J}}_{ext\,i}^T \mathbf{w}_{ext\,i} \quad (2.4)$$

n_{ext} is the number of points where external forces are applied. This number is uncertain *a priori*, but the approach to get it is clarified in section 3.2, as well as the algorithm to compute $\mathbf{w}_{ext\,i}$.

These external forces are modeled as fluid forces in section 2.4, but will not be known to the robot.

Before modeling such forces, some simplifications and assumptions must be done.

2.3 Geometric decomposition

When a robot is simulated in `pybullet`, it is decomposed into diverse links that are connected by joints. These links are meshes built by CAD software and refer to the various rigid pieces of the robot. They can be interpreted as the sections that make up all the limbs.

The list of the links of the robot's legs and their meshes is depicted in Annex A. The general view of the robot looks like this:

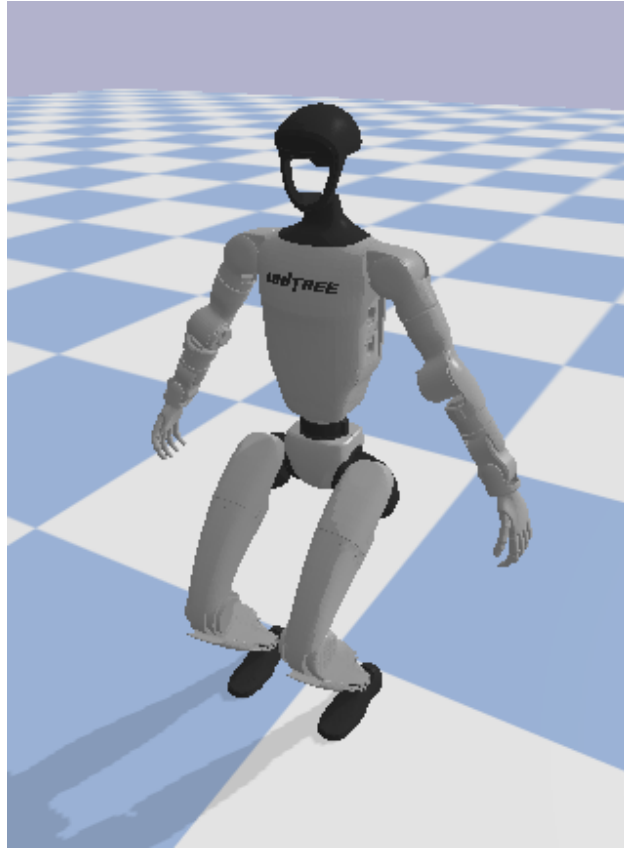


Figure 2.1: General view of the robot model.

The first approach that one thinks is to approximate the different sections of the limb as more well-known geometries, such as cylinders, that have been deeply studied in terms of drag and lift coefficients and under several conditions of yaw or Reynolds numbers. Since the objective of this work is to build a robust controller against perturbations, it is not necessary to know the exact magnitude and direction of the forces as long as the relative order of magnitude between all the forces is correct. Also, the current robot's limbs are not waterproof and differ from a hypothetical sealed design [49]. This final waterproof design could mean either a special suit [34] or a complete redesign of the limbs. In any case, this change of geometry justifies the simplification that is made.

The geometric decomposition model guarantees a correct approximation of the order of magnitude in the different sections of the robot, despite not taking into account the variations of the forces due to the turbulence of the fluid and the complex geometries of the robot.

From figure 2.1, it can be seen that the legs can be treated as yawed cylinders; foot height can

be neglected and forces acting on the feet can be neglected. For simplicity: the analysis is carried out with a maximum water height that does not exceed the height of the pelvis.

The approximated shape of the robot is, therefore:

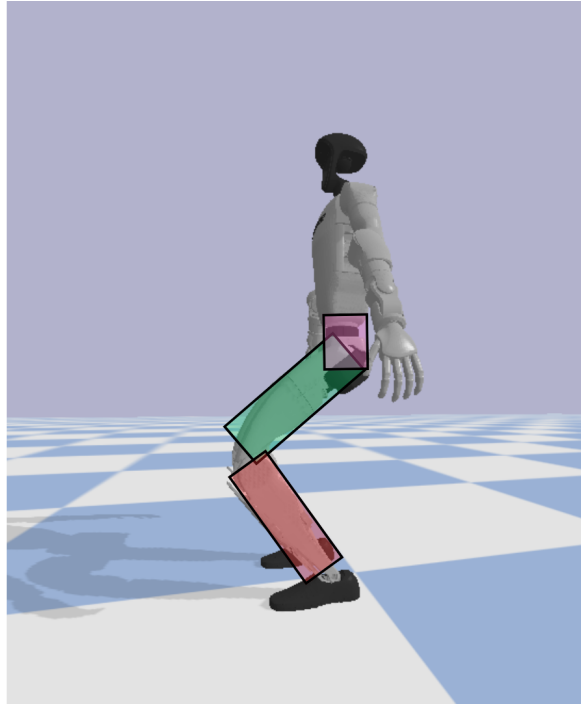


Figure 2.2: Approximation of the leg shape.

Where each section is composed of one or more links:

1. Shins: `left/right_knee_link`
2. Thighs: `left/right_hip_yaw_link + left/right_hip_roll_link + left/right_hip_pitch_link`
3. Pelvis: `pelvis + pelvis_contour_link`

This way, one approximates the shape of the leg to known geometries that can be easily analyzed. Again, it is just the order of magnitude of the forces that matter, so it is a feasible yet simple method to achieve that.

2.4 Fluid forces

Fluid forces are the set of all the forces that the stream is transmitting to the robot. Basically, it is the fluid giving some of its momentum to the robot, which is described by the Navier-Stokes equation:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f} \quad (2.5)$$

and the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (2.6)$$

By considering a spatial and temporal domain, with the appropriate boundary conditions, one can solve equation 2.5 and 2.6 and integrate the pressure along the robot's legs to get the net external force, plus its center of pressure. This equation is, however, extremely complex to solve (so much so that one of the Millennium Problems [30] is to prove whether it has or has not existence and uniqueness of solution). Therefore, current approaches to get approximate solutions for what this equation would yield are:

1. Reduced-order models
2. Computational Fluid Dynamics (CFD)

Given the complexity of the robot's geometry and the impracticality of running CFD cases in real time, reduced-order models must be used. The issue with CFD is that it is extremely computationally heavy if one wants accurate results. Fine boundary conditions are needed [7], but even so, errors and uncertainties still arise if the Reynolds number is high and the boundary layer is turbulent (as they are in this case of study) [32].

So, instead of solving all the solid-fluid interaction monolithically via CFD (where pressure, viscous friction, and buoyancy forces are obtained jointly), reduced-order models demand an explicit differentiation between these forces. Nonetheless, the advantage of these models is that the forces can be computed without the need for excessively complex computation, other than simple formulae.

- On the one hand, hydrostatic buoyancy forces follow the equation:

$$F_{float} = g\rho V \quad (2.7)$$

- On the other hand, dynamic forces due to fluid movement can be described as. From now on, they will be referred as drag forces:

$$F_i = \frac{1}{2}\rho v_i^2 l c_i \quad (2.8)$$

2.4.1 Buoyancy forces

Archimedes' force, or buoyancy, constitutes one of the significant forces acting on the robot. The classical expression for the buoyant force on a fully or partially submerged body is equation 2.7:

$$F_{float} = g\rho V$$

where g denotes the gravitational constant, ρ the density of the surrounding fluid, and V the volume of fluid displaced (i.e., the submerged volume of the body). This force can be computed almost directly, since it does not depend on complex parameters such as a drag coefficient. The only nontrivial element is determining the submerged volume.

For this, two implementations have been proposed:

1. The first one consists on building a uniform random point cloud inside the link's mesh and getting the fraction of submerged volume depending on the number of submerged points. However, this is too heavy computationally because it requires a large amount of points to reduce the error of the estimation. Also, there is no point on generating a huge point cloud for a link that will be likely submerged, such as the foot or the shin.

For these reasons, this method has been developed in annex D but not used in the end.

2. The second one leverages the geometric decomposition that has been described in section 2.3. It consists on assigning an *easy* geometrical figure (cylinders or prisms) to each of the sections, calculating the point where its axis intersects the water height and compute the submerged volume of the known figure.

Both methods can be understood with figure 2.3. Let's consider that the blue line represents water height.

In figure 2.3 (a), the set of the total points in the cloud is the red and blue set. Only red points (23 points) are above the water level, and only blue points (22 points) are below. Therefore, the fraction of volume that is submerged can be computed as $\text{\#submerged points} / \text{\#total points}$; in this case:

$$f_{point\ cloud} = \frac{22}{22 + 23} = 0.49$$

In figure 2.3 (b), the green cylinder can be considered as a good enough approximation of the shin. It is split by the water level, and its submerged volume can be computed as:

$$V_{submerged} = \pi r^2 L_{submerged}$$

With that, the fraction of submerged volume results ¹:

$$f_{cylinder} = \frac{V_{submerged}}{V_{total}} = \frac{L_{submerged}}{L_{total}} = 0.54$$

If the leg is tilted, $L_{submerged}$ is the distance along the cylinder's axis from the bottom of the cylinder to its intersection with the water plane. Due to the symmetry of the cylinder, the approximation of the submerged volume is the same as if it was not tilted.

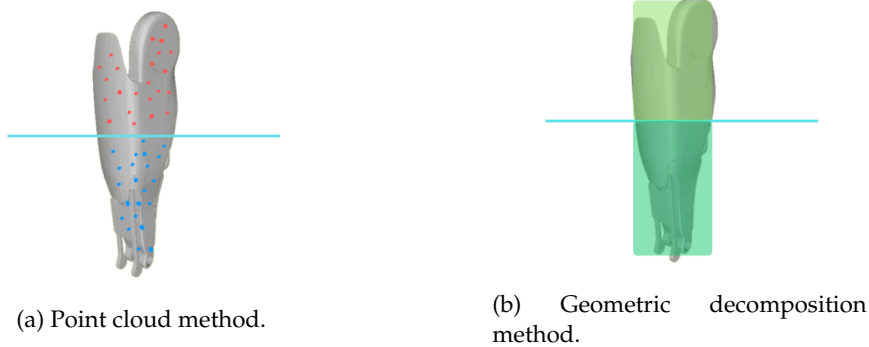


Figure 2.3: Archimedes' force computation methods.

¹These pictures are just examples of how these methods compute the submerged fraction of a link.

2.4.2 Drag forces

Dynamic forces due to fluid movement are the second main force that acts on the robot. Its expression follows equation 2.8:

$$F_i = \frac{1}{2} \rho v_i^2 l c_i$$

Where i refers to the direction of the force, ρ is the fluid's density, v_i is the fluid's velocity in direction i , l is a reference distance or area (if l is a distance, $[F] = \text{N/m}$; if it's an area $[F] = \text{N}$), and c_i is the coefficient for direction i .

The usual convention for the force name in the i -th direction is:

- **Drag** when i matches fluid direction.
- **Lift** when i corresponds to the perpendicular direction of drag in a certain reference plane.
- **Lateral** force closes the trihedron; it's perpendicular to lift and drag.

Back to equation 2.8: ρ , v_i and l are trivial to model. However, c_i is extremely sensitive and depends on a large amount of parameters (Reynolds, shape, roughness...). Normally, lift and drag coefficients for real components are obtained in wind (or water) tunnels, and if the geometry is simple enough, CFD can be used. However, as it was stated, the meshes of the robot's limbs might or might not be the final waterproof versions. Therefore, there's no point in modeling the coefficients for the real limbs in a wind tunnel (even if it is a 3d-printed, reduced version).

Regarding the CFD option, one needs to consider some preliminary conditions to even start thinking about this method as an option to compute the coefficients. First, let's pose the conditions that the robot is going to deal with:

- $\rho = 1000 \text{ kg/m}^3$
- $\mu = 10^{-3} \text{ Pa s}$

These are approximate physical properties for water flowing at 20°C. They can be taken as a good representative of their order of magnitude around that temperature.

Next, a good candidate for a reference distance is the robot's leg diameter. A representative for its order of magnitude is:

- $l = 0.1 \text{ m}$

Lastly, one needs to get an estimate for the water speed that the robot is going to be facing. No more than 3 m/s of stream speed will be considered, so let's take the following velocity as its order of magnitude:

- $v = 1 \text{ m/s}$

With this data, one can compute the approximate Reynolds regime that will be present in the problem:

$$Re = \frac{\rho v l}{\mu} = 10^5 \quad (2.9)$$

This value of Reynolds number places the problem very close to the critical region for external flow around a cylinder, which may trigger a phenomenon called *drag crisis* that can be extremely hard to model.

Drag crisis

When there's flow around an object, there exists a film surrounding the object called the boundary layer, where the fluid transitions from zero speed to the free-stream velocity. Depending on the Reynolds number, this layer can be laminar, turbulent, or it can transition from laminar to turbulent along the object's longitude. Also, unless the object and the conditions are ideal (or the Reynolds number is extremely low), the boundary layer ends up detaching from the object, creating a turbulent wake downstream [28]. The intensity of this turbulent wake is tightly correlated with the final drag that the object generates.

When the object is a cylinder and the flow is external ², three regions can be observed when looking at the Reynolds-Drag chart:

²This happens to all objects, not only cylinders.

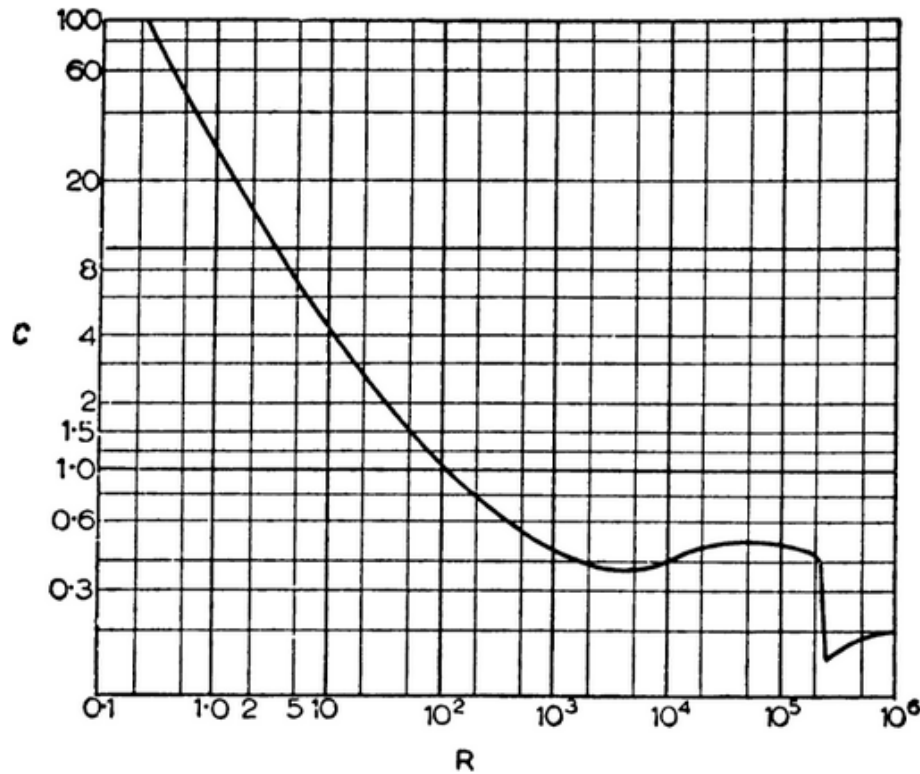


Figure 2.4: Reynolds number (R) versus drag coefficient (C) for external flow around a circular cylinder. Extracted from [35].

This can be understood by how the boundary layer is detached: *at the front there is a laminar boundary layer, then a turbulent layer, and finally the region beyond the line of separation* [35]. The Reynolds number has an important effect on the transition from laminar to turbulent boundary layer, and the intensity of such turbulence determines when it is detached. The later (the more to the rear of the object) it detaches, the narrower the turbulent wake becomes, thus reducing the drag coefficient. This transition from laminar to turbulent boundary layer (and its corresponding detachment shift) happens in a very short range of Reynolds numbers and has a significant impact on drag coefficient.

Modeling a CFD simulation around this regime is extremely difficult, since the slightest change in skin friction, yaw angle, geometry... can have a huge impact. A RANS (adapted to transition flows [36]) or a LES/DNS model could be used to build a CFD case, but the complications they entail are enormous:

- First, extremely careful planning must be carried out in order to build a proper CFD simulation. One would need to build a 3D mesh making sure that it is sufficiently refined and organized, that the wall functions (if used) match the parameters of the simulation...
- Then, one would need to run the first simulation. If the setup is correct enough to get an accurate value of drag, a 3D simulation will take ages to compute if one uses standard software such as Ansys or OpenFOAM [2, 42].
- Finally, the rest of the simulations are needed. The robot is going to be walking and/or moving, so the Reynolds number for the legs is not going to be constant. Plus, when a robot walks, it tilts its limbs. This tilting results in a change of the angle of attack of the object with respect to the flow, which in the end affects its drag, lift and lateral coefficients. Therefore, a set of simulations (for each limb) at different Reynolds and angles of attack must be performed.

Due to the high complexity of such approach, the author has considered that a CFD simulation of the actual (real) robot leg is unfeasible: the effort that it requires would take time away from other tasks, and the computation of external forces is not the main section of this work.

Leveraging the geometric decomposition proposed in the last section, the computation of c_i can be simplified with the aid of experimental data (the amount of CFD required even with this simplification is still unfeasible). This data comes from [9], a study of pressure distribution about yawed cylinders in the subcritical, critical and supercritical Reynolds regimes. Simplifying the legs as yawed cylinders, and integrating the pressure distribution along the circumference, one can get a drag coefficient adimensionalized by the normal velocity ($v_{\perp}$). Since the reference distance is l , the resultant force will be per unit length, so one must integrate the force along the cylinder's axis in order to get a net force.

In `pybullet`, no integration along the cylinder's axis is needed because this alternative method is proposed:

1. The i -th link's velocity is obtained by the corresponding `pybullet` function: v_i . This link is divided into several slices j , each of them will likely see a different water speed.
2. The relative velocity between the slices and the water is computed (water speed is known):

$\mathbf{v}_{rel\,ij} = \mathbf{v}_{water} - \mathbf{v}_i$. It is considered that the magnitude of angular velocity is low and that all points of the link have the same speed.

3. The relative yaw between the slice and the stream is computed: ϕ_{ij} .
4. The drag coefficient is interpolated based on Reynolds number (computed with the normal relative speed $\mathbf{v}_{rel\,ij\perp}$) and relative yaw: $c_{d\,ij}$.
5. The force is computed:

$$\mathbf{F}_{ij\perp} = \frac{1}{2} \rho l_i c_{d\,ij} \mathbf{v}_{rel\,ij\perp} \|\mathbf{v}_{rel\,ij\perp}\|$$

6. Finally, the force is applied in the centroid of the corresponding slice, its torque respect to the center of mass is computed (for practical reasons and for plots displayed in forthcoming sections). Note that this force has already a lift and drag component (lateral force is omitted due to symmetry) when it's projected to the *wind reference*.
7. This process is repeated for all the links, and a resulting external force and torque is obtained.

For an ultimate characterization of the model that has been built, the drag coefficient charts and a water velocity profile are described:

- The drag coefficient is obtained from the following chart:

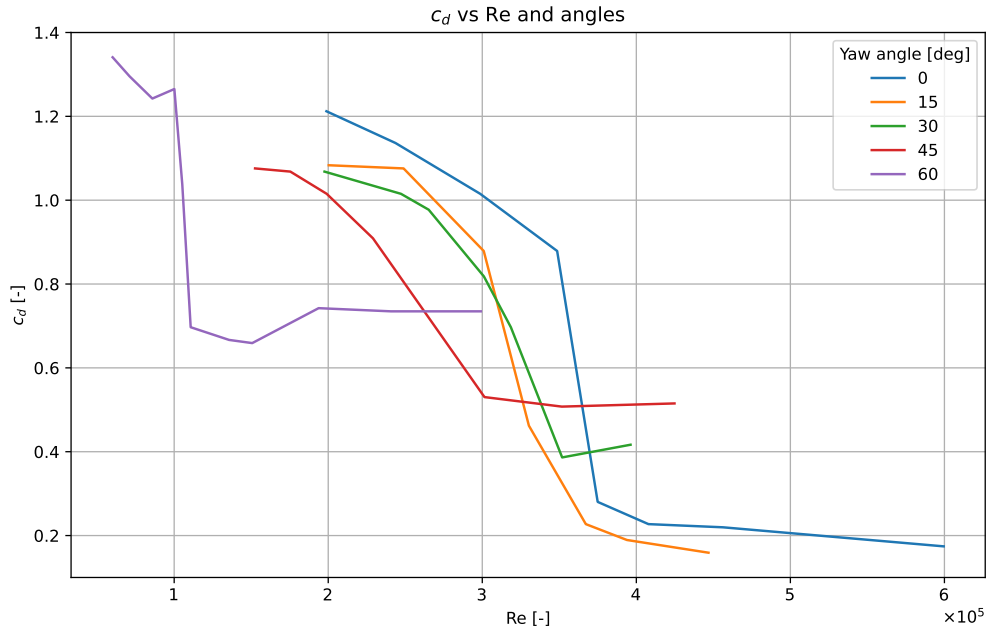


Figure 2.5: Drag coefficient vs Reynolds number at different yaw angles. Extracted from [9].

Which is the result of the integration of pressure around the longitude of the cylinder. The paper runs several experiments where they get a pressure coefficient S at certain angles of the cylinder, as in this picture:

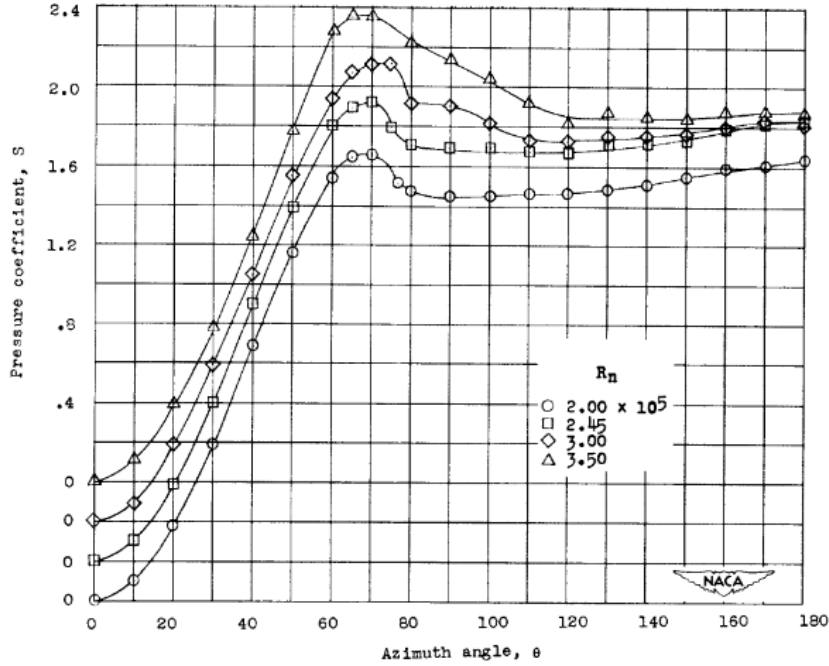


Figure 2.6: Example plot: Pressure coefficient vs azimuth angle at different Reynolds. Yaw: 0 deg. Extracted from [9].

Then, they compute the normal drag coefficient $c_{d\perp}$ as:

$$c_{d\perp} = - \int_0^\pi S \sin(d\theta) = - \int_0^\pi S \cos(\theta) d\theta \quad (2.10)$$

They stated that the best correlation of the parameters came if the Reynolds was computed using the normal velocity to the cylinder's axis instead of the free-stream velocity; and if the resulting force was computed using normal velocity and a drag coefficient based on such. That is why these parameters were used in the computation of the external force.

Finally, in these regimes, pressure forces dominate over viscous forces and the latter can be neglected.

- The water model has also been refined, in the sense that water velocity has not been considered as constant along the depth of the stream. It has been considered, though, as a 2 dimensional approximation in the vertical and stream directions. Fluctuations in the lateral directions have been neglected, though they should be modeled for a more precise model.

Indeed, water profiles in real rivers, streams or channels do not follow constant velocity profiles. And even though they are difficult to characterize because its behavior is complex and does not usually meet 1 or 2 dimensional assumptions, they stick to the following structure (time averaged, in the center region of a steady, fully developed flows in open channels) [8, 14, 17]:

- First, a thin layer is dominated by viscous forces, and follows different velocity laws depending on the smoothness of the wall. In this problem, the wall is considered to be rough and the velocity profile results:

$$U(z) = \frac{u_*}{\kappa} \log \left(\frac{z}{z_0} \right) \quad (2.11)$$

Where u_* is the shear velocity, z_0 is the roughness length of the surface, κ is an experimental constant that has been determined as 0.41 and $\log$ denotes the natural logarithm.

The shear velocity is a magnitude defined as [18]:

$$u_* := \sqrt{\frac{\tau_0}{\rho}} \quad (2.12)$$

Where, τ_0 is the shear stress and can also be computed as [50]:

$$\tau_0 := \rho S_w D g \quad (2.13)$$

S_w is the river slope, D is the average depth and g is the gravity constant. D can be approximated to 0.4. Parallely, the river slope can actually vary depending on the conditions (values between 0.001 and 0.01 can be considered [50]), so the parameter u_* will be left as a control hyperparameter depending on S_w .

z_0 can be computed based on analytical formulae fit to experimental results [8]. For that, one needs to consider the magnitude:

$$Re_* := \frac{k_s u_*}{\nu}$$

Where k_s is the roughness height in meters [19]. For this case, the order of magnitude can be considered as $k_s = 0.1$. According to [8], for a roughness Reynolds of this range of orders of magnitude:

$$z_0 = k_s \left(0.0275 - 0.007 \sqrt{\sin \left(\frac{Re_* - 4}{14} \right) \pi} \right) \quad (2.14)$$

- Then, transition between viscous layer and log-law layer happens, and the behavior of the flow in this area cannot be properly defined by any of the laws.
- Finally, the outer layer is governed by the so-called log-law, regardless of the roughness of the wall:

$$U(u_*, z) = \frac{u_*}{\kappa} \log \frac{z}{k_s} - 2.5 u_* \log \left(\frac{z_0(u_*)}{k_s} \right) \quad (2.15)$$

This equation was additionally modified to account for gravity effects [18] and the dip-phenomenon [1, 25]. This phenomenon consists on a subtle acceleration of the flow right below the free surface. However, these terms complicate too much the equation and will not be considered.

This layer is so dominant (in terms of longitude) with respect to the other two that in practice one can consider this log-law as the only layer present in the model. Therefore, equation 2.15 is the only equation considered for the velocity profile.

For simplicity, and to make the graph continuous, this velocity profile has been extended linearly to intercept height 0 at 0 m/s.

The final equations are:

$$\begin{cases} U(u_*, z) = \frac{u_*}{0.41} \log(10 z) - 2.5 u_* \log(10 z_0(u_*)) & z \geq z_t \\ U(z) = \frac{u_*}{0.41 z_t} z & z < z_t \end{cases} \quad (2.16)$$

such that z_t satisfies:

$$\frac{u_*}{0.41} \log(10 z_t) - 2.5 u_* \log(z_0(u_*)) = \frac{u_*}{0.41 z_t}$$

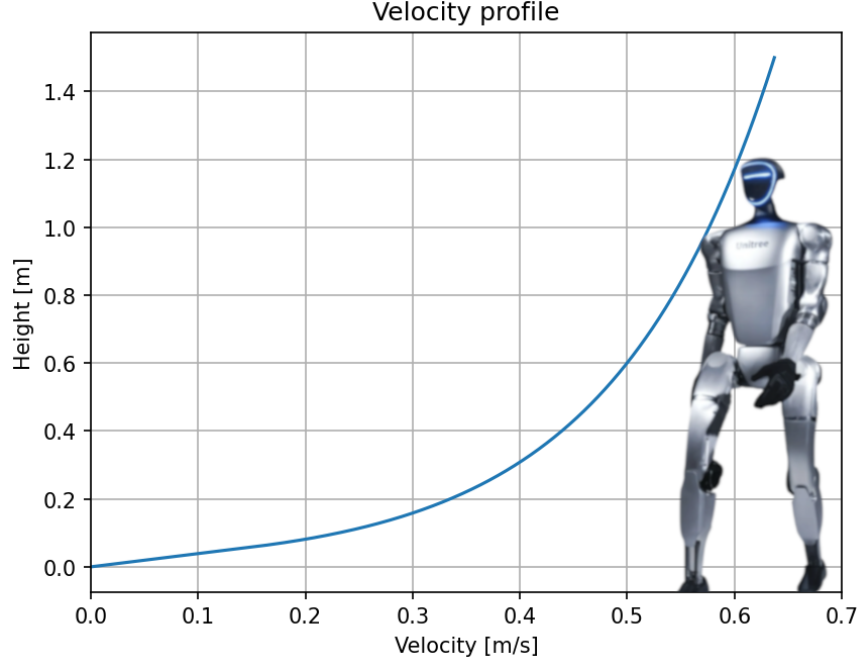


Figure 2.7: Velocity profile for the stream for $S_w = 0.001$. Robot outline extracted from [45].

Chapter 3

External force observation

Once the forces have been modeled with more or less precision, it's time to detect them. The current state of the g1's WBC lacks a mechanism to react to external perturbations, and its way of keeping balance is just using a PD controller to track the position of each limb with respect to a target, with perfectly accurate dynamics and absence of external disturbances. This might work in an ideal environment like a simulation, where every detail has been modeled and considered. However, robots do not operate in optimal environments and often face perturbations and challenges that have not been modeled during simulations. One such example is, of course, a robot deployed in a water stream. The uncertain and fast-changing conditions of this domain require a way of quickly reacting to the changes. Therefore, a more efficient and robust method is essential for maintaining balance.

The first step in reacting to a force is to be able to detect its magnitude and direction. Of course, we, as humans, would not react the same way to a gentle touch or to a violent wave. In the first case, the force is so small that we might not even be able to detect it. The second case, though, might destabilize us and would require an instant reaction to keep our stance.

In robotics, this is equivalent to the need for a disturbance detection mechanism capable of adapting instantly to torque requirements.

The goal of this section is to present the theory behind a momentum observer. A momentum observer is an architecture capable of estimating external forces based on the difference between measured and model-predicted momentum.

3.1 Momentum observer

Let $\mathbf{k}$ represent the following magnitude:

$$\mathbf{k} := \underline{\mathbf{M}}\mathbf{v} \quad (3.1)$$

which is the *generalized moment vector* of the robot.

If one derives equation 3.1:

$$\dot{\mathbf{k}} = \dot{\underline{\mathbf{M}}}\mathbf{v} + \underline{\mathbf{M}}\dot{\mathbf{v}} \quad (3.2)$$

The first term of the equation can be converted into well-known quantities with the formula [22]:

$$\dot{\underline{\mathbf{M}}} = \underline{\mathbf{C}} + \underline{\mathbf{C}}^T \quad (3.3)$$

Substituting this result in equation 3.2, one gets:

$$\dot{\mathbf{k}} = (\underline{\mathbf{C}} + \underline{\mathbf{C}}^T)\mathbf{v} + \underline{\mathbf{M}}\dot{\mathbf{v}},$$

And combining it with equation 2.2:

$$\dot{\mathbf{k}} = \underline{\mathbf{C}}^T\mathbf{v} - \mathbf{G} + \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \boldsymbol{\tau}_{joint} \end{bmatrix} + \boldsymbol{\tau}_{cf} + \boldsymbol{\tau}_{ext} \quad (3.4)$$

So, there is a way to relate the generalized moment and the external forces via the dynamics of the robot, and hence these external forces could be directly computed with a numerical derivation of equation 3.2. However, it needs to be recalled that this implementation is intended to be deployed in real robots, where noise and errors in the measurements are common. A zero-order approach like this is extremely sensitive to noise and abrupt changes and might cause serious stability problems in real environments -or even in simulations-. Therefore, a filter of order n is built to mitigate these effects.

First, let $\hat{\mathbf{k}}$ and $\hat{\boldsymbol{\tau}}_{ext}$ define the estimators of the derivative of the generalized momentum and the external forces, respectively. Note that all of the other terms of equation 3.4 are known:

$$\hat{\mathbf{k}} = \underline{\mathbf{C}}^T \mathbf{v} - \mathbf{G} + \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \boldsymbol{\tau}_{joint} \end{bmatrix} + \boldsymbol{\tau}_{cf} + \hat{\boldsymbol{\tau}}_{ext} \quad (3.5)$$

Also, an estimator of $\mathbf{k}$ can be defined as:

$$\hat{\mathbf{k}} = \int_0^t \hat{\mathbf{k}}(\sigma) d\sigma \quad (3.6)$$

Finally, the initial estimation for $\hat{\boldsymbol{\tau}}_{ext}$ can be set to $\mathbf{0}$, and it can be updated by the law:

$$\hat{\boldsymbol{\tau}}_{ext} = \underline{\mathbf{K}} (\mathbf{k} - \hat{\mathbf{k}}) \quad (3.7)$$

Where $\underline{\mathbf{K}}$ is a gain matrix that needs to be adjusted according to stability criteria and own requirements. Also, note that the continuous equation 3.7 cannot be treated as such in a computational environment. Therefore, numerical integration needs to be done for each timestep.

Note that this law makes sense because it is both consistent and convergent to the expected external force value.

3.1.1 First-order observer: consistency and convergence

Let the following forward Euler method be considered, corresponding to the time discretization of the integral in equation 3.6 (it is ultimately equation 3.7 in discrete form):

$$\hat{\boldsymbol{\tau}}_{ext}^{n+1} = \phi(\hat{\boldsymbol{\tau}}_{ext}^n) = \underline{\mathbf{K}} \left(\mathbf{k}^n - \hat{\mathbf{k}}^{n-1} - \left((\underline{\mathbf{C}}^T \mathbf{v})^n - \mathbf{G}^n + \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \boldsymbol{\tau}_{joint} \end{bmatrix}^n + \boldsymbol{\tau}_{cf}^n + \hat{\boldsymbol{\tau}}_{ext}^n \right) \Delta t \right) \quad (3.8)$$

Here, the discrete consideration of $\hat{\mathbf{k}}$ is, in accordance with forward Euler method applied to equation 3.6:

$$\hat{\mathbf{k}}^n = \hat{\mathbf{k}}^{n-1} + \hat{\mathbf{k}}^n \Delta t \quad (3.9)$$

Its consistency and convergence will be analyzed from the perspective of a stationary iterative

method. Indeed, the objective is to get the value of τ_{ext} that satisfies equation 2.2 under static conditions. In other words, the objective is to find τ_{ext} such that $f(\tau_{ext}) = \mathbf{0}$, with:

$$f(\tau) := \underline{M}\dot{v} + \underline{C}v + \underline{G} - \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \tau_{joint} \end{bmatrix} - \tau_{cf} - \tau \quad (3.10)$$

Although the method depends on non-stationary parameters ($\underline{k}, \underline{C}, \underline{G}...$), the assumption of staticity will be made for simplicity. Of course, this makes sense at this point because the main concern in this chapter is just to show that the algorithm yields a sequence of values that gets closer and closer to the expected external force if no action is taken.

1. It is consistent, and the estimation does not change when it has reached the true value of the external force.

Definition

Definition 3.1.1. Consistency of a method

Given a stationary iterative method $x^{n+1} = \phi(x^n)$ to find the zeros of a function f , it is said to be **consistent** if the roots are the only stationary points:

$$f(x) = 0 \Leftrightarrow \phi(x) = x$$

In order to prove its consistency, one first needs to rearrange equation 3.8 and work with a more tractable expression. Note that equations 3.6 and 3.9 indicate that $\hat{k}^n$ depends on previous states, and this can be somewhat helpful to find a closed form for the iterative method ϕ , as it is described in equation 3.8.

Let us now focus on the start of the computations. The initial guesses for the estimators of $\hat{k}^{-1}$ ¹ and $\hat{\tau}_{ext}^0$ are $\mathbf{0}$ for both. Also, from now on let us consider the following notation, just to simplify:

¹Note that this -1 index is necessary to define $\hat{k}^0$ in the first iteration, as in equation 3.9.

$$\tau_j := \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \tau_{joint} \end{bmatrix} \quad \phi^{n+1} := \hat{\tau}_{ext}^{n+1} \quad (3.11)$$

$$\mathbf{m}^n := \left(\underline{\mathbf{C}}^T \mathbf{v} \right)^n - \mathbf{G}^n + \tau_j^n + \tau_{cf}^n$$

With this notation, equation 3.8 becomes:

$$\phi^{n+1} = \underline{\mathbf{K}} \left(\mathbf{k}^n - \hat{\mathbf{k}}^{n-1} - (\mathbf{m}^n + \phi^n) \Delta t \right) \quad (3.12)$$

So, for the first three iterations:

$$\phi^1 = \underline{\mathbf{K}} (\mathbf{k}^0 - \mathbf{m}^0 \Delta t)$$

$$\phi^2 = \underline{\mathbf{K}} \left[\mathbf{k}^1 - \mathbf{m}^0 \Delta t - (\mathbf{m}^1 + \phi^1) \Delta t \right]$$

$$\phi^3 = \underline{\mathbf{K}} \left[\mathbf{k}^2 - (\mathbf{m}^0 + \mathbf{m}^1 + \phi^1) \Delta t - (\mathbf{m}^2 + \phi^2) \Delta t \right]$$

One can find a pattern here, which follows the general expression:

$$\phi^n = \underline{\mathbf{K}} \left(\mathbf{k}^{n-1} - \Delta t \sum_{i=0}^{n-1} \mathbf{m}^i - \Delta t \sum_{j=1}^{n-1} \phi^j \right) \quad (3.13)$$

This can be easily proven by induction, where the goal is to prove that, based on equation 3.9:

$$\hat{\mathbf{k}}^{n-1} = \Delta t \left(\sum_{i=0}^{n-1} \mathbf{m}^i + \sum_{j=1}^{n-1} \phi^j \right)$$

Indeed:

- The base cases match since:

$$\hat{\mathbf{k}}^0 = \mathbf{m}^0 \Delta t$$

- The induction step holds:

$$\begin{aligned} \hat{\mathbf{k}}^n &= \hat{\mathbf{k}}^{n-1} + \hat{\mathbf{k}}^n \Delta t = \Delta t \left(\sum_{i=0}^{n-1} \mathbf{m}^i + \sum_{j=1}^{n-1} \phi^j + \hat{\mathbf{k}}^n \right) = \Delta t \left(\sum_{i=0}^{n-1} \mathbf{m}^i + \sum_{j=1}^{n-1} \phi^j + \mathbf{m}^n + \phi^n \right) = \\ &= \Delta t \left(\sum_{i=0}^n \mathbf{m}^i + \sum_{j=1}^n \phi^j \right) \end{aligned}$$

Now that equation 3.13 has been proven right, one can build a more compact form of it with telescopic substractions. By subtracting two consecutive terms of the sequence:

$$\phi^n - \phi^{n-1} = \underline{\mathbf{K}} \left(\mathbf{k}^{n-1} - \mathbf{k}^{n-2} - \mathbf{m}^{n-1} \Delta t - \phi^{n-1} \Delta t \right)$$

And rearranging the terms:

$$\phi^n = \underline{\mathbf{K}} \left(\mathbf{k}^{n-1} - \mathbf{k}^{n-2} - \mathbf{m}^{n-1} \Delta t \right) + (\underline{\mathbf{I}} - \Delta t \underline{\mathbf{K}}) \phi^{n-1} \quad (3.14)$$

With this equation, it is now possible to assess the consistency of the method.

So, recalling the staticity assumption, the state of the robot is constant, and so is its generalized moment vector $\mathbf{k}$. Hence:

$$\phi^n = -\Delta t \underline{\mathbf{K}} \mathbf{m}^{n-1} + (\underline{\mathbf{I}} - \Delta t \underline{\mathbf{K}}) \phi^{n-1} = \phi^{n-1} - \Delta t \underline{\mathbf{K}} \left(\mathbf{m}^{n-1} + \phi^{n-1} \right)$$

On the one hand, if $\phi^n = \phi^{n-1}$, the only possibility that yields this result is that $\mathbf{m}^{n-1} + \phi^{n-1} = \mathbf{0}$. Since the generalized moment vector is constant, its derivative is $\mathbf{0}$, and according to equation 3.4, this implies that $\phi^{n-1} = \tau_{ext}$, a root of function 3.10.

On the other hand, if ϕ^{n-1} is a root of function 3.10, this again yields that $\mathbf{m}^{n-1} + \phi^{n-1} = \dot{\mathbf{k}}$, which is $\mathbf{0}$, so $\phi^n = \phi^{n-1}$.

2. It is convergent.

Definition
Definition 3.1.2. *Convergence of a method*

Given a static iterative method $x^{n+1} = \phi(x^n)$ to find the zeros of a function f , it is said to be **convergent** if:

$$\lim_{n \rightarrow \infty} x^n = \alpha$$

being α a value where:

$$f(\alpha) = 0$$

First, let us recall the following proposition:

Proposition
Proposition 3.1.1. *Convergence criterion*

Given a static iterative method $x^{n+1} = \phi(x^n)$ to find the zeros of a function f , where $f(\alpha) = 0$, if

$$\phi'(\alpha) \neq 0 \quad \text{and} \quad |\phi'(\alpha)| < 1$$

the method is (linearly) convergent.

From equation 3.14 ²:

$$\phi'(\tau_{ext}) = \underline{I} - \Delta t \underline{K}$$

Regardless of the value of τ_{ext} . Element-wise, since $\underline{K}$ is a diagonal positive definite matrix, the method is convergent if:

$$|1 - \Delta t k_{ii}| < 1, \quad k_{ii} \neq \Delta t^{-1} \quad \Leftrightarrow \quad k_{ii} \in \left(0, \frac{2}{\Delta t}\right) / \{\Delta t^{-1}\}$$

If $k_{ii} = \Delta t^{-1}$ for some i , the i -th component of the estimated external force would fall out of the hypothesis of the theorem. In this case, the update equation for the i -th component would be:

$$\phi_i^n = k_{ii} \left(\mathbf{k}_i^{n-1} - \mathbf{k}_i^{n-2} - \mathbf{m}_i^{n-1} \Delta t \right) \quad (3.15)$$

²Here, the derivative operator denotes element-wise derivation of the vector.

Where, if the state of the robot is stationary:

$$\phi_i^n = -k_{ii} m_i^{n-1} \Delta t = -m_i^{n-1}$$

Directly yields the value of the external force, again according to equation 3.4.

Therefore, the method is convergent if:

$$k_{ii} \in \left(0, \frac{2}{\Delta t}\right)$$

Thus far, only the first order filter of equation 3.7 has been discussed, and this filter can be upgraded to an arbitrary order.

The method to upgrade the filter is the classical [26]:

$$\gamma_i = \underline{K}_i \int_0^t (-\hat{\tau}_{ext} + \gamma_{i-1}) d\sigma \quad i = 2, \dots, r \quad (3.16)$$

Where γ_1 corresponds to $\hat{\tau}_{ext}$ defined in equation 3.7, and the estimation of the external force is the last step of the filter: $\hat{\tau}_{ext} = \gamma_r$. This scheme must be, again, discretized. Using the same forward Euler scheme as for the first order observer:

$$\begin{aligned} \gamma_i^n &= \gamma_i^{n-1} + \underline{K}_i \left(-\gamma_r^{n-1} + \gamma_{i-1}^n \right) \Delta t \quad i = 2, \dots, r \\ \gamma_1^n &= \underline{K}_1 \left(k^{n-1} - k^{n-2} - m^{n-1} \Delta t \right) + (\underline{I} - \Delta t \underline{K}_1) \gamma_r^{n-1} \end{aligned} \quad (3.17)$$

3.1.2 n-th-order observer: consistency and convergence

Taking up the definitions for *consistency* and *convergence*, it is possible to study such properties of method 3.17.

1. It is consistent. Method 3.17 can be expressed as a discrete linear chain:

$$\bar{\gamma}^n = \underline{C} \bar{\gamma}^n + \underline{A} \bar{\gamma}^{n-1} + \underline{B}^n \quad (3.18)$$

where:

$$(\bar{\gamma}^n)^T = [\gamma_1^T \quad \gamma_2^T \quad \cdots \quad \gamma_r^T]^n$$

and $\bar{\gamma}^0 = \mathbf{0}$.

From equations 3.14 and 3.17, matrices $\underline{A}$ and $\underline{C}$ and vector B can be written as:

$$\underline{A} = \begin{bmatrix} \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{I} - \Delta t \underline{K}_1 \\ \underline{0} & \underline{I} & \underline{0} & \cdots & \underline{0} & -\Delta t \underline{K}_2 \\ \underline{0} & \underline{0} & \underline{I} & \cdots & \underline{0} & -\Delta t \underline{K}_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{I} & -\Delta t \underline{K}_{r-1} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{I} - \Delta t \underline{K}_r \end{bmatrix} \quad B^n = \begin{bmatrix} \underline{K}_1 (k^{n-1} - k^{n-2} - m^{n-1} \Delta t) \\ \mathbf{0} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

$$\underline{C} = \begin{bmatrix} \underline{0} & \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} \\ \Delta t \underline{K}_2 & \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} \\ \underline{0} & \Delta t \underline{K}_3 & \underline{0} & \underline{0} & \cdots & \underline{0} \\ \underline{0} & \underline{0} & \Delta t \underline{K}_4 & \underline{0} & \cdots & \underline{0} \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \Delta t \underline{K}_r & \underline{0} \end{bmatrix}$$

Again, the staticity assumption is made, so the non-stationary indepent term vector B^n becomes:

$$B = \begin{bmatrix} -\underline{K}_1 m^* \Delta t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}$$

Here, m^* denotes the magnitude m defined in equation 3.11 in its stationary state.

Method 3.18 can be rewritten as:

$$\bar{\gamma}^n = (\underline{I} - \underline{C})^{-1} \underline{A} \bar{\gamma}^{n-1} + (\underline{I} - \underline{C})^{-1} B \quad (3.19)$$

Considering method 3.19, with notation $\phi(x^n) = \bar{\gamma}^n$, $x^n = \bar{\gamma}^{n-1}$ and $\bar{\gamma}^*$ the configuration that yields the stationary point of external force, it is needed to prove that:

$$\phi(\bar{\gamma}^*) = \bar{\gamma}^* \quad \Leftrightarrow \quad \tilde{f}(\bar{\gamma}^*) = \mathbf{0}$$

Where $\tilde{f}$ refers to the same dynamics equation as in 3.10. However, the proper definition in this case is:

$$\tilde{f}(\bar{\gamma})^T := \begin{bmatrix} f(\gamma_1)^T & f(\gamma_2)^T & \cdots & f(\gamma_r)^T \end{bmatrix} \quad (3.20)$$

since every step of the observer must converge to the external force value.

Let us start the proof by first supposing that $\tilde{f}(\bar{\gamma}^*) = \mathbf{0}$. In this case, since every γ_i satisfies equation 3.10, the following holds:

$$\gamma_i = -m^* \quad i = 1, \dots, r$$

Also, this proposition needs to be considered:

Proposition

Proposition 3.1.2. *Inverse of a subdiagonal block matrix with commutative blocks*

Let:

$$\underline{M} = \begin{bmatrix} \underline{B}_1 & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} \\ \underline{A}_2 & \underline{B}_2 & \underline{0} & \cdots & \underline{0} & \underline{0} \\ \underline{0} & \underline{A}_3 & \underline{B}_3 & \cdots & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{A}_r & \underline{B}_r \end{bmatrix} \in \mathbb{R}^{nr \times nr}$$

be a matrix where $\underline{A}_i$ and $\underline{B}_i \in \mathbb{R}^{n \times n}$ are square invertible matrices, all of which are pairwise commuting.

Then, there exists a close-form formula for the blocks of its inverse:

$$\{\underline{\mathbf{M}}^{-1}\}_{ij} = \begin{cases} 0 & j > i \\ \underline{\mathbf{B}}_i^{-1} & j = i \\ (-1)^{i-j} \prod_{z=j}^i \underline{\mathbf{B}}_z^{-1} \prod_{z=j+1}^i \underline{\mathbf{A}}_z & j < i \end{cases}$$

The proof is provided in appendix B.

From this proposition, one gets that, blockwise:

$$\{(\underline{\mathbf{I}} - \underline{\mathbf{C}})^{-1}\}_{ij} = \begin{cases} 0 & j > i \\ \underline{\mathbf{I}} & j = i \\ (\Delta t)^{i-j} \prod_{z=j+1}^i \underline{\mathbf{K}}_z & j < i \end{cases}$$

With this, one can use equation 3.19 to check if an iteration over the target yields the same previous result. Performing direct matrix multiplication requires heavy math, which is derived in annex C. In particular, section C.1 describes the elements of $(\underline{\mathbf{I}} - \underline{\mathbf{C}})^{-1} \underline{\mathbf{A}}$ and section C.2 proves that, indeed, an iteration over the target yields the same result.

Therefore, if the dynamics of the problem are satisfied, the point is fixed for method 3.19.

Now, let's prove the other way round. Let's now assume that the point is fixed. The goal is to show that the dynamics are satisfied.

Using equation 3.19, and knowing that the point is fixed, it can be rearranged and rewritten:

$$(\underline{\mathbf{I}} - \underline{\mathbf{C}} - \underline{\mathbf{A}}) \bar{\gamma}^* = \underline{\mathbf{B}} \quad (3.21)$$

The block-elements of the coefficient matrix $\underline{\mathbf{I}} - \underline{\mathbf{C}} - \underline{\mathbf{A}}$ are:

$$\underline{I} - \underline{C} - \underline{A} = \begin{bmatrix} \underline{I} & \underline{0} & \underline{0} & \cdots & \underline{0} & -(\underline{I} - \Delta t \underline{K}_1) \\ -\Delta t \underline{K}_2 & \underline{0} & \underline{0} & \cdots & \underline{0} & \Delta t \underline{K}_2 \\ \underline{0} & -\Delta t \underline{K}_3 & \underline{0} & \cdots & \underline{0} & \Delta t \underline{K}_3 \\ \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ \underline{0} & \cdots & \underline{0} & -\Delta t \underline{K}_{r-1} & \underline{0} & \Delta t \underline{K}_{r-1} \\ \underline{0} & \cdots & \underline{0} & \underline{0} & -\Delta t \underline{K}_r & \Delta t \underline{K}_r \end{bmatrix}.$$

This yields a system of equations where:

$$\gamma_i^* = \gamma_{i-1}^* \quad i = 2, \dots, r$$

Where it can be deduced that:

$$\gamma_r^* = \gamma_{r-1}^* = \cdots = \gamma_2^* = \gamma_1^*$$

The first equation of the system is, however:

$$\gamma_1^* - \gamma_r^* + \Delta t \underline{K}_1 \gamma_r^* = -\underline{K}_1 \underline{m}^* \Delta t \quad \Rightarrow \quad \Delta t \underline{K}_1 \gamma_r^* = -\underline{K}_1 \underline{m}^* \Delta t$$

Since $\underline{K}_1$ is invertible and Δt is a non-zero scalar, the previous expression can be simplified, being the final conclusion:

$$\gamma_r^* = \gamma_{r-1}^* = \cdots = \gamma_2^* = \gamma_1^* = -\underline{m}^*$$

Which proves that, if $\bar{\gamma}^*$ is fixed, it satisfies the dynamics of the system and ultimately yields a correct value of the estimated external force.

2. It is convergent (under certain conditions). Considering method 3.19 and the same notation as in consistency, such method is convergent if and only if the following holds [12]:

Theorem

Theorem 3.1.3. *Convergence of a linear, stationary, multivariable method*

Given an iterative method defined by:

$$\bar{\gamma}^n = \underline{G}\bar{\gamma}^{n-1} + \underline{p}$$

where $\underline{G}$ and $\underline{p}$ are a constant matrix and vector respectively, the method is convergent if and only if:

$$\rho(\underline{G}) < 1$$

Where $\rho(\cdot)$ denotes the spectral radius, i.e.:

$$\rho(\underline{G}) = \max_i (|\lambda_i|)$$

Being $\{\lambda_i\}_i$ its eigenvalues.

So, according to the theorem, method 3.19 is convergent if and only if:

$$\rho \left((\underline{I} - \underline{C})^{-1} \underline{A} \right) < 1$$

But proving this is not an easy task and has not been achieved by the author. Some attempts have been made tho, that can be shown in this report to leave a record, or to give inspiration to future researchers.

- First, matrix norm bounding was attempted. Indeed, if the norm (actually, any norm as long as it is consistent) was bounded to one, the spectral radius would be automatically bounded as well [15]:

Proposition

Proposition 3.1.4. *Bound of spectral radius*

Given $\underline{G} \in \mathbb{C}^{n \times n}$:

$$\rho(\underline{G}) \leq \|\underline{G}\|$$

for any matrix norm that satisfies:

$$\|\underline{G}x\| \leq \|\underline{G}\| \|x\|$$

for all $x \in \mathbb{C}^n$ (consistent norm).

Despite being useful and powerful, this proposition is too restrictive and hard to compute. First, because finding a norm that was bounded by 1 was not possible given the shape of the matrix that was being studied. Norms such as the supremum or manhattan norms (infinity and 1-norms) were used unfruitfully. The definitions of such norms are:

Definition

Definition 3.1.3. *Supremum norm*

$$\|\underline{G}\|_{\infty} := \max_{1 \leq j \leq m} \sum_{i=1}^n |\{\underline{G}\}_{ij}|$$

Definition

Definition 3.1.4. *Manhattan norm*

$$\|\underline{G}\|_1 := \max_{1 \leq j \leq n} \sum_{i=1}^m |\{\underline{G}\}_{ij}|$$

The complexity of the iteration matrix makes it unnecessarily complex to assess the convergence from this point of view, since the presence of absolute values complicates the computations.

Second, it has been seen numerically that matrices whose norm exceeds 1 can still have a spectral radius smaller than such bound. Therefore, bounding the norm to 1 can be too restrictive.

- The next attempt to bound the spectral radius was to directly compute the eigenvalues of the matrix via its characteristic polynomial. For matrix sizes greater than 2, Sarrus rule or Laplace formula were needed to compute the determinant; and again, complexity was a very limiting factor.

As an example, the characteristic polynomial for the iteration matrix for $n = 3$ is shown.

The goal is to calculate the determinant of:

$$\underline{G} := \begin{bmatrix} -\lambda \underline{I} & \underline{0} & \underline{I} - \Delta t \underline{K}_1 \\ \underline{0} & (1 - \lambda) \underline{I} & -\Delta t^2 \underline{K}_1 \underline{K}_2 \\ \underline{0} & \Delta t \underline{K}_3 & -\Delta t^3 \underline{K}_1 \underline{K}_2 \underline{K}_3 + (1 - \lambda) \underline{I} - \Delta t \underline{K}_3 \end{bmatrix} \quad (3.22)$$

If one assumes that the gain matrices are scalar (instead of diagonal), this matrix is blockwise scalar, so the computations can be simplified via the Kronecker product [27].

Definition

Definition 3.1.5. Kronecker product

Given a pair of matrices $\underline{A} \in \mathbb{C}^{m \times n}$ and $\underline{B} \in \mathbb{C}^{p \times q}$, the Kronecker between them is defined as the block matrix:

$$\underline{A} \otimes \underline{B} := \begin{bmatrix} \{\underline{A}\}_{11} \underline{B} & \cdots & \{\underline{A}\}_{1n} \underline{B} \\ \vdots & \ddots & \vdots \\ \{\underline{A}\}_{m1} \underline{B} & \cdots & \{\underline{A}\}_{mn} \underline{B} \end{bmatrix} \in \mathbb{C}^{mp \times nq}$$

From this definition, the following theorem arises [27]:

Theorem

Theorem 3.1.5. Eigenvalues of a block matrix

Let $\underline{A} \in \mathbb{C}^{n \times n}$ and $\underline{B} \in \mathbb{C}^{m \times m}$.

If the set of eigenvalues for $\underline{A}$ is $\sigma(\underline{A}) = \{\lambda_i \mid i = 1, \dots, n\}$ and the set of eigenvalues for $\underline{B}$ is $\sigma(\underline{B}) = \{\mu_j \mid j = 1, \dots, m\}$, then the set of eigenvalues of $\underline{A} \otimes \underline{B}$ is $\sigma(\underline{A} \otimes \underline{B}) = \{\lambda_i \mu_j \mid i = 1, \dots, n \ j = 1, \dots, m\}$ including algebraic multiplicities.

With the definition of Kronecker product, one can conclude that the matrix $\underline{G}$ defined in equation 3.22 can be rewritten as:

$$\underline{G} = \tilde{\underline{G}} \otimes I_v \quad (3.23)$$

Where v is the size of each of the γ_i , i.e., 33; and:

$$\tilde{\underline{G}} = \begin{bmatrix} -\lambda & 0 & 1 - \Delta tk_1 \\ 0 & 1 - \lambda & -\Delta t^2 k_1 k_2 \\ 0 & \Delta tk_3 & -\Delta t^3 k_1 k_2 k_3 + 1 - \lambda - \Delta tk_3 \end{bmatrix} \quad (3.24)$$

According to Theorem 3.1.5 eigenvalues of $\underline{G}$ are the eigenvalues of $\tilde{\underline{G}}$ with algebraic multiplicity 33. So, finally, the characteristic polynomial of $\tilde{\underline{G}}$ (with which the eigenvalues can be computed) is:

$$p(\lambda) = -\lambda [\lambda^2 + \lambda(\Delta t^3 k_1 k_2 k_3 + \Delta tk_3 - 2) + (1 - \Delta tk_3)] = 0 \Leftrightarrow$$

$$\lambda = 1 - \frac{\Delta tk_3}{2} - \frac{\Delta t^3 k_1 k_2 k_3}{2} \pm \frac{1}{2} \sqrt{(\Delta tk_1 k_2 k_3 + \Delta tk_3)^2 - 4\Delta t^3 k_1 k_2 k_3}, \quad \lambda = 0$$

Not only this is very complex to solve, but it also requires numerical techniques to do so. Moreover, finding a close formula for every observer order (matrix size) is a very challenging -if not impossible- task. The author has considered that arriving to this point is enough and the complexity makes this approach not worthy.

- The last option that has been explored is the use of the Sherman-Morrison formulae. According to appendix C.1, the shape of the iteration matrix is this:

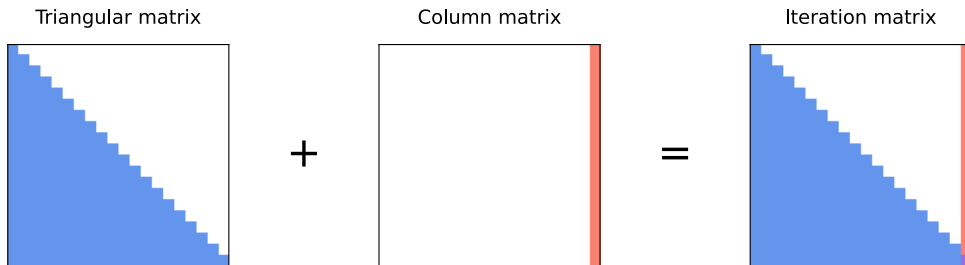


Figure 3.1: Shape of the iteration matrix

Where every matrix is blockwise diagonal. Again, if one considers constant gains for each gain matrix, the iteration matrix becomes blockwise scalar and so does its decomposition into triangular and column.

So, the simplified iteration matrix $\tilde{\underline{G}}$ can be decomposed into:

$$\tilde{\mathbf{G}} = \mathbf{T} + \mathbf{C}$$

A triangular and column matrix. So, it can be thought as a triangular plus a 1-rank update.

The Sherman-Morrison formula states [44]:

Theorem

Theorem 3.1.6. Sherman-Morrison formula

Let $\mathbf{A} \in \mathbb{C}^{n \times n}$, $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$ and β an arbitrary scalar. Assume that $\sigma := 1 + \beta \mathbf{v}^T \mathbf{A}^{-1} \mathbf{u} \neq 0$. Then:

$$(\mathbf{A} + \beta \mathbf{u} \mathbf{v}^T)^{-1} = \mathbf{A}^{-1} - \frac{\beta}{\sigma} \mathbf{A}^{-1} \mathbf{u} \mathbf{v}^T \mathbf{A}^{-1}$$

From which the following lemma for the determinant can be deduced [44]:

Lemma

Lemma 3.1.7. Formula for determinant of 1-rank update

Given all the elements of the theorem, if $\mathbf{A}$ is invertible:

$$|\mathbf{A} + \beta \mathbf{u} \mathbf{v}^T| = |\mathbf{A}| (1 + \beta \mathbf{v}^T \mathbf{A}^{-1} \mathbf{u})$$

If it is not invertible, or as a general case, denoting $\tilde{\mathbf{A}}$ as its adjoint:

$$|\mathbf{A} + \beta \mathbf{u} \mathbf{v}^T| = |\mathbf{A}| + \beta \mathbf{v}^T \tilde{\mathbf{A}} \mathbf{u}$$

Applying this lemma to the case of study, the following correspondence can be established:

$$\mathbf{A} = \mathbf{T} - \lambda \mathbf{I}$$

$$\mathbf{u} = \begin{bmatrix} \{\mathbf{C}\}_{1n} \\ \{\mathbf{C}\}_{2n} \\ \vdots \\ \{\mathbf{C}\}_{nn} \end{bmatrix} \quad \mathbf{v}^T = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \end{bmatrix}$$

And from the shape of the iteration matrix, $\text{diag}(\underline{T})$ can be considered as a ones vector, except for its first element, which is zero. That is, $\text{diag}(\underline{A}) = \{-\lambda, 1 - \lambda, \dots, 1 - \lambda\}$

So, from the general formula:

$$|\underline{A} + \underline{u}\underline{v}^T| = \lambda(1 - \lambda)^n + \underline{v}^T \underline{\tilde{A}} \underline{u}$$

The last step is to compute the adjoint matrix to obtain the last term of the equation. Sadly, and even though only the last row of the adjoint matrix is needed due to the shape of the $\underline{v}$ vector, the lack of special properties of $\underline{A}$ makes it unnecessarily complex, and not worth it, to compute.

So, since all of the three attempts to algebraically determine the bound for the spectral radius have failed (in terms of unnecessary complexity), the proposed solution is numerical. That is, instead of finding a general formula for the conditions of convergence, the observer would compute its dominant eigenvalue via the power method (in practice, and since the simulation and the observer are built in python, the command used will be `eig`) beforehand, and run if and only if this obtained eigenvalue is smaller than one [24].

3.2 Force estimation

The result of the described observer is a vector τ_{ext} , which estimates the force applied in each generalized coordinate. This values are necessary and will be taken into account in forthcoming sections. In addition to this generalized external force vector, actual values for external wrenches are needed in certain parts of the robot. Specifically, since the problem focuses on external forces caused by streams, precise estimation of perturbations is needed in each section of the leg. In order to keep up the way the problem has been modeled, external wrenches are to be computed in each of the geometries shown in section 2.3.

Let's recall equation 2.4:

$$\tau_{ext} = \sum_{i=1}^{n_{ext}} \underline{J}_{ext i}^T \underline{w}_{ext i}$$

So, the goal is to find a set of external wrenches $\{\underline{w}_{ext i}\}_i$ such that:

$$\boldsymbol{\tau}_{ext} = \begin{bmatrix} \boldsymbol{J}_{ext\,1}^T & \cdots & \boldsymbol{J}_{ext\,n_{ext}}^T \end{bmatrix} \begin{bmatrix} \boldsymbol{w}_{ext\,1} \\ \vdots \\ \boldsymbol{w}_{ext\,n_{ext}} \end{bmatrix} = \boldsymbol{J}_{ext} \bar{\boldsymbol{w}}_{ext} \quad (3.25)$$

One could pseudo-invert the matrix of jacobians to get:

$$\boldsymbol{J}_{ext}^* \boldsymbol{\tau}_{ext} = \bar{\boldsymbol{w}}_{ext}$$

Note that $\boldsymbol{J}_{ext} \in \mathbb{R}^{33 \times 6 \cdot n_{ext}}$.

This would yield a set of wrenches, one for each force application point that is considered. Note that the torques would be with respect to that point. So, if one wanted to refer the external net wrench with respect to another point j , the following transformation would need to be performed:

$$\boldsymbol{w}_{ext\,j} = \sum_{i=1}^{n_{ext}} \{ \boldsymbol{w}_{ext\,i} + \boldsymbol{f}_{ext\,i} \times (\boldsymbol{p}_j - \boldsymbol{p}_i) \} \quad (3.26)$$

Where $\boldsymbol{f}_{ext\,i}$ corresponds to the force components of $\boldsymbol{w}_{ext\,i}$, and $\boldsymbol{p}_j, \boldsymbol{p}_i$ correspond to the reference point and the force application point, respectively.

Chapter 4

Whole Body Control

So far, an external force observer has been built, and its stability properties have been analyzed. Now, it is time to implement these observed forces into the robot's controller. For that, one first needs a deep understanding of how a Whole Body Controller (WBC) works and how the external forces play a key role in its tasks to improve stability.

This section aims to describe how the employed WBC works, explore its current limitations, and propose patches and improvements to enhance the stability control of the robot.

4.1 Operating principles of the WBC

According to [48, 52], whole body control is a type of control approach used in robotics which unifies each locomotion task under a single optimization algorithm. That is, instead of considering each different movement separately, it solves a problem that simultaneously controls the whole behavior of the robot, possibly with different task priorities. This leads to the consideration of hierarchical control, where some actions have more importance than others and must be prioritized.

4.1.1 The rpc framework

The framework that has been used in this project is UT Austin's (and, precisely, Human Centered Robotics Lab's) rpc [4, 5]. This code has integrated two different approaches regarding task prioritization: *Implicit Hierarchical Whole-body Controller* (IHBWC) and *Whole-body Impulse Controller* (WBIC).

On the one hand, IHWBC solves a complete QP problem for each timestep using both floating base and joint accelerations ($\dot{v} = [\ddot{x}_b^T \ \dot{\omega}_b^T \ \ddot{q}^T]^T$) and contact wrenches ($w_c = [w_{c1}^T \ w_{c2}^T]^T$) as decision variables. It establishes physical restrictions regarding the compliance of dynamics equations and torque and cone friction limitations, but task priorities are *soft* and are balanced through weights. This is useful when flexible task management wants to be addressed, as in this case.

On the other hand, WBIC solves a QP (also for each timestep) concerning just the floating base accelerations and contact wrenches, after a stage of null-space projection to ensure a strict fulfillment of higher-prioritized tasks. It also establishes physical constraints.

In this work, IHWBC it is therefore adopted: a single weighted QP that natively integrates dynamics, friction, torque limits, and where $\hat{\tau}_{ext}$ can be implemented easier. WBIC is left as future work when strict lexicographic priorities or MPC-based dynamic gaits are required.

4.1.2 IHWBC

The formulation of the QP problem that is being solved in IHWBC is the following:

$$\left\{ \begin{array}{l} \min_{x=[\dot{v}^T \ w_c^T]^T} \frac{1}{2} x^T \underline{G} x + g_0^T x \\ \\ s.t : \\ \underline{C}_E^T x + c_E = 0 \\ \underline{C}_I^T x + c_I \geq 0 \end{array} \right. \quad (4.1)$$

Where three clear parts can be distinguished, that must be visited to properly understand the elements involved in the optimization problem and its potential improvements:

1. Cost function.
2. Equality constraints.
3. Inequality constraints.

Cost function

The controller sets a list of tasks and assigns a reference to each of them. The list of tasks is:

- Right and left hands position and orientation.
- Centroidal angular momentum.
- Torso and pelvis orientation.
- Center of mass position.
- Upper body configuration.

Each of the tasks has a certain reference assigned: $\ddot{\mathbf{x}}_{tref}$ that is computed by a combination of feed-forward and PD control:

$$\ddot{\mathbf{x}}_{tref} = \ddot{\mathbf{x}}_{tff} + K_{tp}(\mathbf{x}_{tref} - \mathbf{x}) + K_{td}(\dot{\mathbf{x}}_{tref} - \dot{\mathbf{x}}) \quad (4.2)$$

Parallely, each of the reference $\mathbf{x}_t$ is a function of the generalized space ¹:

$$\mathbf{x}_t = \mathbf{f}_t(\mathbf{p}) \quad \mathbf{p} = \begin{bmatrix} \mathbf{x}_b \\ \phi_b \\ \mathbf{q} \end{bmatrix} \quad (4.3)$$

So, differentiating equation 4.3, and due to the chain rule:

$$\dot{\mathbf{x}}_t = \underline{\mathbf{J}}_t \dot{\mathbf{p}} = \underline{\mathbf{J}}_t \mathbf{v} \quad (4.4)$$

Differentiating again:

$$\ddot{\mathbf{x}}_t = \dot{\underline{\mathbf{J}}}_t \mathbf{v} + \underline{\mathbf{J}}_t \dot{\mathbf{v}} \quad (4.5)$$

So, one can consider that the cost assigned to each of the tasks is the error (under a certain norm) of the reference, and the analytically differentiated magnitude:

$$e_t = \frac{1}{2} \|\dot{\underline{\mathbf{J}}}_t \mathbf{v} + \underline{\mathbf{J}}_t \dot{\mathbf{v}} - \ddot{\mathbf{x}}_{tref}\|_{W_t} \quad (4.6)$$

¹Note that $\dot{\mathbf{p}} = \mathbf{v}$.

In this framework, the employed norm is the classical weighted 2-norm. This gives the sense to $\underline{W}_t$, since:

$$\|\underline{u}\|_W = \underline{u}^T \underline{W} \underline{u}$$

So, rewriting equation 4.6 in matrix form:

$$e_t = \frac{1}{2} \dot{\underline{v}}^T \underline{J}_t^T \underline{W}_t \underline{J}_t \dot{\underline{v}} + (\underline{J}_t \underline{v} - \ddot{\underline{x}}_{tref})^T \underline{W}_t \underline{J}_t \underline{v} + const. \quad (4.7)$$

This error is computed for each task and can be stacked due to linearity. This will be shortly clarified.

A regularization term based on the inertia matrix is added, which favors smooth solutions. This term is general, in the sense that it is computed just once, and not for every task:

$$\frac{1}{2} \lambda_{\dot{\underline{v}}} \dot{\underline{v}}^T \underline{M} \dot{\underline{v}} \quad (4.8)$$

Finally, two force tasks regarding the wrenches are considered. These forces are not something that the robot can control explicitly (contrary to the joint torques), but in the solution space there exist infinite combinations of forces and torques that satisfy the dynamics. These tasks ensure more controllability during transitions and improve the friction margins in the wrench cones (which will be defined shortly).

Again, to these force tasks, regularization is added to guarantee smooth solutions:

$$e_{tf} = \frac{1}{2} (\underline{w}_c - \underline{w}_{cref})^T \underline{W}_{tf} (\underline{w}_c - \underline{w}_{cref}) + \frac{1}{2} \underline{w}_c^T \underline{\Lambda}_f \underline{w}_c \quad (4.9)$$

In order to recover the matrix form of 4.1, these previous equations can be stacked:

$$\underline{G} = \begin{bmatrix} \sum_t \underline{J}_t^T \underline{W}_t \underline{J}_t + \lambda_{\dot{\underline{v}}} \underline{M} & \underline{0} \\ \underline{0} & \underline{W}_{tf} + \underline{\Lambda}_f \end{bmatrix}$$

$$\underline{g}_0^T = \begin{bmatrix} \sum_t (\underline{J}_t \underline{v} - \ddot{\underline{x}}_{tref})^T \underline{W}_t \underline{J}_t & -\underline{w}_{cref}^T \underline{W}_{tf} \end{bmatrix}$$

Equality constraints

Equality constraints refer to the compliance of the dynamics equation presented in 2.2²:

$$\underline{M}\dot{\underline{v}} + \underline{C}\underline{v} + \underline{G} = \begin{bmatrix} \underline{0} \\ \underline{0} \\ \underline{\tau}_{joint} \end{bmatrix} + \underline{\tau}_{cf}$$

However, in practice, just the floating base dynamics are considered as equality constraints. Therefore, after manipulating equation 2.2 to shape it as in 4.1:

$$\begin{bmatrix} \underline{S}_n \underline{M} & -\underline{S}_n \underline{J}_c^T \end{bmatrix} \begin{bmatrix} \dot{\underline{v}} \\ \underline{w}_c \end{bmatrix} + \underline{S}_n \underline{h} = 0 \quad (4.10)$$

Where:

$$\underline{S}_n = \begin{bmatrix} \underline{I}_6 & \underline{0}_{6 \times 27} \end{bmatrix} \in \mathbb{R}^{6 \times 33}$$

is the selection matrix for the DOFs corresponding to the floating base, and:

$$\underline{h} = \underline{C}\underline{v} + \underline{G}$$

accounts for non-linear effects (coriolis and gravity).

And where, according to equation 2.3, contact wrenches have been stacked $\underline{w}_c = \begin{bmatrix} \underline{w}_{c1}^T & \underline{w}_{c2}^T \end{bmatrix}^T$ to get:

$$\underline{\tau}_{cf} = \begin{bmatrix} \underline{J}_{c1}^T & \underline{J}_{c2}^T \end{bmatrix} \underline{w}_c = \underline{J}_c^T \underline{w}_c \quad (4.11)$$

Finally:

$$\underline{C}_E^T = \begin{bmatrix} \underline{S}_n \underline{M} & -\underline{S}_n \underline{J}_c^T \end{bmatrix} \quad \underline{c}_E = \underline{S}_n \underline{h}$$

²For now, $\underline{\tau}_{ext}$ is omitted, since it was not implemented in the framework.

Inequality constraints

As they are currently implemented in IHWBC, inequality constraints refer just to contact wrench cone compliance [11]. The robot's foot is a rectangle with four points of contact:

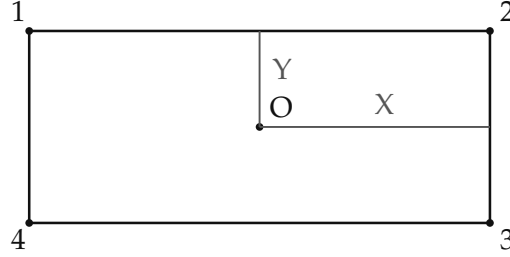


Figure 4.1: Foot shape

Where a certain force $\mathbf{f}_i = [f_{ix}, f_{iy}, f_{iz}]^T$ is applied at every contact point: $\{1, 2, 3, 4\}$. According to Coulomb's friction model:

$$0 < f_{iz} \quad (4.12)$$

$$|f_{ix}|, |f_{iy}| \leq \mu f_{iz} \quad (4.13)$$

Where μ is the friction coefficient.

Compliance with the wrench cone means that there exists a wrench $\mathbf{w}_O = [f_x \ f_y \ f_z \ \tau_x \ \tau_y \ \tau_z]^T_O$ with respect to the origin O that satisfies equations 4.12 and 4.13 and that is equivalent to the force system of four point forces, one per each contact point.

In other words, when the wrench cone condition is fulfilled, the equivalent force system keeps the local center of pressure inside the stability boundary of the foot, and it does not either rotate or slip.

The equations that are included now to satisfy the contact wrench cone are, with respect to the point O :

$$|f_x| \leq \mu f_z \quad (4.14) \quad 0 < f_z \quad (4.16)$$

$$|\tau_x| \leq Y f_z \quad (4.17)$$

$$|f_y| \leq \mu f_z \quad (4.15) \quad |\tau_y| \leq X f_z \quad (4.18)$$

(4.19)

$$f_z \leq f_{z\max} \quad (4.20)$$

where:

$$\tau_{z\min} = -\mu(X + Y)f_z + |Yf_x - \mu\tau_x| + |Xf_y - \mu\tau_y|$$

$$\tau_{z\max} = \mu(X + Y)f_z - |Yf_x + \mu\tau_x| - |Xf_y + \mu\tau_y|$$

Equations 4.14 and 4.15 ensure the condition of no slipping (coulomb friction). Equation 4.16 assures that the foot is not *attracted* to the floor. Equations 4.17 and 4.18 bound the position of the foot's local center of pressure. Equation 4.19 avoids yaw-slippage. Finally, equation 4.20 sets a limit for the vertical force that the foot can make.

These equations can be summarized with the following system (one for each contact). Note that, for now, this system corresponds to the matrices $\underline{C}_I^T$ and c_I , but the notation is changed because of practical reasons:

$$\underbrace{\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & \mu & 0 & 0 & 0 \\ -1 & 0 & \mu & 0 & 0 & 0 \\ 0 & 1 & \mu & 0 & 0 & 0 \\ 0 & -1 & \mu & 0 & 0 & 0 \\ 0 & 0 & Y & 1 & 0 & 0 \\ 0 & 0 & Y & -1 & 0 & 0 \\ 0 & 0 & X & 0 & 1 & 0 \\ 0 & 0 & X & 0 & -1 & 0 \\ Y & X & (X+Y)\mu & -\mu & -\mu & 1 \\ Y & -X & (X+Y)\mu & -\mu & \mu & 1 \\ -Y & X & (X+Y)\mu & \mu & -\mu & 1 \\ -Y & -X & (X+Y)\mu & \mu & \mu & 1 \\ -Y & -X & (X+Y)\mu & -\mu & -\mu & -1 \\ -Y & X & (X+Y)\mu & -\mu & \mu & -1 \\ Y & -X & (X+Y)\mu & \mu & -\mu & -1 \\ Y & X & (X+Y)\mu & \mu & \mu & -1 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{bmatrix}}_U + \begin{bmatrix} f_x \\ f_y \\ f_z \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix}_O + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -f_{z \max} \end{bmatrix}}_u \geq \mathbf{0} \quad (4.21)$$

Note that these equations assume that the wrench is computed in the center of the foot O . This

is how they are implemented so far, so if the reference point were to be displaced, the equations should vary.

4.2 Improvements of the controller

So far, the equations of the current WBC lacked very important features to carry out a feasible study:

1. First, the external forces were not included in the robot's dynamics, thus producing biased equations in the presence of disturbances (the entirety of simulations of this study).
2. Second, contact wrench was referenced at the true foot center, instead of the real position of the ankle, where the actual wrench was applied. This caused instability and limited the range of operation of the robot in the presence of strong need of force compensation.
3. Finally, torque equations were not well-built and encountered computational errors when the joint limits were imposed. These limits are necessary and avoid actuators to work over their capabilities.

The rest of the section introduces and patches these three issues.

4.2.1 Inclusion of external forces

External forces affect the dynamics of the robot as presented in equation 2.2:

$$\underline{M}\dot{\underline{v}} + \underline{C}\underline{v} + \underline{G} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \tau_{joint} \end{bmatrix} + \tau_{cf} + \tau_{ext}$$

The deduction of equality constraints is exactly analogous to the one presented in section 4.1.2. The only change is that the magnitude $\underline{h}$, which accounted for the coriolis and gravitational non-linear effects, now also accounts for the external generalized torque:

$$\underline{h} = \underline{C}\underline{v} + \underline{G} - \tau_{ext}$$

This external generalized torque is fed from the observer architecture built in section 3.1.

Cost functions and wrench cone inequalities remain unaffected by this change; however, the

quadratic problem is now deeply different and the dynamics have been seriously changed with this slight modification.

4.2.2 Offset of the wrench cone

As it was posed in the presentation of inequality constraints (end of section 4.1.2), if the reference point where the contact wrench is computed is not the exact center of the foot, efficiency is lost. This is because two of the parameters that are passed to the constraint builder are the foot's half-length and half width. Let's say that the foot is 0.17×0.05 meters (length $\times$ width), which it actually is. In that case, if the reference point was the center of the foot, its half-length and half width would be 0.085 and 0.025 respectively. However, if the reference point was not the center of the foot, but the same values for half-length and width were passed, the feasible region for the local center of pressure could fall outside the actual support polygon, making the robot lose its stability.

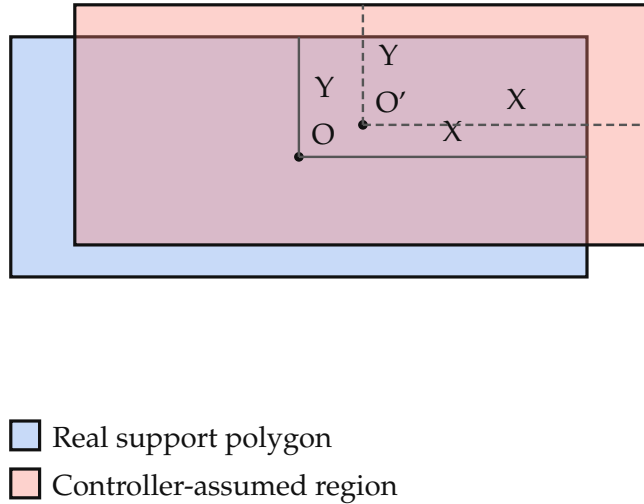


Figure 4.2: Reference offset

Therefore, restrictions 4.14 to 4.20 must be updated to consider this offset. These restrictions with respect to the foot's geometric center are already well implemented and work, so one could think that they should be leveraged in some way. Such constraints bound the components of a certain wrench w_O that is computed with respect to the geometric center. So, the wrench at the reference point $w_{O'}$ can be moved to the actual center (to apply the restrictions there) via a simple geometric translation:

The force components $\mathbf{f}_O = \mathbf{f}_{O'} = \begin{bmatrix} f_x & f_y & f_z \end{bmatrix}^T$ remain the same, but the torque components are moved:

$$\boldsymbol{\tau}_O = \boldsymbol{\tau}_{O'} + \mathbf{f}_{O'} \times \overrightarrow{O'O}$$

Therefore:

$$\mathbf{w}_O = \begin{bmatrix} \mathbf{f}_O \\ \boldsymbol{\tau}_O \end{bmatrix} = \begin{bmatrix} \underline{\mathbf{I}}_3 & \mathbf{0} \\ \underline{\mathbf{L}} & \underline{\mathbf{I}}_3 \end{bmatrix} \begin{bmatrix} \mathbf{f}_{O'} \\ \boldsymbol{\tau}_{O'} \end{bmatrix} := \underline{\mathbf{T}} \mathbf{w}_{O'} \quad (4.22)$$

Where $\underline{\mathbf{L}}$ is a matrix such that $\underline{\mathbf{L}} \mathbf{f}_{O'} = \mathbf{f}_{O'} \times \overrightarrow{O'O}$. If one considers $\overrightarrow{O'O} = \begin{bmatrix} dx & dy & dz \end{bmatrix}^T$:

$$\underline{\mathbf{L}} = \begin{bmatrix} 0 & dz & -dy \\ -dz & 0 & dx \\ dy & -dx & 0 \end{bmatrix}$$

The actual wrench $\mathbf{w}_{O'}$ can now be considered in the QP problem with real constraints by merging equations 4.21 and 4.22:

$$\underline{\mathbf{U}} \underline{\mathbf{T}} \mathbf{w}_{O'} + \mathbf{u} \geq \mathbf{0} \quad (4.23)$$

4.2.3 Torque limiting

Other restriction that is normally considered when a WBC controller is built is the implementation of torque limiting. This avoids the controller from thinking it has an arbitrary range of power to compute the robot's locomotion. This is not only useful, but also necessary if the robot is going to be deployed in a real-life scenario.

Experimentally, it has been observed in simulations that, without torque limiting, the controller reacts in an extreme and chaotic way if the robot is falling or losing its posture. In neither case does this reaction revert the situation, and the robot ends up going down.

There are some critical situations, however, where the robot is still standing in a very unstable equilibrium. Sometimes, the controller applies a little bit of extra force in certain actuators that manage to get the situation back to normal (and reach a more stable equilibrium).

In order to: 1) avoid the simulation from crashing and 2) respect real torque constraints and be faithful to the resources that the robot can actually offer, torque constraints are added (keeping in mind the existence of external force).

Recalling equation 2.2:

$$\underline{M}\dot{\underline{v}} + \underline{C}\underline{v} + \underline{G} = \begin{bmatrix} \underline{0} \\ \underline{0} \\ \underline{\tau}_{joint} \end{bmatrix} + \underline{\tau}_{cf} + \underline{\tau}_{ext}$$

One can multiply both sides of the equation by the selection matrix for the DOFs corresponding to the joint torques, and rewrite the contact forces as a product of the contact jacobian and wrenches, as in 4.10:

$$\underline{S}_a \{ \underline{M}\dot{\underline{v}} + \underline{C}\underline{v} + \underline{G} \} = \underline{\tau}_{joint} + \underline{S}_a \{ \underline{J}_c^T \underline{w}_c + \underline{\tau}_{ext} \} \quad (4.24)$$

Therefore, keeping the notation that $\underline{h} = \underline{C}\underline{v} + \underline{G} - \underline{\tau}_{ext}$:

$$\underline{\tau}_{joint} = \begin{bmatrix} \underline{S}_a \underline{M} & -\underline{S}_a \underline{J}_c^T \end{bmatrix} \begin{bmatrix} \dot{\underline{v}} \\ \underline{w}_c \end{bmatrix} + \underline{S}_a \underline{h}$$

Since $\underline{\tau}_{min} \leq \underline{\tau}_{joint} \leq \underline{\tau}_{max}$, the following constraints can be formulated:

$$\begin{bmatrix} -\underline{S}_a \underline{M} & \underline{S}_a \underline{J}_c^T \end{bmatrix} \begin{bmatrix} \dot{\underline{v}} \\ \underline{w}_c \end{bmatrix} - \underline{S}_a \underline{h} + \underline{\tau}_{max} \geq 0 \quad (4.25)$$

$$\begin{bmatrix} \underline{S}_a \underline{M} & -\underline{S}_a \underline{J}_c^T \end{bmatrix} \begin{bmatrix} \dot{\underline{v}} \\ \underline{w}_c \end{bmatrix} + \underline{S}_a \underline{h} - \underline{\tau}_{min} \geq 0 \quad (4.26)$$

And now, adding these torque constraints and the wrench cone equations, the final set of inequality constraints can be built:

$$\underbrace{\begin{bmatrix} \underline{\mathbf{0}} & \underline{\mathbf{U}\mathbf{T}} \\ -\underline{\mathbf{S}_a}\underline{\mathbf{M}} & \underline{\mathbf{S}_a}\underline{\mathbf{J}_c^T} \\ \underline{\mathbf{S}_a}\underline{\mathbf{M}} & -\underline{\mathbf{S}_a}\underline{\mathbf{J}_c^T} \end{bmatrix}}_{\underline{\mathbf{C}}_l^T} \begin{bmatrix} \dot{\mathbf{v}} \\ \mathbf{w}_c \end{bmatrix} + \underbrace{\begin{bmatrix} \mathbf{u} \\ -\underline{\mathbf{S}_a}\mathbf{h} + \tau_{max} \\ \underline{\mathbf{S}_a}\mathbf{h} - \tau_{min} \end{bmatrix}}_{\mathbf{c}_l} \geq 0 \quad (4.27)$$

Chapter 5

Force compensation

Mathematically, the robot should now be able to account for external forces and be resilient and able to keep its stance in the presence of external perturbations, at least when it must not perform locomotion tasks. However, this is not true if the center of mass positioning task remains unaltered. If the task of center of mass task is still wanted to be accomplished, and the center of mass remains in the same position, the ZMP will forcefully fall out of the support polygon and the robot will lose stability. This applies for static and locomotion tasks. This will be analyzed in section 6.

This section aims to present the necessary theory and patches to include and compensate the external force in the algorithms.

5.1 Linear Inverted Pendulum

Let $\{C\}$ denote the center of mass frame of the robot. Its orientation is the same as the floating base $\{B\}$, and its origin is located in its center of mass. Let $\mathbf{x}_c$, $\dot{\mathbf{x}}_c$ and $\ddot{\mathbf{x}}_c$ define the position, velocity and acceleration of such a point with respect to $\{W\}$.

According to Newton's Second Law, the dynamics of the robot's center of mass can be modeled as:

$$\ddot{\mathbf{x}}_c = \frac{1}{m} \mathbf{F}_c \quad (5.1)$$

Where $\mathbf{F}_c$ denotes the total net force acting on the center of mass.

So, there exists an infinite number of force trajectories that can integrate $\ddot{x}_c$ and lead to the desired initial and final positions. If this force was controlled, say, by the presented feed-forward and PD law (like the tasks), then a very important physical constraint could be violated: the wrench cone. And even though the controller guarantees the compliance of the wrench cone via the inequality constraints, what would be the point of planning a trajectory with force tasks that are never going to be achieved? This can lead to serious discrepancies, instabilities, and ultimately to catastrophic failures. This is why the LIPM was posed.

The LIMP model assumes simplified robotic dynamics that are physically representative and computationally tractable, and states four hypotheses [20]:

1. The center of mass stays at a constant height, z_c and moves in a constant horizontal plane. This implies that the vertical component of the leg force has to exactly compensate the robot's weight. Also, this helps to linearize the motion equations.
2. The robot is a point mass.
3. Therefore, there is no torque around the center of mass (this implies that the leg force passes through the CoM).
4. By definition, the *zero-tilting moment point* (ZMP) has no horizontal torque component, so the leg force passes through the ZMP [10].

The definition of the ZMP is the point on the support plane where the net ground reaction force does not generate a moment in the horizontal plane. These assumptions turn the robot into an inverted pendulum:

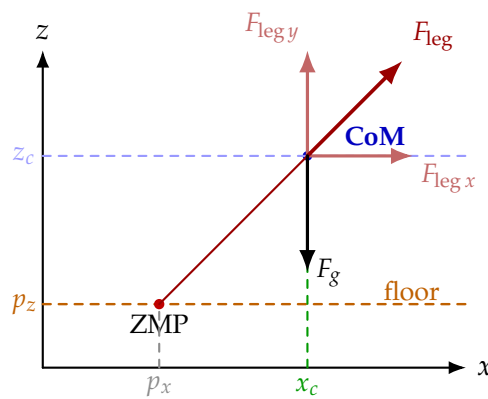


Figure 5.1: LIPM formulation. Extracted from [20].

And with them, the following equations can be formulated:

$$\begin{aligned}\ddot{x}_c &= \frac{g}{z_c - p_z} (x_c - p_x) \\ \ddot{y}_c &= \frac{g}{z_c - p_z} (y_c - p_y)\end{aligned}\tag{5.2}$$

$$z_c = \text{const.}$$

Where $\mathbf{p} = \begin{bmatrix} p_x & p_y & p_z \end{bmatrix}$ is the position of the ZMP.

If one defines $b := \sqrt{z_c/g}$ and considers $p_z = 0$, the equations result:

$$\begin{aligned}\ddot{x}_c &= \frac{1}{b^2} (x_c - p_x) \\ \ddot{y}_c &= \frac{1}{b^2} (y_c - p_y)\end{aligned}\tag{5.3}$$

$$z_c = \text{const.}$$

Finally, multiplying the mass on both sides:

$$\mathbf{F}_c = \frac{m}{b^2} (\mathbf{x}_c - \mathbf{p})\tag{5.4}$$

Locomotion or static control then consists of determining a coherent and feasible trajectory for the ZMP. Note that, as described in section 4.1.2, the compliance of wrench cone constraints is equivalent to the ZMP being contained in the support polygon. Therefore, a necessary condition for stable gait is that the ZMP must always remain in this space. By themselves, these equations are inherently unstable (pendulum dynamics), and it is convenient to reformulate the approach in order to build a robust gait controller, also taking into account external forces (that are not implicitly considered in this model). This is why the Divergent Component of Motion is proposed.

5.2 Divergent Component of Motion

The DCM is a physical magnitude defined as [31]:

$$\xi := x_c + b\dot{x}_c \quad (5.5)$$

Its projection on the XY plane can be interpreted as the point where the ZMP would need to be placed to compensate for the motion and come to a complete stop. After rearranging and differentiating equation 5.5, one can relate the ZMP and DCM dynamics as:

$$\dot{x}_c = -\frac{1}{b}(x_c - \xi) \quad (5.6)$$

$$\dot{\xi} = \frac{1}{b}(\xi - p) \quad (5.7)$$

Now, one would like to consider external forces and not model the equations around the ZMP (which does not take them into account). For that, one could consider equation 5.4 and update it. In this case, instead of F_c representing leg forces, now one can consider F_{ext} as a more general concept that includes every force acting on the center of mass (except gravity). However, now p is not a reference point that one can use, so the *Enhanced Centroidal Momentum Pivot* (eCMP) is introduced. This is the point that satisfies:

$$F_{ext} = \frac{m}{b^2}(x_c - r_{ecmp}) \quad (5.8)$$

And can or cannot be in the floor. Note that if there are no external forces acting on the robot and the change of angular momentum is neglected, $p = r_{ecmp}$.

Finally, in order to also consider gravity in the equation, one can define the *Virtual Repellent Point* (VRP) as:

$$r_{vrp} := r_{ecmp} + \begin{bmatrix} 0 \\ 0 \\ z_c \end{bmatrix}$$

And get the final equation:

$$\mathbf{F}_c = \frac{m}{b^2}(\mathbf{x}_c - \mathbf{r}_{vrp}) \quad (5.9)$$

Accounting for the effects of every force that acts on the center of mass. And with that equation, get the final relation between the DCM and the VRP:

$$\dot{\xi} = \frac{1}{b}(\xi - \mathbf{r}_{vrp}) \quad (5.10)$$

5.3 External forces

With these dynamics, (VRP repelling the DCM, DCM attracting the center of mass), one can robustly build reference trajectories for the robot to follow, thus generating gait patterns. This is how the rpc module works and how it is described in [20, 21, 31]. In these papers, and in rpc, these trajectory references are built without considering external forces (just taking into account gravity and leg forces). So, what if an external force was applied to the robot?

Well, if one considers reference trajectories for the DCM and the VRP:

$$\xi_{ref}(t) \quad \mathbf{r}_{vrp ref}(t)$$

By linearity, an external, perturbative force applied on the center of mass would offset these trajectories [31]:

$$\xi_{shift}(t) = \xi_{ref}(t) - \frac{b^2}{m}\mathbf{F}_{pert}(t) \quad (5.11)$$

$$\mathbf{r}_{vrp shift}(t) = \mathbf{r}_{vrp ref}(t) - \frac{b^2}{m}\mathbf{F}_{pert}(t) \quad (5.12)$$

However, it is not so easy to implement because the rpc works in a particular way where this equation is not useful. Indeed, when the robot is instructed to walk, the controller performs the following set of actions:

1. First, it precomputes references for every footstep that is going to be taken.
2. Then, it uses those footsteps references to compute the trajectories of DCM and VRP. It considers the footsteps as waypoints for VRPs, joins them with smooth curves and then com-

putes the reference for DCM using equation 5.10.

3. Next, it obtains a reference trajectory for the center of mass with equation 5.7.
4. Finally, it stores the trajectories and uses the reference for the center of mass to feed the controller.

The point is, all the trajectories are precomputed before even starting to walk (if it walks at all). Also, it is not possible to store a shifted trajectory when the future forces are not known.

What it is proposed, then, is to compute the walking trajectory and update the center of mass desired values at every instant. Let's consider an external net wrench acting on the center of mass $w_{ext}(t) = \begin{bmatrix} F_{ext}^T(t) & M_{ext}^T(t) \end{bmatrix}^T$. In chapter 3, it has been shown that this wrench can be observed and estimated as $\hat{w}_{ext}(t)$. Neglecting the effect of the torques [31], one can apply the observed external force to the references in real time. According to equation 5.6, a shift in the DCM will cause a shift in the reference for the center of mass. If one makes the assumption of negligible external force derivative (quasistatic force), the center of mass reference trajectory can be also updated by adding the force offset:

$$x_{cshift}(t) = x_{cref}(t) - \frac{b^2}{m} \hat{F}_{ext}(t) \quad (5.13)$$

Then, this reference is fed to the center of mass task, and the controller is responsible for fulfilling it. The center of mass velocity reference is not updated. If the derivative of the force were considered, then a different approach must be followed to fulfill the ODE, but it is reserved for future work. For now, this is the way it is implemented.

5.4 Limitations

It is important to address the theoretical limitations of such a model. If one considers the pendulum model, but this time with external forces, the center of mass is going to be displaced from the vertical of the origin by a factor of $-b^2/m$, as shown in equation 5.13. This offset generates a moment with respect to the foot (where the reaction wrench is computed), as depicted in the following picture:

Therefore, the foot force reaction is $\mathbf{F}_r = -\mathbf{F}_c$. The reaction moment is obtained by multiplying the lever arm by the force:

$$\mathbf{M}_r = \mathbf{x}_c \times \mathbf{F}_c$$

So, the reaction wrench is:

$$\mathbf{w}_r = \begin{bmatrix} \mathbf{F}_r^T & \mathbf{M}_r^T \end{bmatrix}^T \quad (5.14)$$

If one plugs this wrench into equation 4.23, constraints on the maximum force that the robot can physically withstand are obtained. Solving this set of inequations analytically is not trivial, even though it is linear. So, in order to determine the physical limit of a perturbation, the result of this system can be checked (with actual values of μ , X and Y) to address its feasibility. This will be an useful tool in the forthcoming section.

Chapter 6

Experiments

Previous chapters have established the theoretical background and the necessary software modifications for a bipedal robot to estimate and counteract external disturbances. This last chapter aims to empirically validate the effectiveness of such implementations via `pybullet` simulations. So, a series of experiments will be carried out with the following objectives:

1. Validate the force observer presented in chapter 3: study the precision in magnitude and direction of the observed forces.
2. Evaluate the static postural control: check the compliance of cone wrench, torque and slip-page constraints.
3. Analyze the performance of the robot in locomotion tasks.

Finally, the forces described in section 2.4 will be compared to point forces, in order to validate the viability of this whole pipeline in flooded environments, which is the final aim of this project.

Experiments in sections 6.1, 6.2, 6.3 and 6.4 have been carried out with an observer of second order, with constant gains of 100 for every entry; standard values of task gains (as defined in [4]) and, unless otherwise stated, a value of friction coefficient of $\mu = 0.7$.

6.1 Valid force detection

First, one would like to assess if the forces that the robot can detect are consistent and coherent. For that, two experiments will be carried out. First, no force will be applied to the robot. Second, a point force of known magnitude and direction will be applied. The estimated external wrench

(force, plus moment with respect to center of mass) will be computed as shown in section 3.2, and its error will be compared with the theoretical values. This force will be applied at a known point, namely, the base of the robot. This point does not match the center of mass, but the distance between these two points is small enough to approximate both points as the same.

6.1.1 No external force

After running an experiment without external perturbations, one can observe two phases. First, a transient state exists, where the observed force (unlike the true one) is non-zero. This happens because the robot does not start the simulation in the initial state (Double Support Stand Up-DSSU), but rather in a preliminary state (Duble Support Balance-DSB) that needs to transition to the nominal state (DSSU). In this first state (DSB), the robot is touching the ground with a force distribution that is not the desired one, and neither is its initial position. Due to these changes in contact forces and stance, the observer needs to converge, which represents this initial transient state. After it, the observer effectively converges to the zero-force values.

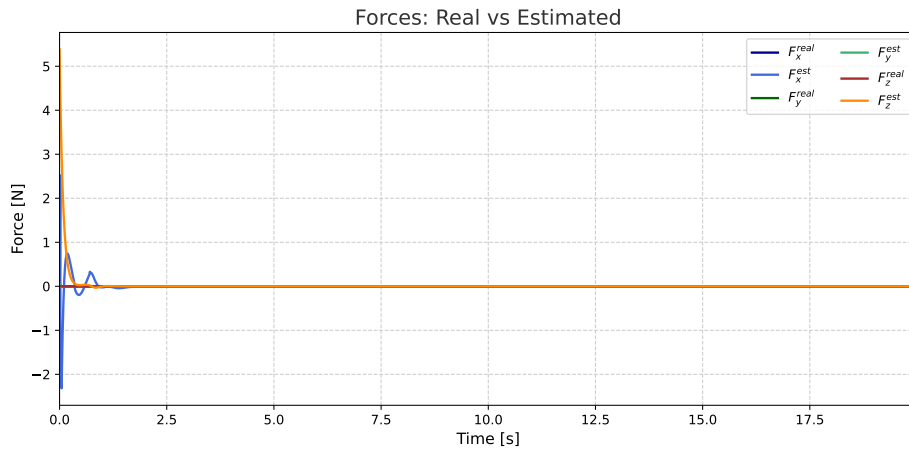


Figure 6.1: No external force - Real vs estimated force at CoM.

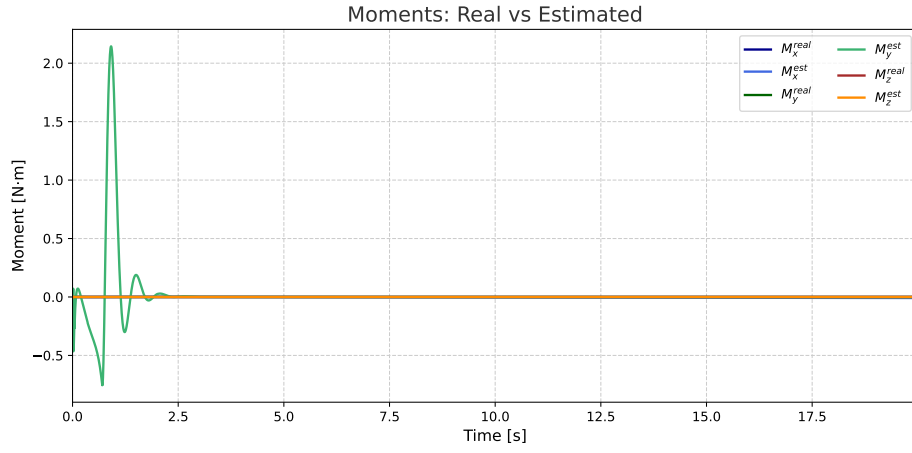


Figure 6.2: No external force - Real vs estimated moment at CoM.

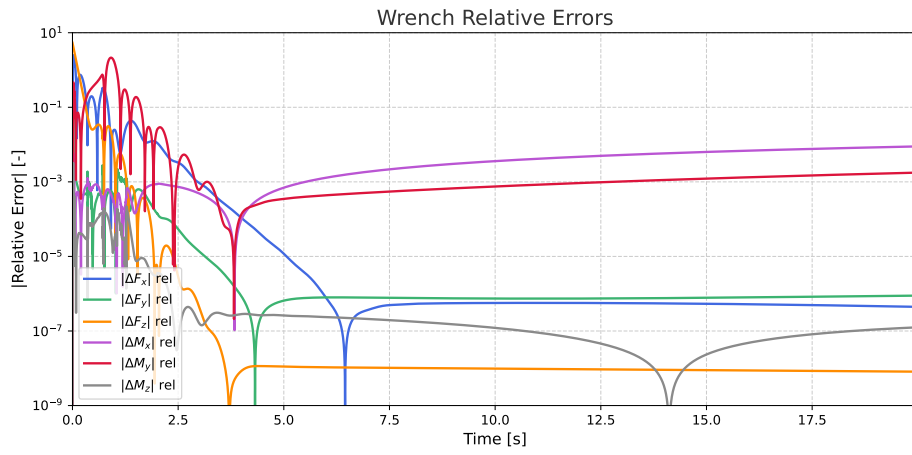


Figure 6.3: Relative error between real and estimated magnitudes.

6.1.2 Known external force

Now, it is the moment of testing the accuracy of the observer for a known force. For that, a known force of known magnitude is applied at the base of the robot. The applied force is: $\mathbf{F} = \begin{bmatrix} 30 & 20 & 0 \end{bmatrix}$ N, in order to study the accuracy in both directions. Again, the transient state is observed, as well as the convergence to the real values. The error is considered to be so small as to be neglected and to consider the experiment a success. This force will be applied with an arctangent profile (since a sudden impulse of that magnitude makes the simulation no longer quasistatic, and the speed of the observer is not enough to compensate for the force in time):

$$F_i(t) = \frac{2}{\pi} F_i \arctan(t - t_0) \quad ^1$$

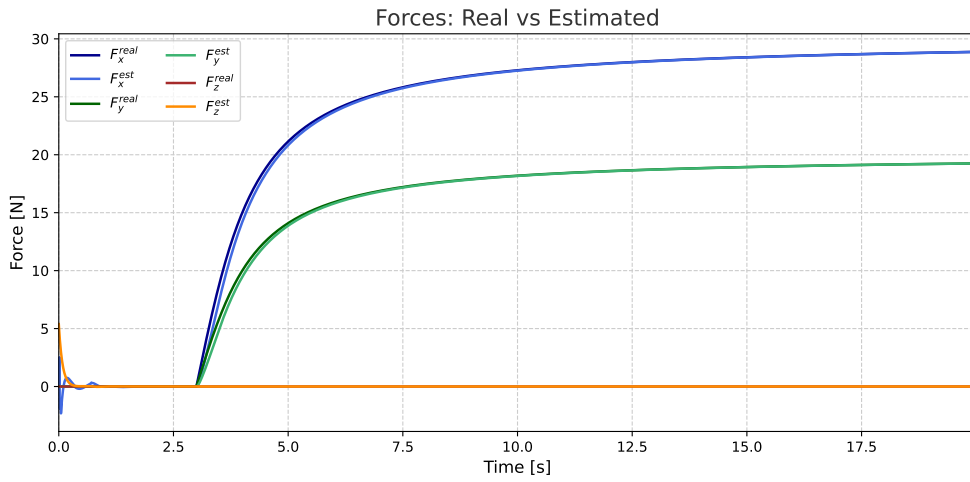


Figure 6.4: Known external forces - Real vs estimated force at CoM.

¹ $t_0 = 3$ s

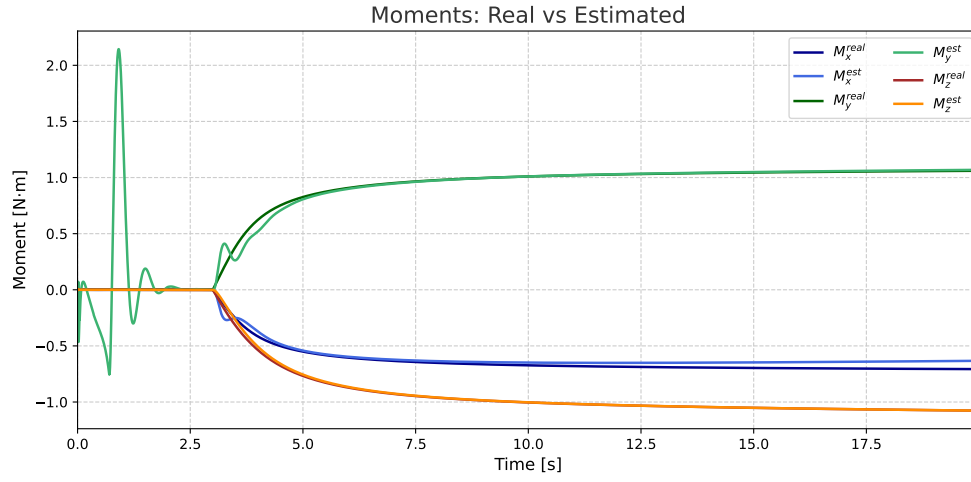


Figure 6.5: Known external forces - Real vs estimated moment at CoM.

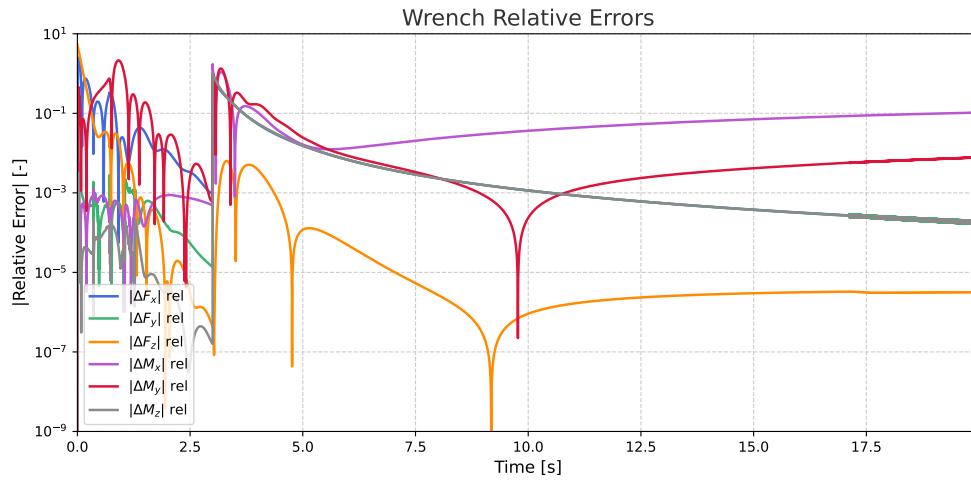


Figure 6.6: Relative error between real and estimated magnitudes.

The maximum relative error is obtained for M_x (order of 10%), while the rest remain under 1%. The presence of such error is blamed on the offset of the application point and the center of mass. This will be further addressed in forthcoming experiments.

Also, one can plot the center of mass to check that the robot is indeed considering and accurately following the offset tasks.

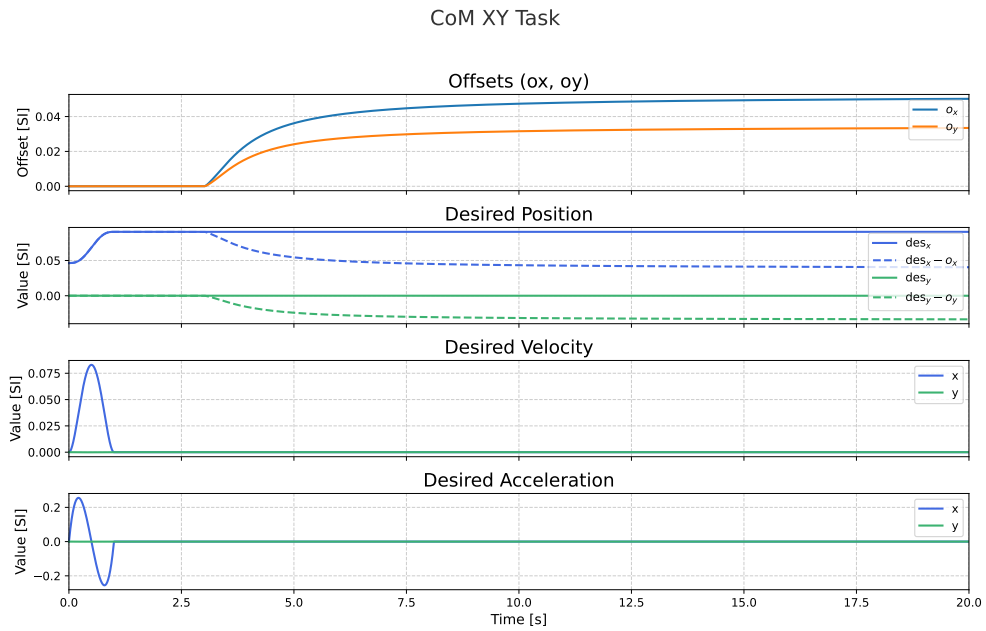


Figure 6.7: CoM Task - Offsets and references.

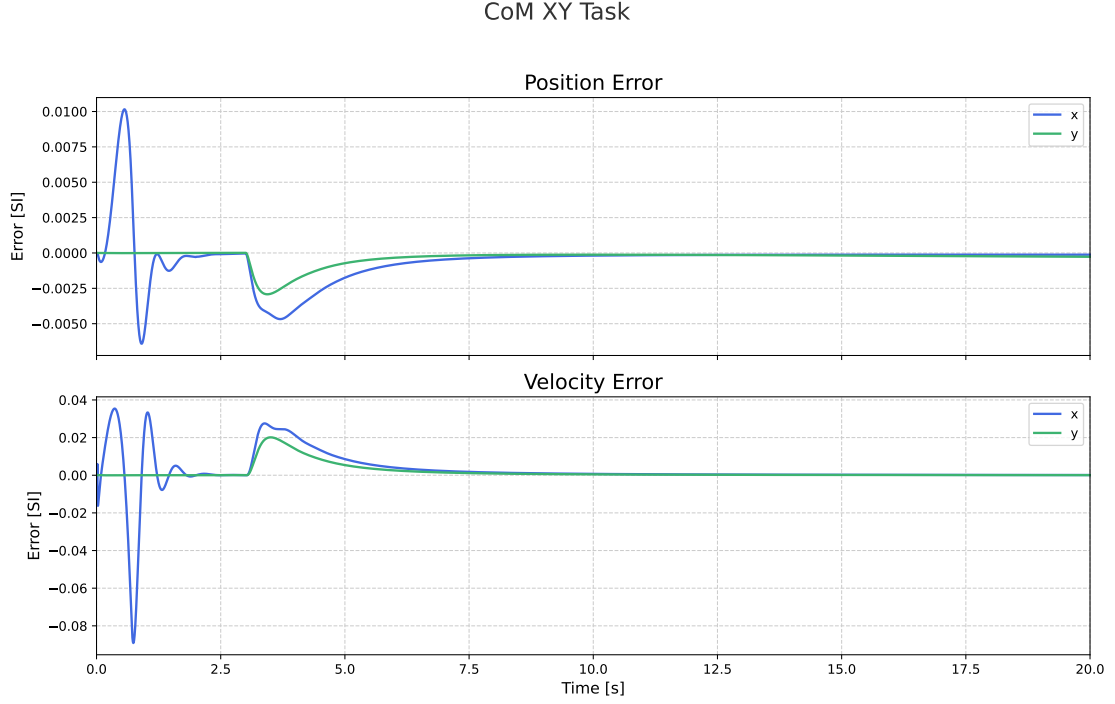


Figure 6.8: CoM Task - Absolute errors.

6.2 Static stability and limits

This next set of experiments aims to validate the assumptions made in section 5.4. If one plugs 5.14 into equation 4.21, with certain values of F_{ext} and μ , one can get a feasible region for the external force that can be theoretically applied to the robot without reaching the limits.

6.2.1 Theoretical limits

For the values of mass of the g1 robot (35.11 kg), the standard value for $\mu = 0.7$; and considering that its center of mass is aligned vertically with the contact, the theoretical limits that are obtained are:

$$\mathbf{F}_{ext\,xy} = \left[\pm 239.39, \pm 239.39 \right] \text{ N}$$

Which corresponds to the slippage inequalities. However, if one performs analysis to validate

such limits, the results that are obtained are much lower. For pure forces in X and Y directions, the limits are:

$$F_{ext\,x} \in [-75, +184] \text{ N} \quad F_{ext\,y} \in [-73, +73] \text{ N}$$

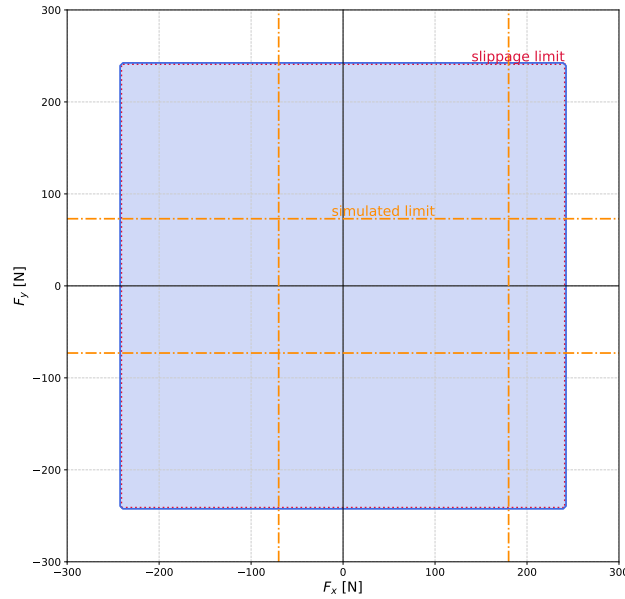


Figure 6.9: Simulated limits for $\mu = 0.7$.

These theoretical limits do not match the simulated ones, and the reason is joint limits. Indeed, when the robot leans to counteract the external force, there exists a physical limitation that avoids the robot from leaning over a certain angle. This physical limit (that is provided by pybullet) is not considered by the controller, leading to discrepancies and instability. In fact, the first warning of wrench cone compliance starts when the joints reach this physical limit. In order to consider this issue, modifications to the QP problem must be proposed, but it falls out of the scope of this work.

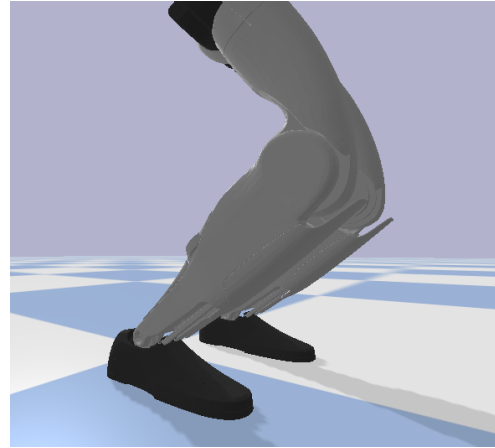


Figure 6.10: The instant when warnings start corresponds to the shin's angle limit. $F_x = -75 \text{ N}$.

6.2.2 Reduction of friction factor

There is a way of studying this issue more in depth, and it is reducing the value of μ until the feasible region is smaller than the region limited by the joint limits. If one reduces it to $\mu = 0.1$, it is theoretically impossible for the robot to withstand a force greater than the maximum friction force; hence, slipping.

In this case, the theoretical limits are:

$$\mathbf{F}_{ext\,xy} = \left[\pm 33.33, \pm 33.33 \right] N$$

the limits when the controller showed the first warning (collapsing afterward) were:

$$F_{ext\,x} \in \left[-32, +33 \right] N \qquad F_{ext\,y} \in \left[-31, +31 \right] N$$

These limits almost identically match the theoretical ones; the mismatches can be blamed on the fact that the model does not fully represent the dynamics of the robot; since (even though the controller tries to keep the reaction forces on both legs equal) there are slight discrepancies in leg forces that make one foot collapse first. Anyway, the results show that the model is feasible and that the limits are strict.

6.2.3 No momentum observer

The final test of this section justifies the use of the observer. The experiment is carried out without the momentum observer. According to 4.21, a theoretical limit of

$$\mathbf{F}_{ext\,xy} = \left[-33, +51 \right] \times \left[\pm 9 \right] N$$

can be achieved. This is far less than the limits that have been obtained experimentally. Fortunately, simulation yields slightly more optimistic results (even though they are still worse than when the observer is active):

$$F_{ext\,x} \in \left[-28, +50 \right] N \qquad F_{ext\,y} \in \left[-39, +39 \right] N$$

This increase in performance (in Y-axis) comes from a reaction of the robot. It does not stay still and, by changing its posture, redistributes the forces in a way that delays the collapse of the

equations. The hypothesis of pendulum along X-axis makes more sense, so as it was expected, behavior in this axis is more correlated to the theoretical predictions.

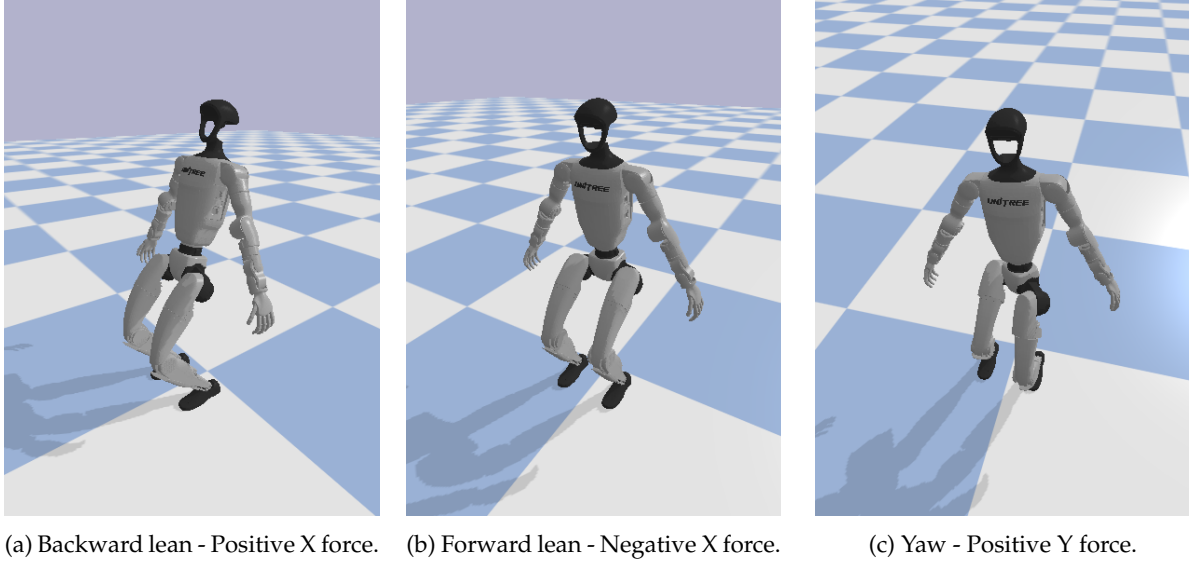


Figure 6.11: Reactions of the robot.

In order to make the simulations conditions more similar to the hypothesis on Y-axis, a new set of states has been programmed. These states allow the robot to lift one leg and stand on just one leg.

The behaviors about forces on X direction are the same. However, there exist new limits concerning the Y axis. Now, the equations fail from 8 N onwards. This mismatch can be due to an offset on the center of mass (that starts slightly displaced from the vertical of the contact). Anyway, the results are taken as valid, and the model is considered to be loyal to the simulation.

Either way, the limits without the external observer are less than 50% to those with the momentum observer, which justifies the use of such architecture.



Figure 6.12: New programmed state of one-leg-stance.

6.3 Locomotion performance

Now, the task is to make the robot move under perturbations. For the moment, they are known, constant, and applied to the center of mass.

6.3.1 Swaying

The first test consists of addressing the robot's ability to counteract the forces while moving. For now, this movement does not involve the feet and just requires the center of mass to sway. The act of swaying and its parameters were programmed beforehand by [4] and are not corrected at all. Three experiments are carried out: force on positive X and Y directions (and combined), with active and inactive observer; and a reference case to track an expected amount of error. In order to be just, a force of 20 N of magnitude (bearable by all situations) is applied. For every simulation (except the reference), sway command is called on $t_c = 5s$. Forces are applied during 5 seconds, followed by a recovery period of 5s (while swaying command is still active). Simulation ends on $t_f = 45s$.

No force. Reference

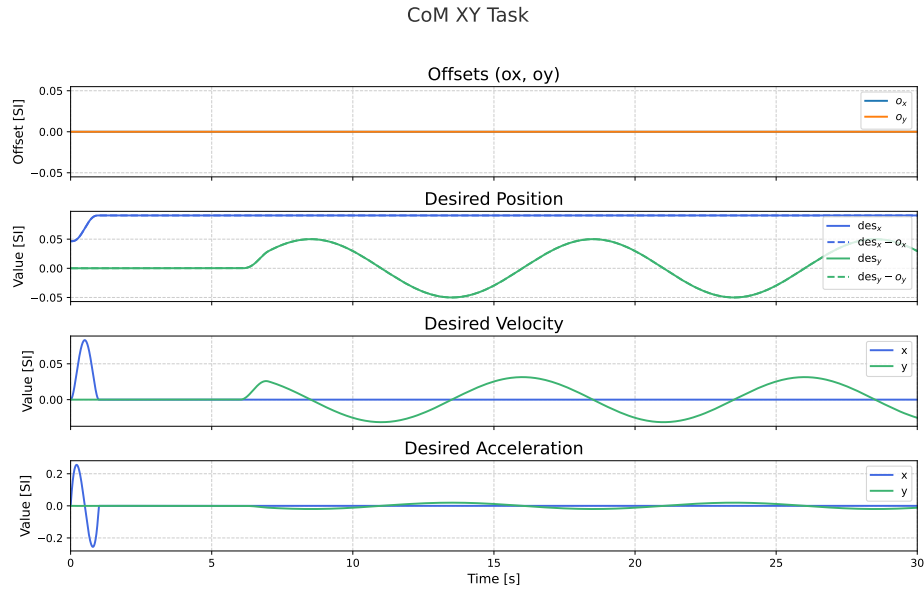


Figure 6.13: CoM Task - References. Reference case.

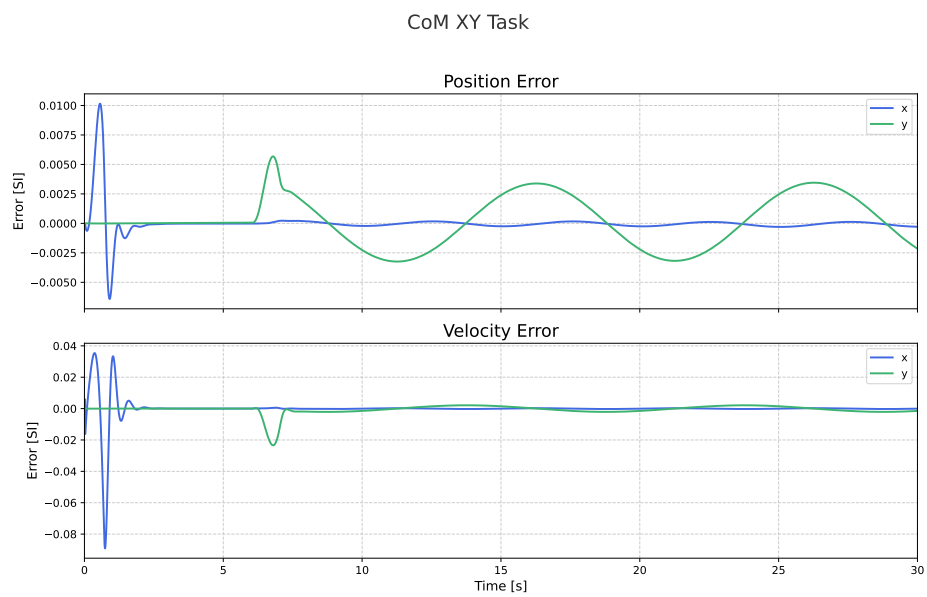


Figure 6.14: CoM Task - Absolute errors. Reference case.

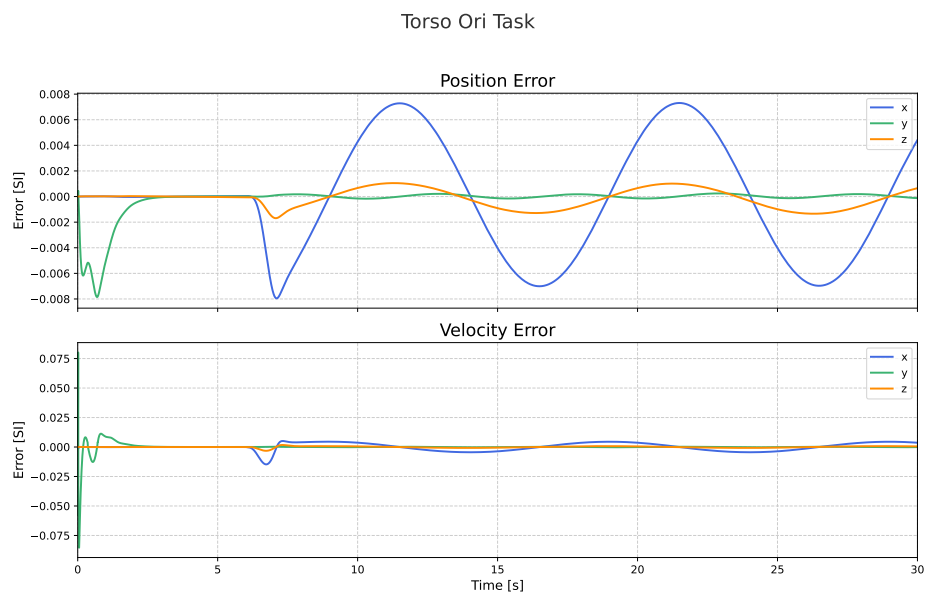


Figure 6.15: Torso Orientation Task - Absolute errors. Reference case.

Forces applied. Active observer

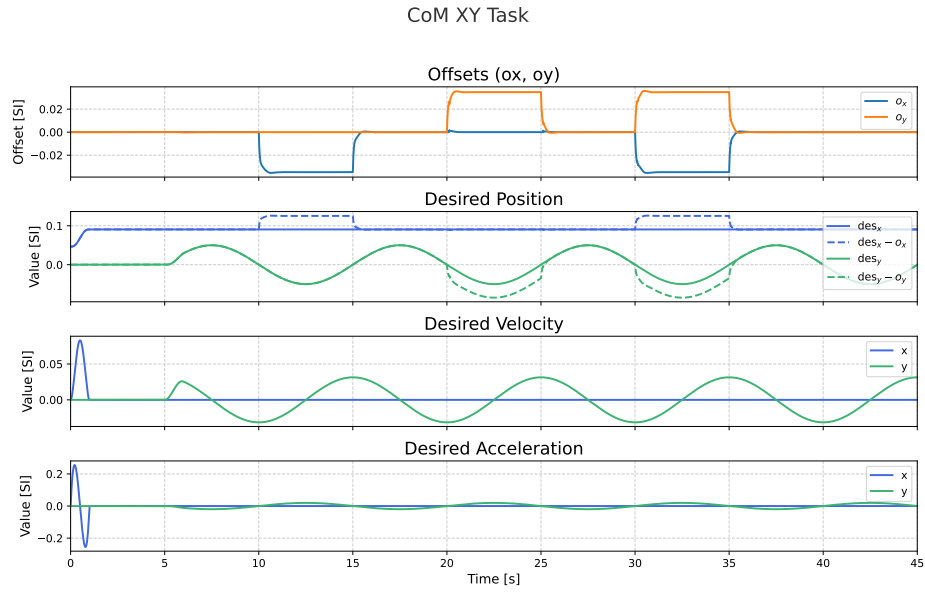


Figure 6.16: CoM Task - References. Forces applied and active observer.

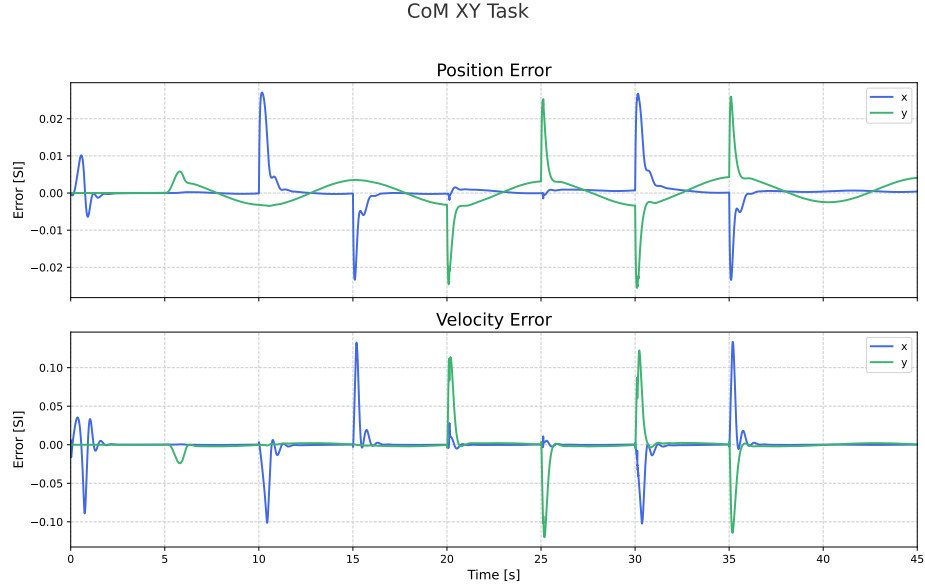


Figure 6.17: CoM Task - Absolute errors. Forces applied and active observer.

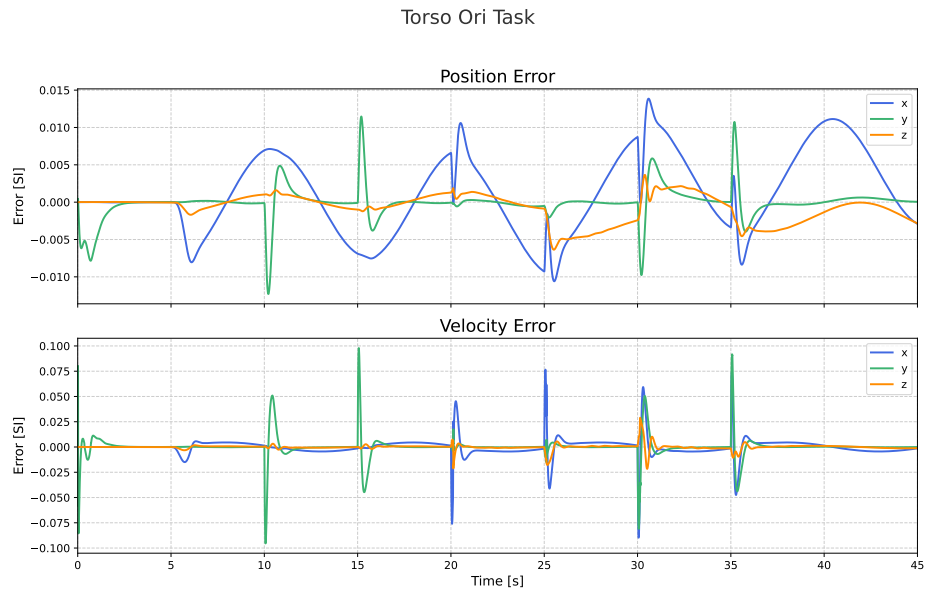


Figure 6.18: Torso Orientation Task - Absolute errors. Forces applied and active observer.

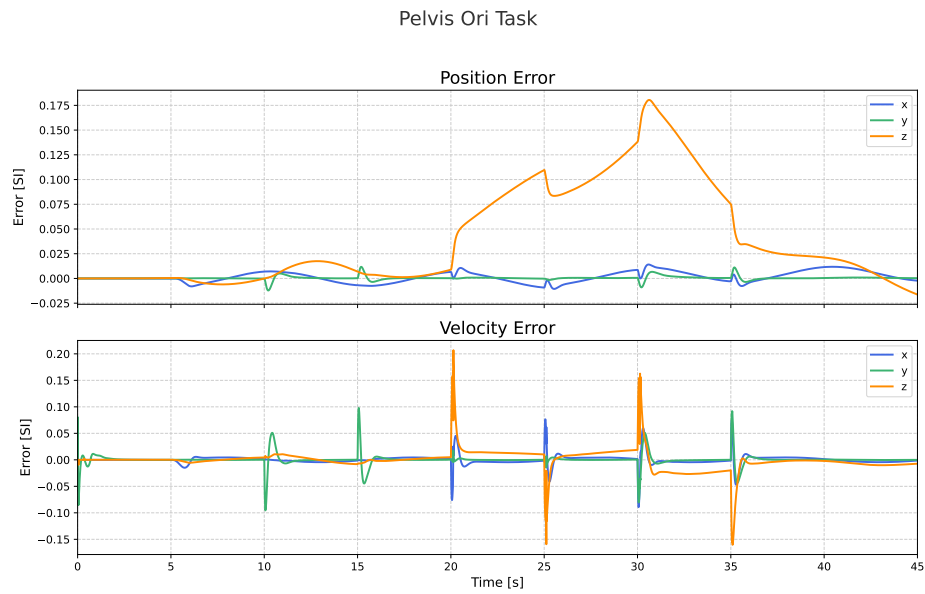


Figure 6.19: Pelvis Orientation Task - Absolute errors. Forces applied and active observer.

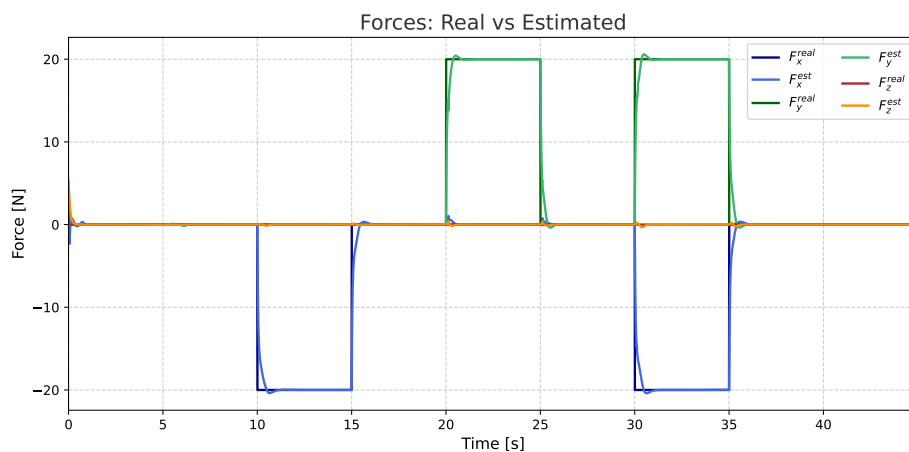


Figure 6.20: Estimated external forces. Forces applied and active observer.

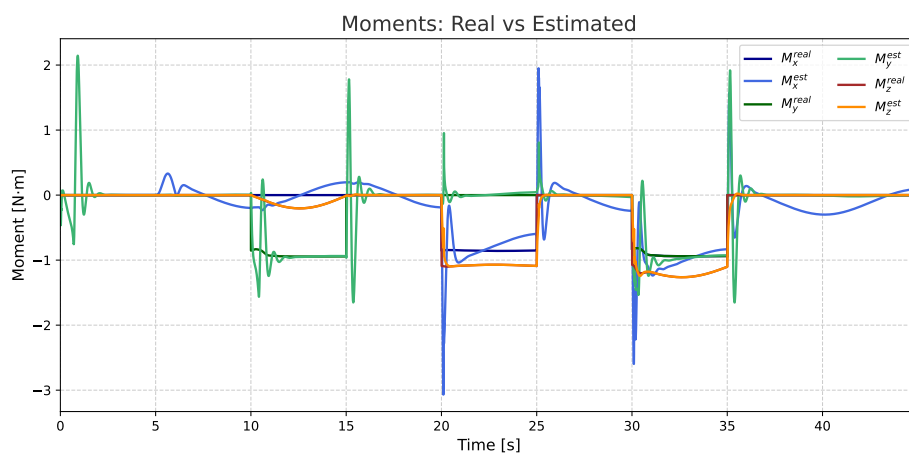


Figure 6.21: Estimated external moments. Forces applied and active observer.

Forces applied. Inactive observer

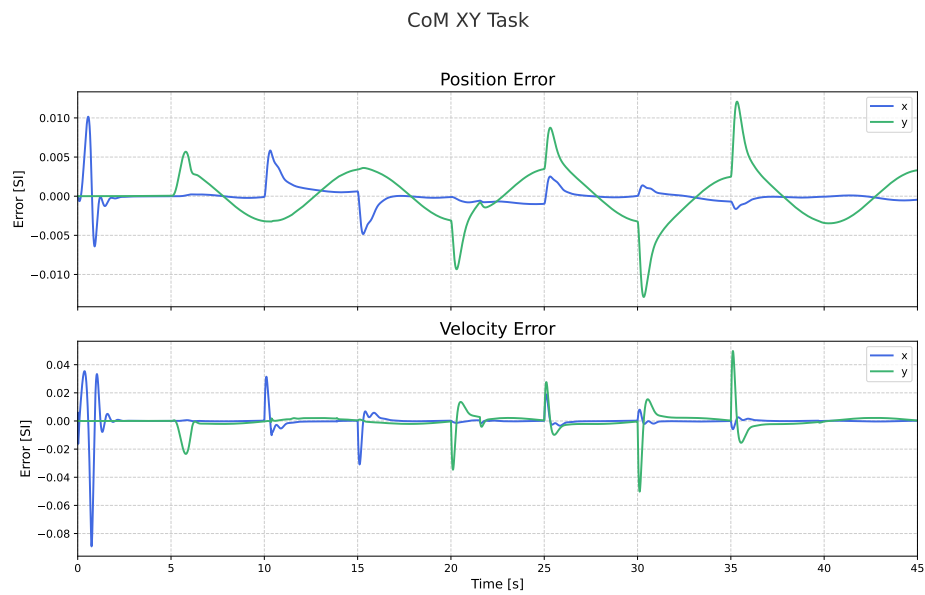


Figure 6.22: CoM Task - Absolute errors. Forces applied and inactive observer.

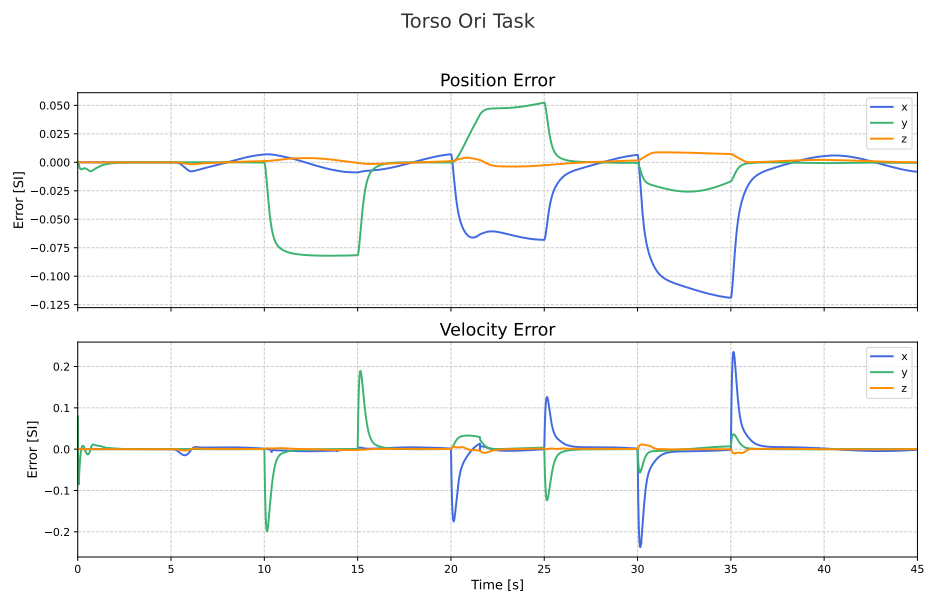


Figure 6.23: Torso Orientation Task - Absolute errors. Forces applied and inactive observer.

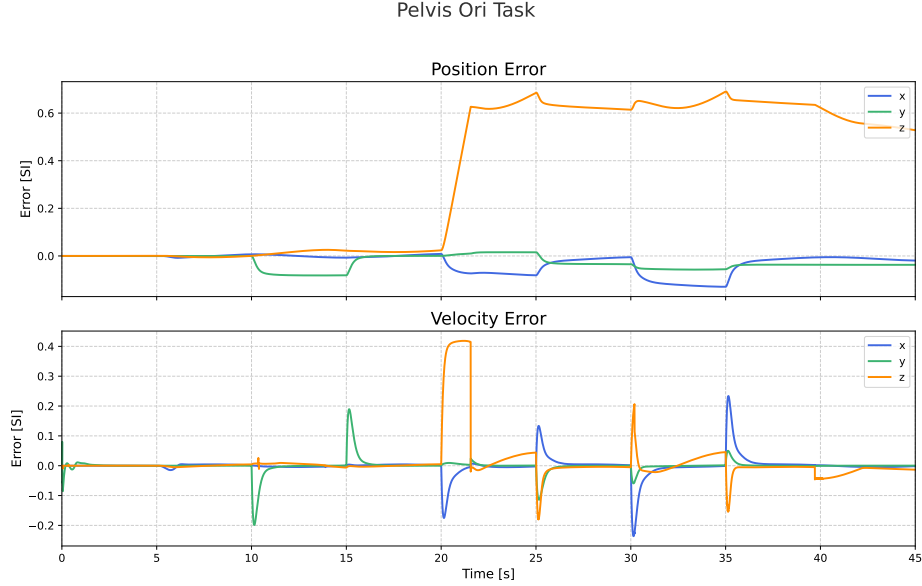


Figure 6.24: Pelvis Orientation Task - Absolute errors. Forces applied and inactive observer.

- First and foremost, the reference case has been plotted in order to see how much error exists in a perturbation-free environment. Forces and moments are being estimated correctly, even when the robot is moving. The slight errors in moment come from the change of angular momentum of the robot when it sways (the case is not perfectly quasistatic), which can be seen in the error tracking of the CoM and Torso Orientation tasks (6.14, 6.15).
- Second, offsets (6.16) are applied as expected, and the center of mass is following its trajectory with the same amount of error as in the reference case, as seen in 6.17.
- Next, center of mass tracking, as well as torso and pelvis orientation tasks are not effectively computed when the observer is not active, since the robot is fighting a force that it does not understand. Not only the error is bigger, but also it is stationary and does not go back to zero (6.22, 6.23, 6.24).
- Finally, as it was seen in previous experiments, the impact of the observer goes beyond main tasks (center of mass) and keeps the robot upright. It is observed that, without the observer, the error in the orientation is high and is not compensated at all (6.23, 6.24).

6.3.2 Walking

The final step is to verify the ability to walk in the presence of perturbations. First, a controlled constant force has been applied to the robot's base, and its operational range has been characterized. This means, the maximum force that the robot can face while stably taking steps has been empirically obtained. For the current implementation of walking patterns, without the observer, the operational range lies between:

$$F_{ext\,x} \in [-11, +16] \text{ N} \qquad F_{ext\,y} \in [-4, +4] \text{ N}$$

With the observer active, these values jump to:

$$F_{ext\,x} \in [-20, +33] \text{ N} \qquad F_{ext\,y} \in [-12, +12] \text{ N}$$

Which means an increase of bearable force of 81% and 106% in negative and positive X direction respectively, and 200% in both Y directions.

However, this is still way lower than the static limit presented in section 6.2.1. This mismatch comes from the hypothesis of quasistaticity: the fact that the robot moves makes the observer change its state, and even though there is a filter that reduces the pikes in force changes, it is not enough. The observer is constantly trying to converge to a changing state, and this can lead to instability. Also, since the robot walks by precomputing a trajectory, slight errors in tracking can lead to accumulations that make the gait unstable in the long term. It is also needed to say that the values presented as *limit* correspond to values for which the robot was able to take several steps (precisely, six) without showing signs of instability (false steps, excessive leans...). Without this strict consideration, the robot was able to take steps up to 50 N, but failing catastrophically afterward.

This final experiment shows the ability of the robot to walk under discontinuous perturbations, smoothly tracking external forces and adapting to offsets, but also showing the problems that arise when estimating moments due to the continuous movement and changes of angular momentum of the robot.

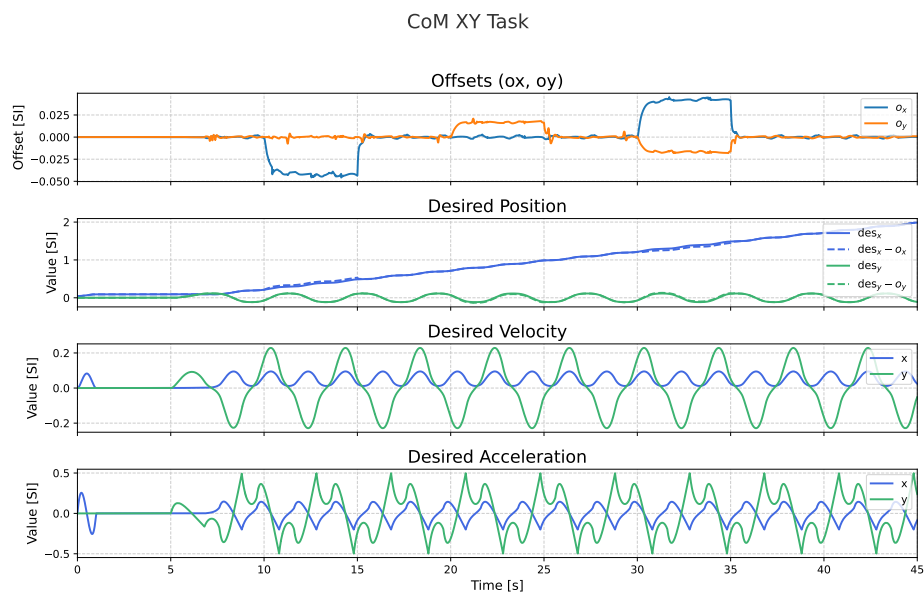


Figure 6.25: CoM Task - References. Walking task.

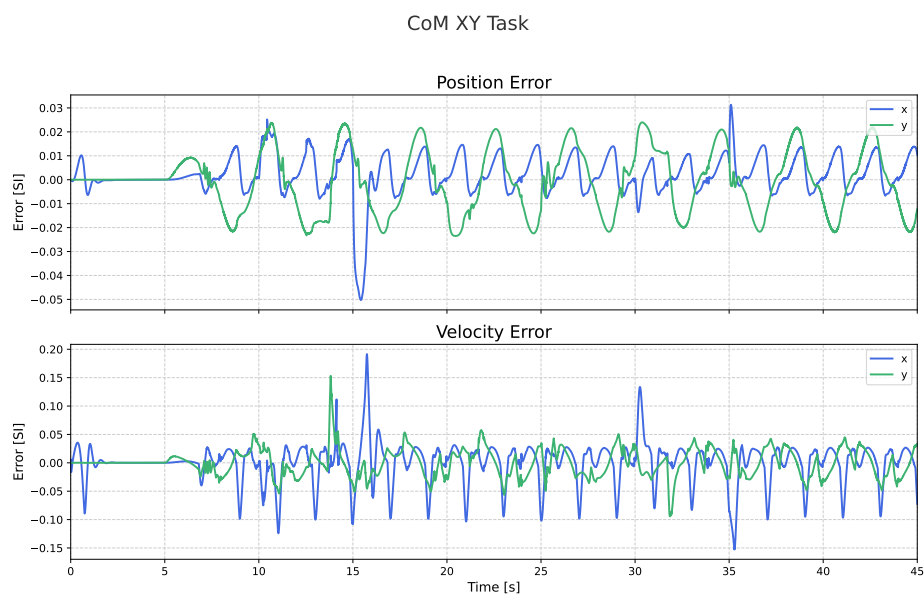


Figure 6.26: CoM Task - Absolute errors. Walking task.

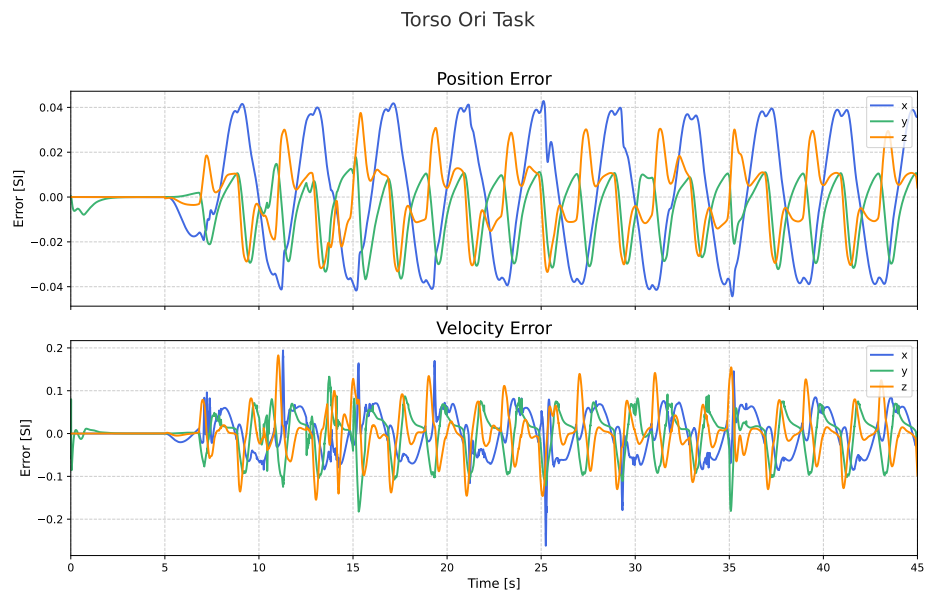


Figure 6.27: Torso Orientation Task - Absolute errors. Walking task.

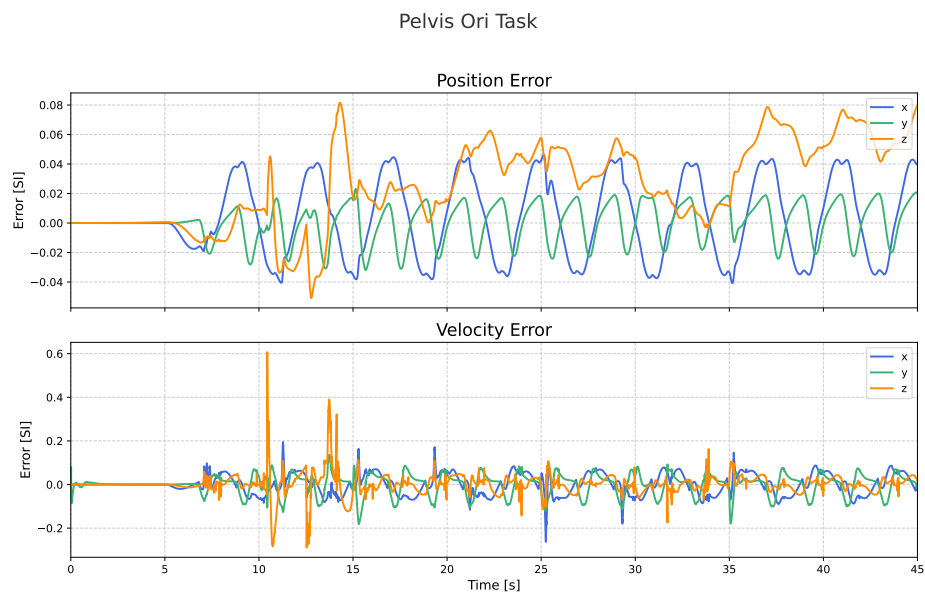


Figure 6.28: Pelvis Orientation Task - Absolute errors. Walking task.

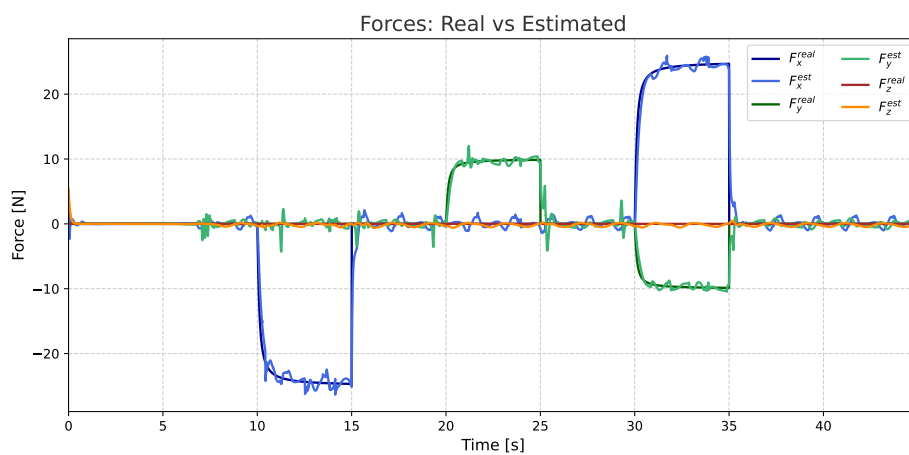


Figure 6.29: Estimated external forces. Waking task.

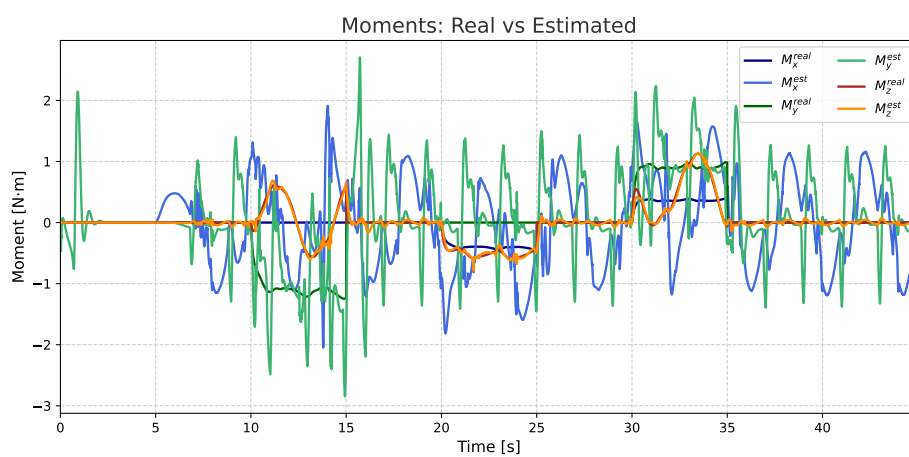


Figure 6.30: Estimated external forces. Walking task.

6.4 Disperse forces

At the moment, all the tests have been carried out applying forces on the base of the robot, following the hypothesis of negligent angular momentum on its base. However, it is important to know if the observer can also get the external force (or, at least, the equivalent force-momentum on the center of mass) when the force is applied in different places. For that, this first set of experiments tries to determine the power of the observer when it needs to estimate point or distributed forces acting away from the center of mass.

6.4.1 Known point forces

Recalling equation 3.25, there are n_{ext} jacobians involved in the recovery of the external force, each of them codifying one application point. It has been observed experimentally that the result of the net wrench force acting on the center of mass is independent of the amount of jacobians present in the formula. Fortunately, this external estimated wrench matches exactly the real wrench applied to the robot, transported to the center of mass. For example, the following test shows two point forces being applied to two different points of the legs.

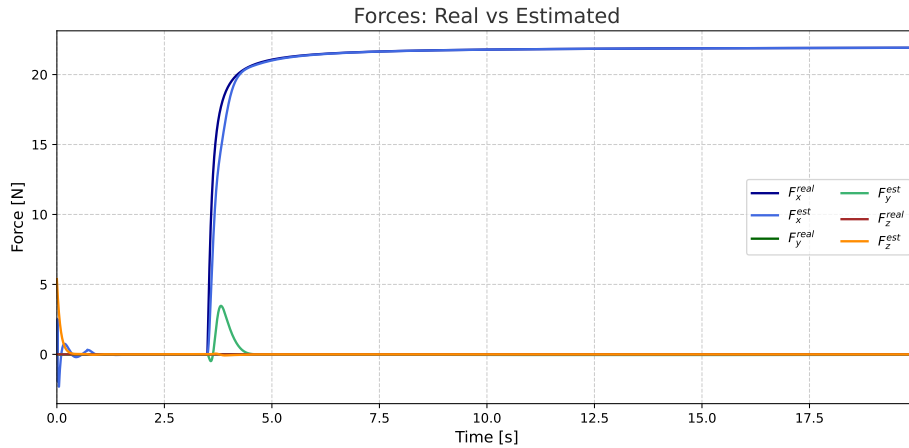


Figure 6.31: Disperse force - Real vs estimated force at CoM.

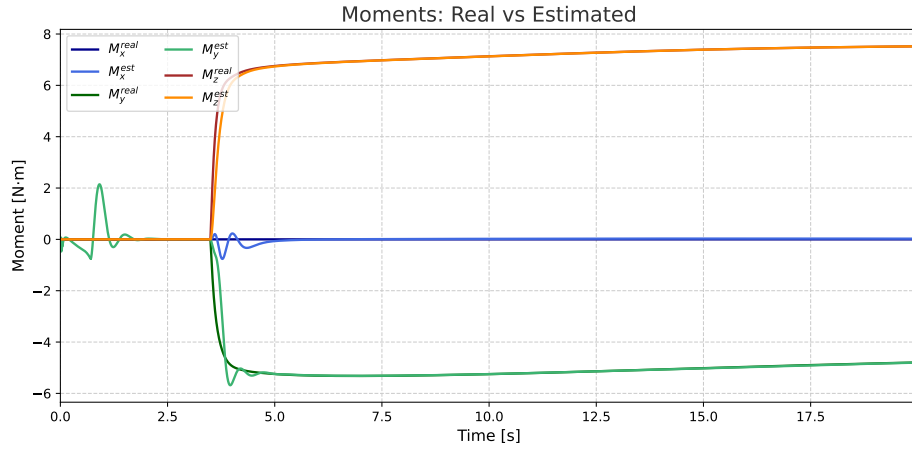


Figure 6.32: Disperse force - Real vs estimated moment at CoM.

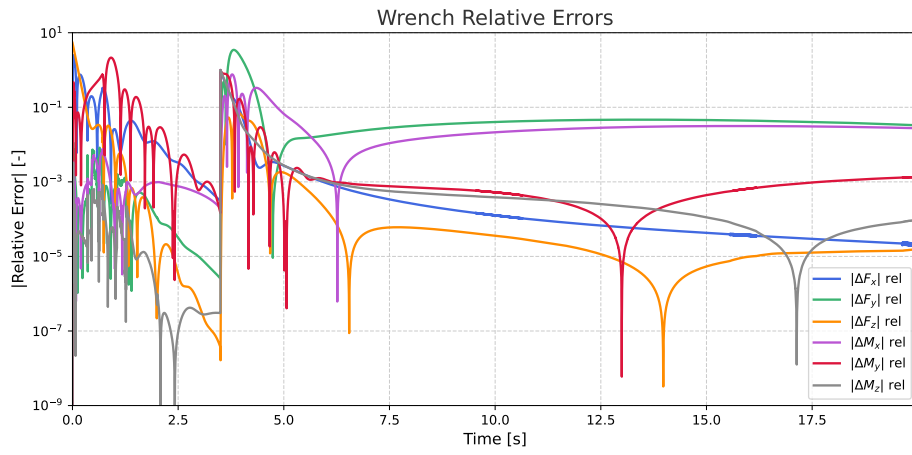


Figure 6.33: Relative error between real and estimated magnitudes.

The problem may come because this estimation is very truthful regarding the center of mass, but might not be representative about *where* the force is actually being applied. If one checks the resultant force referenced to each of the links:

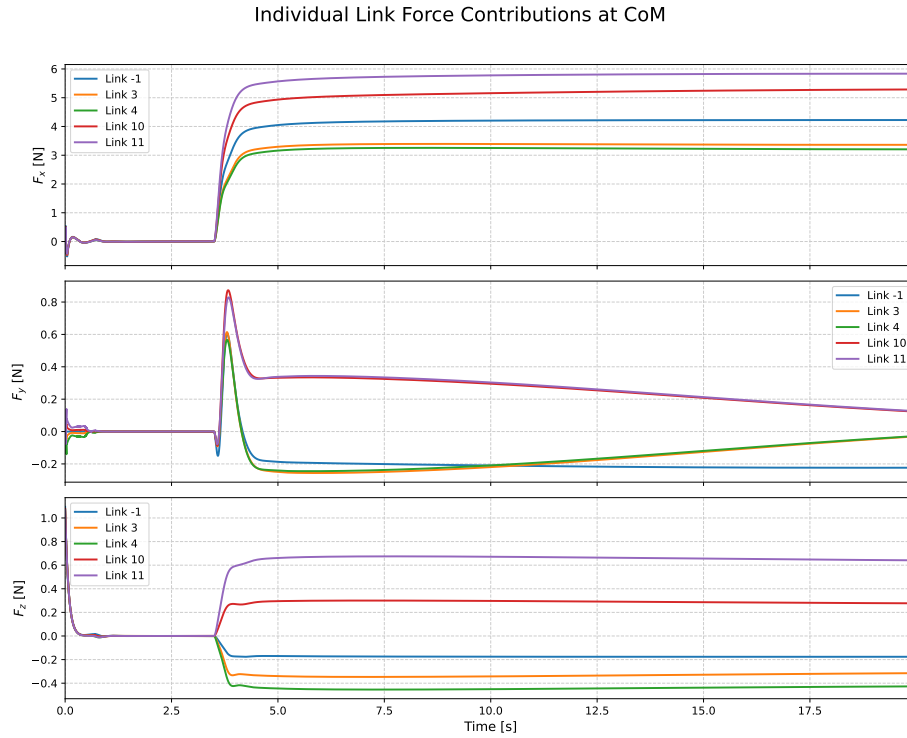


Figure 6.34: Disperse force - Individual force contributions at each link.

Every link that is being used in the computation is being assigned a value of force that the controller thinks is being applied to it. However, in this case, link -1 (pelvis, the base) is not subjected to any force, and it still considers a force on it. This should not be a problem, because the point of interest is the center of mass, and as long as the force there is being computed precisely, the center of mass tasks updater will work correctly.

6.4.2 Distributed forces

Let this final test show the power of the observer in front of distributed forces along the robot's legs. Again, the number of jacobians involved in the recovery of the force is pointless, but either way, the experiment is run using three contact points per leg plus the pelvis.

This test force that is applied to the robot is the model recovered from section 2.4, and it takes into account drag and buoyancy forces.

As expected, the robot is able to detect the net wrench acting on the center of mass, which is the crucial key to jump to the following and final section. In the end, it is the same as detecting multiple (integrated) forces and moments, which has been solved and successfully achieved in experiment 6.4.1. Forces acting on each of the detected points are still not precise.

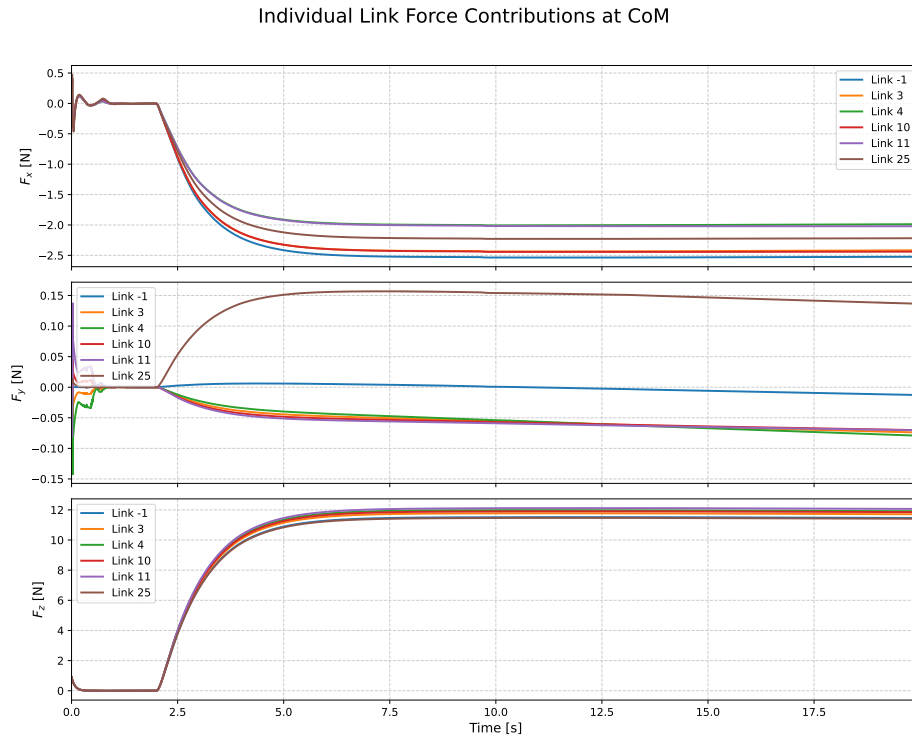


Figure 6.35: Distributed forces - Individual contributions and each link.

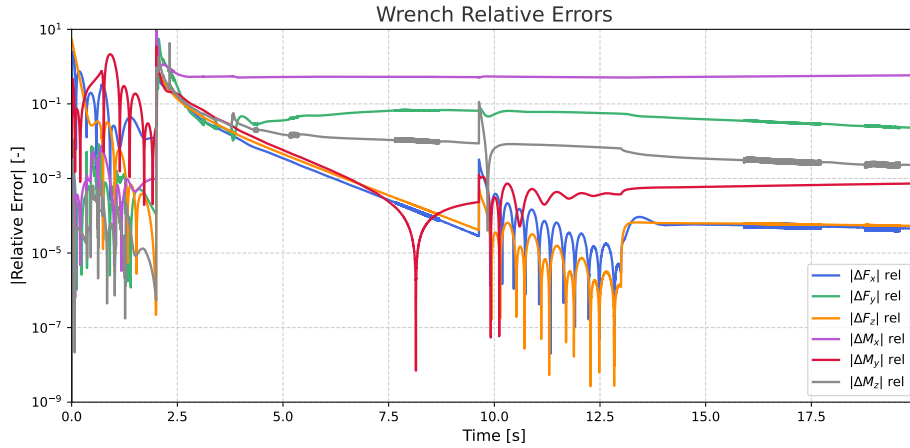


Figure 6.36: Relative error between real and estimated magnitudes.

6.4.3 Walking in streams

Finally, experiments regarding all the elements presented along this project are carried out. The observer (3.1) is going to track the external forces (2.4); the whole body control algorithm is going to adapt to these perturbations and is going to replan its tasks to take into account these effects (4.1.2, 5.5).

At first glance, for reduced values of speed current and low water height, the robot is able to walk properly and without any inconvenience. However, as the forces and water height increase, some discrepancies arise and the robot is not able to walk anymore.

If one recalls the hypothesis that the walker assumed:

1. Negligent angular momentum about the center of mass.
2. Negligent torque acting on the center of mass.
3. The robot's leg compensates the robot's weight.

All of them are, in a way or another, unfulfilled. First, and since the robot is walking, it generates angular momentum about the center of mass. Second, external forces are acting on the legs, that have a strong lever arm to the center of mass and thus generate momentum. Third, the robot's leg compensates vertical force, but this force is now damped by buoyancy forces. Contradictorily, more vertical force in the reaction relaxes the constraints for the cone compliance, hence the heavier the robot, the more stable its gait (against perturbations of the same magnitude). Since the

virtual weight of the robot is now reduced, its ability to remain stable also is drastically affected.

Even though forces are still being estimated accurately, the fact that they do not match the hypotheses anymore makes the robot unable to perform as precisely as when the forces were applied on its center of mass.

Also, it has been observed empirically that a reduction of the height of the center of mass is not helpful (since it enhances buoyancy forces), and an increase on its height is not either (since it is more prone to reaching singularities in the legs). A good balance has been found with $z_c = 0.62\text{ m}$. What really showed an increase in performance was a change in step length. It was changed from 0.1 m to 7 cm .

Anyway, one can plot the regions that are safe for the robot to walk (with this control architecture and gait configuration):

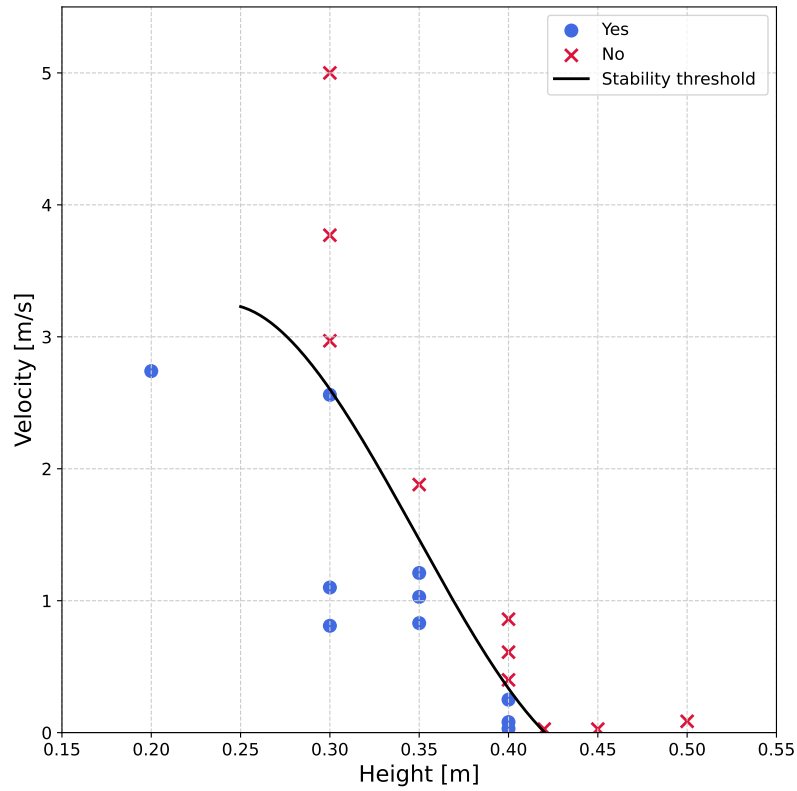


Figure 6.37: Stability threshold for current architecture and gait configuration.

Chapter 7

Summary and future work

This project has addressed the idea of robots walking in a very particular environment: floods and streams. The main objective was to build a simulation environment with enough fidelity to mimic these forces, as well as an architecture able to detect and react to external perturbations.

The first key contribution has been the modeling of hydrodynamic forces. Due to the unfeasibility of real-time CFD, a reduced order model has been developed. This model decomposes the robot's legs into more well-known and studied geometries, which help characterize aerodynamic coefficients in the complex flux regimes that this problem poses.

The second feature has been the implementation and analysis of an external force observer. Its theoretical convergence and consistence has been mathematically proven, and its ability to precisely estimate wrenches acting on the center of mass of the robot has been studied, with satisfactory results. Experiments show that the observer is robust, working either for point forces and distributed forces, laying the foundation for further compensation strategies.

From this estimation, improvements on the existing WBC framework have been introduced: external force inclusion in dynamics equation, ankle offset considerations and torque limiting have been successfully implemented and tested. Not only these features have made the robot more loyal to reality, but they also have drastically improved its stability.

Finally, an active force compensation strategy has been built, modifying the center of mass' trajectory in real time. This strategy is based on LIPM and DCM theory models. Experimental results are convincing: an increase of more than 100% on withstandable force in static conditions. The

final goal of the project, walking under river-like forces, validates the effectivity and integrity of the implemented architecture. Finally, a stability threshold has been proposed, confirming that the proposed framework can be effective and worked upon.

Despite the optimistic results, this project opens the door to numerous branches of research and development:

The used models of LIP and DCM assume hypotheses that are partially unfulfilled when the robot is immersed in the fluid. Priority lines of work would be to reformulate the trajectory generators to consider an explicit external wrench, and not just force.

Physical limits must be considered in the WBC quadratic problem in some sort of way, in order not to fight against them when the robot reaches physical limits.

The observer loses a lot of efficiency when it works during highly dynamic movements. This contradicts the quasistatic assumption. Hence, an observer able to model moment changes derived from locomotion would reduce estimation peaks and improve stability in high speed regimes or for more abrupt perturbations (and reactions).

Finally, in order to further test these results, a real platform could be used to validate the proposed theoretical and simulated results.

Author's contribution to the code can be found in <https://github.com/v2812ic/Aug9>.

Chapter 8

Methodology and planning

This chapter shows the methodology and schedule followed to develop this project.

The original idea for my stay in HCRL was the study and characterization of high-acceleration actuators. Due to initial conversations with my advisor and other members of the lab about the scope of the project, I knew that this was not going to get me motivated and switched to something more *softwarey*. After spending over a month reviewing literature about machine and reinforcement learning, and arriving to the conclusion that I needed to start working on something, I came up with this specific idea, that could be worked upon to an infinite extent, and could be developed in many ways. Again, after defining the idea with my advisor and some lab partners, I concluded that I could work on disturbance rejection, applied to a very specific problem such as fluid force rejection.

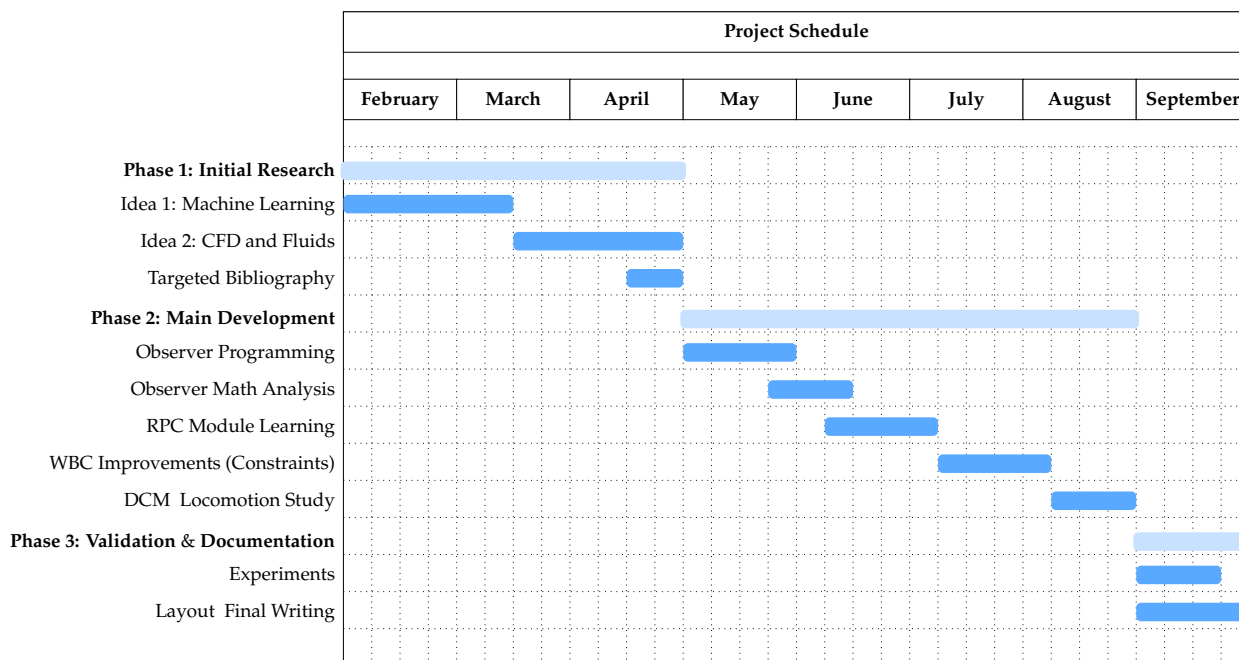
Four tasks were set from the start of the project that wanted to be accomplished:

- Characterize fluid forces acting on the robot.
- Build a momentum observer able to detect them.
- Update the Whole Body Controller to make it more robust against perturbations, first when the robot did not perform locomotion tasks.
- Update the locomotion tasks to take into account these forces.

In the end, tasks 3 and 4 were built together, although task 3 was first reached as it did not require a full understanding of the DCM models.

It was planned that the construction of the observer and the simulation environment would take at least two months; updates on the controller would take at least a month; and the study of locomotion tasks two more. Finally, data gathering and documentation would be the main focus on the last month.

However, self-taught learning of the rpc framework was not an easy task and delayed the schedule in the update and implementation of the improvements of the WBC. And, even though the main tasks were achieved and all the intended studies were carried out, I discovered further improvements (such as the quasistatic or the zero-moment-about-CoM assumptions), that could have been enhanced and further developed, had I had more time.



Chapter 9

Sustainability report

This chapter analyzes sustainability of this work from an environmental, social and economic perspectives, following CFIS and UPC guidelines.

9.1 Sustainability matrix

9.1.1 Environmental impact

This project has been entirely computational, and therefore it has not generated any residues itself, other than CO₂ due to computer usage. The amount of time dedicated to the project can be estimated in 30 hours per week during 8 months, giving a total of roughly 960 hours. The devices that have been used throughout the project have been an HP Envy 15-ep0000nx laptop, plus a couple of monitors depending on the moment of the stay (sometimes I could sit in a desk with monitors, sometimes I could not). This laptop has a battery of 83 Wh [29], and the average cycle was two recharges per day. Considering that a monitor has a power consumption of around 40 W, and that, in average, one of them was on while I was working, one can estimate a power consumption of 67.6 W. Times the amount of hours dedicated: 73.9 kWh. The city of Austin set an upper limit of 508 lbs of CO₂ per MWh generated [3], so, after some multiplication, the amount of CO₂ generated throughout the project, just due to energy consumption, has been around 17 kg CO₂e.

On top of this, the project has been a collaboration between UPC and UT Austin, and I have traveled between Spain (Malaga) and the United States (Austin) twice. According to [39], a return flight via these two cities generates 4.7 T CO₂ per passenger. Two trips equal 9.4 T CO₂.

In total, the amount of CO₂ emitted related to this project has been 9.4 T, coming almost everything from the intercontinental flights.

Other aspects that can be considered are the amortization of the computer and monitors, but since no information is provided by the manufacturers regarding production costs, it is omitted. Also, transport impact during the stay can also be neglected, since the main transport mean has been either on foot or by bike. This bike was purchased in a benefic shop: *Yellow bike project* that contributes to circular economy, and has been given away to a friend after its use.

The environmental risks that this project entails come from many places:

- First, tests must be carried out, needing a platform (namely, a robot) to carry out experiments and controlled environments to do them. Catastrophic failures in the experiments could lead to full hardware replacement.
- Also, failures during actual operation can convert the robot into a technological residue, with the risk of water and land being contaminated by its batteries.

However, and optimistically, the positive impact that the robot can have is of great consideration, since it could be able to complete missions concerning dams, oleoducts... where a quick response (that a human might not be able to provide) can save or prevent a bigger catastrophe.

Finally, the impact of the fabrication of test platforms and hardware has not been considered as it falls out of the scope of the project.

9.1.2 Economic impact

The economic costs can be divided into human and material costs. First, I have dedicated around 960 hours to the project. These hours do not have any cost associated, since I have conducted the project as my Final Degree Thesis and I was not explicitly paid. However, I have received two grants: one from Fundació Mir-Puig and another one from Generalitat de Catalunya that have kindly helped with the expenses associated to life in Austin.

As to material costs, the computers and monitors were already there and nothing was purchased in relation with the project. Anyway, if one considers that the price of the computer was 1500 \$, that it was purchased when I started the degree and that it has an expected lifespan of 3 more years (in total, 8), the cost associated to this fact is 125\$. The only tangible cost associated to the

thesis has been the subscription to Overleaf for the last two months (42\$ in total) to write the contents when they started to get extensive. This could have been avoided had I used local versions of $\LaTeX$, but since a huge portion of the contents were already drafted, and I prefer this online version, I opted to stay with this option.

The material costs associated to tests and development in real platforms may vary depending on hardware used or failures. If this idea is, indeed, successful and is propitiously implemented and becomes a startup, the economic impact can even become positive.

9.1.3 Social impact

The development of the project has had important development of my competencies in many areas: robotics, fluids, control... The international collaboration with UT Austin has propitiated scientific and cultural exchange.

This project utilizes impersonal technical language, by any mean sexist.

If finally executed, the project can help many collectives: people affected by floods, rescue teams, maintenance workers... However, and in a very long time span, it might move professionals away from their jobs. I hope that, if this finally happens (which is definitely not the intention), it is in favor of safety or quick response that a limited amount of human beings cannot bear. Nonetheless, this is all speculative, since the idea has just been posed and many, many more advances need to be made before any deployment is seen.

9.2 Ethical implications

This project is the opening door for a branch of investigation regarding robots operating and moving in hazardous environments, such as flooded areas. The principal motivation is protection of human life, by safely performing reconnaissance, exploration or rescue tasks.

This can carry ethical implications in many ways. First, this robust locomotion technologies could be implemented in military robots, that need to move and carry out missions concerning human ethics. Also, if these methods were implemented in autonomous robots, a very interesting debate will arise examining decision-making capabilities of a machine in life or death situations.

Finally, this work respects ethical and integrity codes of UPC.

9.3 Relationship with the Sustainable Development Goals

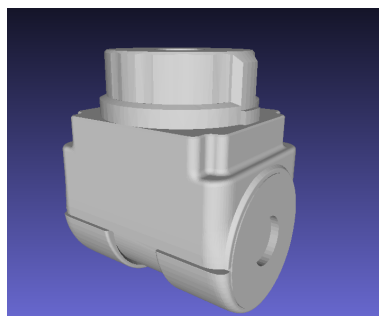
According to [41] proposed by the United Nations, this project contributes mainly to the following sustainable development objectives:

- **9. Industry, Innovation and Infrastructure.** In particular, it fits **9.5: *Enhance scientific research, upgrade the technological capabilities of industrial sectors [...].*** Indeed, this work aims to explore a very particular research branch with the objective to better understand and dominate the behavior of robots operating under river forces.
- **11. Sustainable Cities and Communities.** Particularly, **11.5 and 11.c: *Significantly reduce the number of deaths and the number of people affected and substantially decrease the economic losses relative to gross domestic product caused by disasters, including water-related disasters[...], Substantially increase the number of cities and human settlements adopting and implementing integrated policies and plans towards inclusion, resource efficiency, mitigation and adaptation to climate change, resilience to disasters, and develop and implement, in line with the Sendai Framework for Disaster Risk Reduction 2015-2030, holistic disaster risk management at all levels.*** The initial idea (and ultimate goal) is to study methods to implement in robots whose mission is going to be to operate and move in hazardous environments, providing help to emergency crew in SAR missions, among others.
- **13. Climate Action.** **13.1: *Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries.*** Indeed, until we are not able to control climate, we will need a way to quickly and resiliently react when disaster strikes.

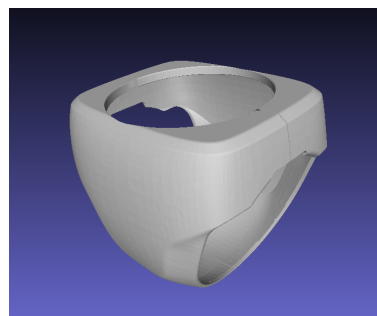
Appendix A

Links of the robot

Here, a decomposition of the leg into its links is shown. Each link is considered to belong to a certain section of the leg.



(a) pelvis



(b) pelvis_contour_link

Figure A.1: Pelvis links.

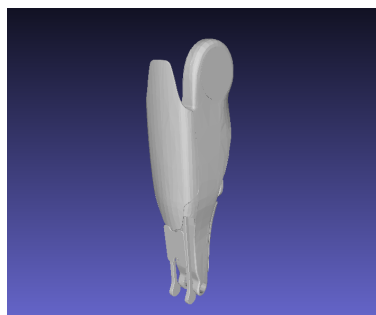
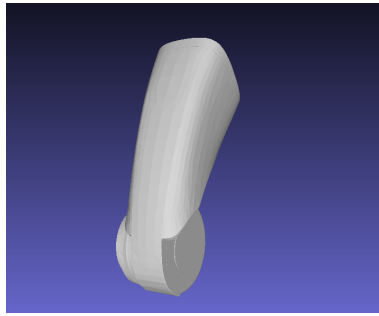
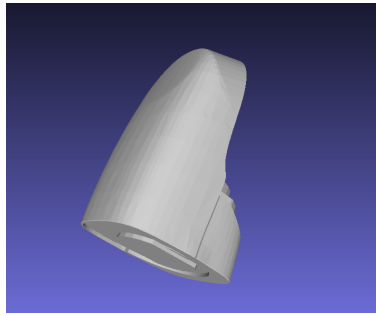


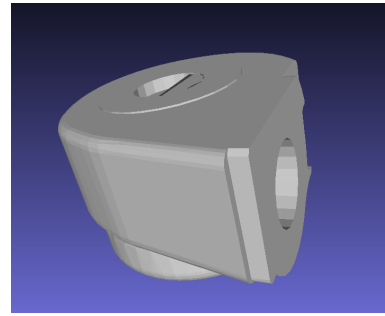
Figure A.2: Shin link: knee_link.



(a) hip_yaw_link



(b) hip_roll_link



(c) hip_pitch_link

Figure A.3: Thigh links.

Appendix B

Inverse of a subdiagonal block matrix

A proof for proposition 3.1.2 is provided. Given the presented shape of matrix $\underline{M}$, it is shown that:

$$\{\underline{M}^{-1}\}_{ij} = \begin{cases} 0 & j > i \\ \underline{B}_i^{-1} & j = i \\ (-1)^{i-j} \prod_{z=j}^i \underline{B}_z^{-1} \prod_{z=j+1}^i \underline{A}_z & j < i \end{cases}$$

It is proven by direct multiplying.

- The first row ($i = 1$) matches trivially.
- The formula also holds for the following rows:

$$\{\underline{M}\underline{M}^{-1}\}_{ij} = \underline{B}_i \{\underline{M}^{-1}\}_{ij} + \underline{A}_i \{\underline{M}^{-1}\}_{i-1j}$$

If $j > i$, none of the blocks of $\underline{M}^{-1}$ that appear in the sum are non-zero. So:

$$\{\underline{M}\underline{M}^{-1}\}_{ij} = \underline{0} \quad j > i$$

If $j = i$, only the first addend is non-zero. Specifically, it is the identity:

$$\{\underline{M}\underline{M}^{-1}\}_{ij} = \underline{I} \quad j = i$$

If $j = i - 1$ both terms appear:

$$\left\{ \underline{M} \underline{M}^{-1} \right\}_{ij} = -\underline{B}_i \underline{B}_{i-1}^{-1} \underline{B}_i^{-1} \underline{A}_i + \underline{A}_i \underline{B}_{i-1}^{-1}$$

which is clearly zero, since every pair of matrices commutes.

Finally, for a general case of $j < i$:

$$\begin{aligned} \left\{ \underline{M} \underline{M}^{-1} \right\}_{ij} &= \underline{B}_i (-1)^{i-j} \prod_{z=j}^i \underline{B}_z^{-1} \prod_{z=j+1}^i \underline{A}_z + \underline{A}_i (-1)^{i-j-1} \prod_{z=j}^{i-1} \underline{B}_z^{-1} \prod_{z=j+1}^{i-1} \underline{A}_z \\ &= (-1)^{i-j} \underline{A}_i \underline{B}_i \underline{B}_i^{-1} \prod_{z=j}^{i-1} \underline{B}_z^{-1} \prod_{z=j+1}^{i-1} \underline{A}_z - (-1)^{i-j} \underline{A}_i \prod_{z=j}^{i-1} \underline{B}_z^{-1} \prod_{z=j+1}^{i-1} \underline{A}_z \end{aligned}$$

Again, this is zero due to the commutativity of the matrices.

Appendix C

Derivation of complex matrix operations

This section proves the operations carried out in section 3.1.2. For notational convenience, the following block-triangular matrix

$$\underline{P} := (\underline{I} - \underline{C})^{-1}$$

that replaces every occurrence of $(\underline{I} - \underline{C})^{-1}$. In particular, the expressions shown are:

- $\underline{P}\underline{A}$,
- $\underline{P}\underline{A}\bar{\gamma}^* + \underline{P}\underline{B}$.

C.1 PA

Blockwise, the entries of $\underline{P}$ satisfy:

$$\{\underline{P}\}_{ij} = \begin{cases} 0 & j > i \\ \underline{I} & j = i \\ (\Delta t)^{i-j} \prod_{z=j+1}^i \underline{K}_z & j < i \end{cases}$$

The blocks of $\underline{A}$ are defined by:

$$\{\underline{\mathbf{A}}\}_{ij} = \begin{cases} \underline{\mathbf{I}} & i = j & i \neq 1, r \\ \underline{\mathbf{I}} - \Delta t \underline{\mathbf{K}}_1 & i = 1 & j = r \\ -\Delta t \underline{\mathbf{K}}_i & 2 \leq i \leq r-1 & j = r \\ \underline{\mathbf{I}} - \Delta t \underline{\mathbf{K}}_r & i = r & j = r \\ \underline{\mathbf{0}} & \text{otherwise} \end{cases}$$

The blockwise multiplication rule states:

$$\{\underline{\mathbf{PA}}\}_{ij} = \sum_{s=1}^r \{\underline{\mathbf{P}}\}_{is} \{\underline{\mathbf{A}}\}_{sj} \quad (\text{C.1})$$

The resulting blocks are obtained as follows ¹:

- **First column** ($j = 1$). Since $\{\underline{\mathbf{A}}\}_{i1} = \underline{\mathbf{0}}$,

$$\{\underline{\mathbf{PA}}\}_{i1} = \underline{\mathbf{0}}$$

- **Intermediate columns** ($j = 2, \dots, r-1$). The only non-zero block in the j -th column of $\underline{\mathbf{A}}$ is $\{\underline{\mathbf{A}}\}_{jj} = \underline{\mathbf{I}}$; therefore

$$\{\underline{\mathbf{PA}}\}_{ij} = \{\underline{\mathbf{P}}\}_{ij} \quad j = 2, \dots, r-1$$

- **Last column** ($j = r$). By equation C.1, and the definition of $\underline{\mathbf{A}}$:

$$\{\underline{\mathbf{PA}}\}_{ir} = \sum_{s=1}^r \{\underline{\mathbf{P}}\}_{is} \{\underline{\mathbf{A}}\}_{sr} = \{\underline{\mathbf{P}}\}_{i1}(\underline{\mathbf{I}} - \Delta t \underline{\mathbf{K}}_1) + \{\underline{\mathbf{P}}\}_{ir}(\underline{\mathbf{I}} - \Delta t \underline{\mathbf{K}}_r) - \Delta t \sum_{s=2}^{r-1} \{\underline{\mathbf{P}}\}_{is} \underline{\mathbf{K}}_s \quad (\text{C.2})$$

The property of $\underline{\mathbf{P}}$ being triangular is helpful here, since it establishes a limit in the amount of addends. Let's break this sum into three different cases:

1. **First row** ($i = 1$). In this case, the only existent block of $\underline{\mathbf{P}}$ is $\{\underline{\mathbf{P}}\}_{11} = \underline{\mathbf{I}}$. Therefore:

$$\{\underline{\mathbf{PA}}\}_{1r} = \underline{\mathbf{I}} - \Delta t \underline{\mathbf{K}}_1$$

¹For sizes greater than 1 ($r > 1$).

2. **Intermediate rows** ($i = 2, \dots, r-1$). Equation C.2 can be simplified to:

$$\begin{aligned} \{\underline{PA}\}_{ir} &= \{\underline{P}\}_{i1}(\underline{I} - \Delta t \underline{K}_1) - \Delta t \sum_{s=2}^{i-1} \{\underline{P}\}_{is} \underline{K}_s - \Delta t \underline{K}_i \\ &= (\underline{I} - \Delta t \underline{K}_1)(\Delta t)^{i-1} \prod_{z=2}^i \underline{K}_z - \Delta t \sum_{s=2}^{i-1} \underline{K}_s (\Delta t)^{i-s} \prod_{z=s+1}^i \underline{K}_z - \Delta t \underline{K}_i \end{aligned}$$

3. **Last row** ($i = r$). In this last case, it is just needed to add the last term to the sum:

$$\{\underline{PA}\}_{rr} = (\underline{I} - \Delta t \underline{K}_1)(\Delta t)^{r-1} \prod_{z=2}^r \underline{K}_z - \Delta t \sum_{s=2}^{r-1} \underline{K}_s (\Delta t)^{r-s} \prod_{z=s+1}^r \underline{K}_z + (\underline{I} - \Delta t \underline{K}_r)$$

Note that the sum from $s = 2$ to $i - 1$ only makes sense if $i > 2$. When computing the second row ($i = 2$), this addend is nonexistent.

C.2 $\underline{P}\underline{A}\gamma + \underline{P}\underline{B}$

The intention of computing $\underline{P}\underline{A}$ is no other than to address the consistency of method 3.19. Here, a part of the proof of such consistency is proposed.

Let's first recall that $\gamma_i = m^*$, since the dynamics of the system are supposed to be satisfied. Therefore, the goal is to show that:

$$-\underline{P}\underline{A} \begin{bmatrix} m^* \\ m^* \\ \vdots \\ m^* \end{bmatrix} + \underline{P}\underline{B} = - \begin{bmatrix} m^* \\ m^* \\ \vdots \\ m^* \end{bmatrix} \quad (\text{C.3})$$

First, it is easy to state that:

$$\underline{P}\underline{B} = \begin{bmatrix} -\underline{K}_1 m^* \Delta t \\ -\underline{K}_1 \underline{K}_2 m^* \Delta t \\ \vdots \\ -(\prod_{s=1}^r \underline{K}_s) m^* (\Delta t)^r \end{bmatrix} = - \begin{bmatrix} \Delta t \underline{K}_1 & \underline{0} & \cdots & \underline{0} \\ \underline{0} & (\Delta t)^2 \underline{K}_1 \underline{K}_2 & \cdots & \underline{0} \\ \vdots & \vdots & \ddots & \vdots \\ \underline{0} & \underline{0} & \cdots & (\Delta t)^r \prod_{s=1}^r \underline{K}_s \end{bmatrix} \begin{bmatrix} m^* \\ m^* \\ \vdots \\ m^* \end{bmatrix}$$

So, equation C.3 can be rewritten as:

$$- \left(\underline{P}\underline{A} + \begin{bmatrix} \Delta t \underline{K}_1 & \underline{0} & \cdots & \underline{0} \\ \underline{0} & (\Delta t)^2 \underline{K}_1 \underline{K}_2 & \cdots & \underline{0} \\ \vdots & \vdots & \ddots & \vdots \\ \underline{0} & \underline{0} & \cdots & (\Delta t)^r \prod_{s=1}^r \underline{K}_s \end{bmatrix} \right) \begin{bmatrix} m^* \\ m^* \\ \vdots \\ m^* \end{bmatrix} = - \begin{bmatrix} m^* \\ m^* \\ \vdots \\ m^* \end{bmatrix} \quad (\text{C.4})$$

Where, according to the problem structure, it is needed to show that the sum of all the blocks in a block-row of the matrix is the identity. That way, equation C.3 would ultimately hold.

- **First row.** Trivial, since:

$$\underline{I} - \Delta t \underline{K}_1 + \Delta t \underline{K}_1 = \underline{I}$$

- **Intermediate rows.** Only the sum of the block-rows of $\underline{PA}$ is studied:

$$\begin{aligned}
 \sum_{s=1}^r \{\underline{PA}\}_{is} &= \sum_{s=2}^{i-1} \{\underline{PA}\}_{is} + \{\underline{PA}\}_{ii} + \{\underline{PA}\}_{ir} \\
 &= \sum_{s=2}^{i-1} (\Delta t)^{i-s} \prod_{z=s+1}^i \underline{K}_z + \underline{I} + (\underline{I} - \Delta t \underline{K}_1) (\Delta t)^{i-1} \prod_{z=2}^i \underline{K}_s - \\
 &\quad - \sum_{s=2}^{i-1} \underline{K}_s (\Delta t)^{i-s+1} \prod_{z=s+1}^i \underline{K}_z - \Delta t \underline{K}_i \\
 &= \dots \\
 &= \sum_{s=2}^{i-1} \left\{ (\Delta t)^{i-s} \prod_{z=s+1}^i \underline{K}_z (\underline{I} - \Delta t \underline{K}_s) \right\} - \Delta t \underline{K}_i + \\
 &\quad + (\Delta t)^{i-1} \prod_{z=2}^i \underline{K}_z - (\Delta t)^i \prod_{z=1}^i \underline{K}_z + \underline{I}
 \end{aligned}$$

Note that the sum in this expression is telescopic and can be simplified. Let's start by defining each term of the sum as:

$$\begin{aligned}
 \underline{D}_{is} &:= (\Delta t)^{i-s} \prod_{z=s+1}^i \underline{K}_z (\underline{I} - \Delta t \underline{K}_s) \\
 &= (\Delta t)^{i-s} \prod_{z=s+1}^i \underline{K}_z - (\Delta t)^{i-s+1} \prod_{z=s}^i \underline{K}_z
 \end{aligned}$$

If one also defines:

$$\underline{E}_{is} := (\Delta t)^{i-s} \prod_{z=s+1}^i \underline{K}_z$$

the following relation is obtained:

$$\underline{D}_{is} = \underline{E}_{is} - \underline{E}_{i\ s-1}$$

Consequently, adding all the addends $\underline{D}_s$ from $s = 2$ to $s = i - 1$:

$$\sum_{s=2}^{i-1} \underline{D}_{is} = \underline{E}_{i\ i-1} - \underline{E}_{i1} = \Delta t \underline{K}_i - (\Delta t)^{i-1} \prod_{z=2}^i \underline{K}_z$$

yielding that, the final result for the sum of the blocks of the i -th block-row ($i \neq r$) of $\underline{PA}$ is:

$$\sum_{s=1}^r \{\underline{PA}\}_{is} = \underline{I} - (\Delta t)^i \prod_{z=1}^i \underline{K}_z \quad i \neq r \quad (\text{C.5})$$

- **Last row.** Again, only the sum of the block-rows of $\underline{PA}$ is studied.

$$\begin{aligned} \sum_{s=1}^r \{\underline{PA}\}_{rs} &= \sum_{s=2}^{r-1} \{\underline{PA}\}_{rs} + \{\underline{PA}\}_{rr} \\ &= \sum_{s=2}^{r-1} (\Delta t)^{i-s} \prod_{z=s+1}^r \underline{K}_z + (\underline{I} - \Delta t \underline{K}_1) (\Delta t)^{r-1} \prod_{z=2}^r \underline{K}_z - \\ &\quad - \Delta t \sum_{s=2}^{r-1} \underline{K}_s (\Delta t)^{r-s} \prod_{z=s+1}^r \underline{K}_z + (\underline{I} - \Delta t \underline{K}_r) \\ &= \dots \\ &= \sum_{s=2}^{r-1} \left\{ (\Delta t)^{i-s} \prod_{z=s+1}^i \underline{K}_z (\underline{I} - \Delta t \underline{K}_s) \right\} + (\Delta t)^r \prod_{z=2}^r \underline{K}_z - \\ &\quad - (\Delta t)^r \prod_{z=1}^r \underline{K}_z + \underline{I} - \Delta t \underline{K}_r \end{aligned}$$

Since the sum is the same telescopic one as for intermediate rows, the above expression can be simplified, and the final result of the sum of the blocks of the r -th block-row of $\underline{PA}$ is:

$$\sum_{s=1}^r \{\underline{PA}\}_{rs} = \underline{I} - (\Delta t)^r \prod_{z=1}^r \underline{K}_z \quad (\text{C.6})$$

So, the sum of the blocks of the i -th row of $\underline{PA}$ follows the equation:

$$\sum_{s=1}^r \{\underline{PA}\}_{is} = \underline{I} - (\Delta t)^i \prod_{z=1}^i \underline{K}_z \quad i = 1, \dots, r \quad (\text{C.7})$$

Finally, the block-rows of the matrix in equation C.4 directly sum the identity matrix. This implies that equation C.4 indeed holds.

Appendix D

Cloud point method for Archimedes' force

Even though the method was not used in the end, the author considers that its utility might be useful for more complex geometries and/or scenarios, and wants to leave it in the report.

It is assumed that the water height is equal to h . To determine the robot's submerged volume, the structure is decomposed into individual links, and the submerged fraction of each link is analyzed separately. For each link i , the buoyant force is

$$F_{\text{float},i} = g \rho V_i f_i \quad (\text{D.1})$$

where V_i is the total volume of link i (obtained via `pybullet`) and f_i its submerged fraction. Thus, the remaining task is to estimate f_i .

A cloud of N points $\{\mathbf{x}_{k,i}\}_{k=1}^N$ is generated uniformly at random within link i . The indicator variable

$$\mathbb{1}_i(z_{k,i} < h)$$

is defined to be 1 if the z -coordinate of $\mathbf{x}_{k,i} = (x_{k,i}, y_{k,i}, z_{k,i})$ in world coordinates falls below the water level h , and 0 otherwise. Its success probability equals 1 for a fully submerged link, 0 for a fully emerged link, and f_i for a partially submerged one. Consequently, the number of submerged points in link i follows

$$\mathbb{X}_{N,i} := \sum_{k=1}^N \mathbb{1}_i(z_{k,i} < h) \sim \text{Bin}(N, f_i), \quad (\text{D.2})$$

and the estimator for the fraction of the submerged volume:

$$\hat{f}_{N,i} := \frac{1}{N} \mathbb{X}_{N,i} \quad (\text{D.3})$$

is unbiased and consistent, since

$$\mathbb{E}[\hat{f}_{N,i}] = \frac{1}{N} \mathbb{E}[\mathbb{X}_{N,i}] = f_i.$$

An error bound and confidence level are obtained by invoking the Central Limit Theorem:

Theorem

Theorem D.0.1. Central Limit Theorem

Let $(X_i)_{i=1}^n$ be IID random variables with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Define

$$S_n = \frac{1}{\sigma\sqrt{n}} \sum_{i=1}^n (X_i - \mu).$$

Then for all $x \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \Pr(S_n \leq x) = \Phi(x),$$

where Φ is the CDF of the standard normal distribution.

By application of this theorem to $\hat{f}_{N,i}$, one obtains the approximation

$$\hat{f}_{N,i} \approx \mathcal{N}\left(f_i, \frac{f_i(1-f_i)}{N}\right), \quad (\text{D.4})$$

from which confidence intervals may be constructed.¹

For a confidence level $1 - \alpha$, such an interval is obtained as:

$$\hat{f}_{N,i} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{f}_{N,i} (1 - \hat{f}_{N,i})}{N}} \quad (\text{D.5})$$

¹Note that Theorem D.0.1 is applicable to this estimator because $\mathbb{1}_i(z_{k,i} < h)$ satisfy the conditions of such Theorem.

Therefore:

$$|\hat{f}_{N,i} - f_i| \leq z_{1-\alpha/2} \sqrt{\frac{\hat{f}_{N,i} (1 - \hat{f}_{N,i})}{N}} \quad (\text{D.6})$$

The right-hand side of this equation trivially reaches a maximum when $\hat{f}_{N,i} = 0.5$, so naturally:

$$|\hat{f}_{N,i} - f_i| \leq \frac{z_{1-\alpha/2}}{2\sqrt{N}} \quad (\text{D.7})$$

Finally, if one establishes a bound for the absolute error of the estimation ϵ , the necessary N to achieve such error is, given a confidence level $1 - \alpha$:

$$N \geq \frac{z_{1-\alpha/2}^2}{4\epsilon^2} \quad (\text{D.8})$$

It is not possible to obtain a value of N dependent on the relative error in a non-overcomplicated way. This would imply the precomputation of a point cloud to make a first estimation of the parameters, which would make this process repetitive and heavy. It is considered that the bound of an absolute error is a good enough approximation. Recall that the precision in the modeling of the forces is sought only in the order of magnitude of such forces.

The point of application $\mathbf{x}_F$ of the buoyant force is approximated using the same point cloud. Since the resultant buoyant force acts at the centroid of the submerged volume,

$$\mathbf{x}_F = \frac{1}{N} \sum_{k=1}^N \mathbf{x}_k \mathbb{1}(z_k < h). \quad (\text{D.9})$$

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